

Reproducing the LHS 1140 b helium line with a solved wind: how far EXHALE gets, and what is missing

1. Goal, and the answer up front

The objective is to reproduce the measured He I 10830 Å transit line of LHS 1140 b with EXHALE, which solves the wind rather than imposing it, and to establish what — if anything — has to be added to the physics for the model to match the observation.

Result. The *amount* of absorption can be matched, but the composition it takes is not one number: it splits in two according to what is placed at the lower boundary. Above a prescribed metal-free scalar base the measured equivalent width is reached at He/H = 1.63; above a lower atmosphere solved as a photochemical column and handed over as a profile, C, N and O included, it is reached at He/H = 9.05. Both runs have binary H/He element diffusion active and both sit at the same constant $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ — a stated number, and stated with no pressure dependence: what that assumption carries is measured in Sect. 11.3 — so what separates the two values is the base and not the mixing. The *width* is not matched over either base, and the velocity it asks for is the same number every time. Solving the lower atmosphere instead of prescribing it does not remove the match — it moves it.

What this revision changed, and what it did not. Every composition quoted here is lower than the value the earlier draft of this memo carried, by a nearly uniform factor. The cause is one rate coefficient: the He(2^3S) destruction rate on atomic hydrogen was the Taylor et al. (2025) two-branch fit, which jumps by 1.5724 at 4000 K, and it has been replaced by the García Muñoz (2025) closed form, which is continuous (Sect. 2.5). The new coefficient is smaller everywhere in this wind, so the metastable level lives longer, the line is stronger at fixed composition, and every crossing moves down. Measured on re-solved composition ladders rather than scaled from a slope, the five crossings are 0.74–0.81 of what they were, and the local exponent $d \log(\text{He}/\text{H})/d \log K_{zz}$ is the same to within 0.01–0.02 at every decade. *The structure of the result — that the composition this line implies is conditional on the assumed spectrum and on what is placed at the lower boundary, and only weakly on the assumed mixing — survives unchanged; what moved is the level, by about 20 per cent.* The ratios between the branches are the ones the earlier draft had: the two bases still differ by a factor 5.6, the well-mixed and $K_{zz} = 10^9$ crossings by 3.8, and the two proxy spectra by 9.

(i) The amount is always matchable, so what the line returns is a conditional pair rather than a measurement.

The prescribed-base branch. With the base a scalar state carrying no metals, every combination of assumed spectrum and assumed mixing reaches the observed red-pair equivalent width at some helium abundance, and the conditions stack inside this branch. In the well-mixed limit the crossing is He/H = 0.4245 on the GJ 1132 proxy spectrum and 0.0479, below solar, on the GJ 699 one — the second stand-in for LHS 1140's unobserved XUV, 5.1 times brighter, which the paper also uses. With binary H/He element diffusion active — helium then settles out of the base unless an eddy term lifts it — it rises again, to 3.81 wherever $K_{zz} \leq 10^4 \text{ cm}^2 \text{ s}^{-1}$ does nothing against the molecular coefficient, 1.6261 at the adopted $K_{zz} = 10^9$ and 0.847 at 10^{11} (Sect. 8.4). The dependence on the eddy coefficient is weak and monotonic, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all; the dependence on the assumed spectrum is a factor 9.

The solved-column branch. Replacing that base with a photochemical column — Photochem with its climate step solved, handed to EXHALE as a profile at a $1\ \mu\text{bar}$ match, at the same $K_{zz} = 10^9$ and with the same diffusion active — brings C, N and O into the base, and the same reservoir then absorbs about $2.6\times$ less (Sect. 11.6, re-measured on the current coefficient). A ladder of eleven solutions on this base, each closed on its own elemental fluxes, has two rungs that straddle the measurement directly, 9.01 below it and 9.11 above, and crosses it at $\text{He}/\text{H} = 9.048$ (8.465–10.736 on the measurement's 1σ interval) (Sect. 11.7).

Neither number is an independent prediction of the model. Each is the reservoir the observed line picks out over its own base, and the escape rate — which is an output, not a fit — barely moves along the way. The flux closure does not change that on the profile branch. It returns $\text{He}/\text{H} = 2.09235$ at the match from trial fluxes a factor of ten apart, single-valued to 5.5×10^{-6} (Sect. 11.), but at $K_{zz} = 10^9$ the eddy term homogenizes the column all the way to the microbar match, so only 2×10^{-4} of that answer comes from the escape flux: the closure hands back the reservoir it was given. And the reservoir it was given, 2.09, was itself a value fitted to this same line over the earlier metal-free scalar base. What the closure adds is that a composition is consistent with a photochemical lower atmosphere carrying its own C, N and O, not that it is measured. Every rung of the ladder satisfies the closure at residuals of 2.3–4.9% against its 5% tolerance; the line is what selects one.

Why the same helium absorbs differently over the two bases. The base metals are a thermostat setting, and helium is the other half of the same thermostat. The photochemical column's C, N and O cool the wind — T_{max} from 5498 to 3943 K — and although the metastable peak itself rises, 50.8 to $57.7\ \text{cm}^{-3}$, the cooler wind does not carry the population outward, so the metastable column above $2R_p$ falls by 2.5 and the line weakens by about the same factor, 2.6. Raising the reservoir reheats the wind: at $\text{He}/\text{H} = 10.31$ the mean mass per particle at $2R_p$ is $2.53 m_{\text{H}}$ against 1.50, the same deposited energy is shared among fewer and heavier particles, the coolant is diluted with the hydrogen it is referred to, and T_{max} returns to 5914 K — past the metal-free 5498. The ladder is therefore closer to a thermal trade between base metal content and helium content than to a determination of composition, and this one line cannot separate the two (Sect. 11.6, Fig. 17).

The escape rate is flat on every axis, and end to end the composition spans a factor 189. $\log_{10}(\dot{M}/\text{g s}^{-1}) \simeq 7.7\text{--}7.8$ on the GJ 1132 proxy across composition, diffusion and eddy coefficient alike, 7.39–7.40 across the whole closed ladder, and 8.58 on the GJ 699 proxy — there within 0.05 dex of what the paper itself retrieves on that same proxy (Sect. 10.). Against that, the composition the line implies runs from 0.0479 to 9.05 over choices this measurement cannot arbitrate: strongly conditional on the assumed spectrum and on what is placed at the lower boundary, weakly and monotonically on the assumed mixing. Any composition read off this line — the paper's $\text{H}:\text{He} \lesssim 10^{-3}$ as much as any value here — inherits those conditions.

(ii) The width is not reached on any axis, and the deficit is invariant. At the equivalent-width crossing the profile is about 3.3 times too deep and 3.3 times too narrow, and no composition fixes it, because composition changes how much helium absorbs, not how the absorption is spread in velocity. Measured directly — convolving the solved profile with a Gaussian and asking which one reproduces the observed line — the missing spread is $22.22\ \text{km s}^{-1}$ FWHM ($\sigma = 9.4\ \text{km s}^{-1}$) in the well-mixed limit, $22.23\ \text{km s}^{-1}$ with diffusion and the adopted K_{zz} , $21.95\ \text{km s}^{-1}$ for the flux-closed solution on the solved photochemical column, and $21.97\ \text{km s}^{-1}$ on the $5.1\times$ brighter spectrum — a spread of

1.3 per cent over four solutions that differ by a factor 189 in composition (Sects. 8.2, 8.4, 11.7 and 10.). Supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, with nothing tuned to them; what produces it is unidentified. It is not thermal — the solved gas cools below 1000 K over the line-forming region — and not bulk outflow, which reaches 1.2 km s^{-1} at $20R_p$ on the fainter spectrum and 3.6 on the brighter. Read as turbulence it is $\mathcal{M} = 2.4\text{--}4.2$ against the sound speed these solutions themselves carry, supersonic on every convention and well above what the literature is willing to assume. The one mechanism that would supply width without a free parameter — the adiabatic-cooling metastable bump of Schulik & Owen (2025) — is present in these solutions and measured, but sits at $0.25\text{--}1.5 \text{ km s}^{-1}$ (Sect. 8.6). The shortfall is in the energy and momentum budget, not in the composition, and it is a published problem rather than a defect of this setup (Sect. 8.).

The paper's own p-winds fit does not reproduce the line either. It is broader than the wind solved here, because its temperature is imposed at 5000–6000 K where the line forms, but at FWHM $0.575 \text{ \AA}$ with its turbulence term on (0.458 without it) it is still 1.5 times narrower than the measured 0.841, and it overshoots the blue component threefold. Adding broadening does not rescue it: its absorbing power is 24% below the measurement, so no convolution can reach the observed depth and width at once. *The width is a shortfall the two codes share*, which is what the audit of the measurement side (Sect. 8.3) leaves as the reading: the measurement is not at fault, and neither one-dimensional steady-state wind reproduces it.

A caveat that attaches to the escape rate alone. These solutions stay subsonic out to $30R_p$, so their mass flux is set at the outer boundary rather than at a critical point, and the gas is already collisionless there ($\text{Kn} = 1.4\text{--}2.4$): the $\dot{M}$ quoted throughout is *unvalidated in the collisional sense*, while the He 10830 layer at $1.1\text{--}3R_p$ sits at $\text{Kn} \leq 0.019$ and the line analysis is untouched (Sect. 12.).

Four systematics ride on the absolute numbers, and none of them is caught by a convergence test. They are stated here because they bound how far any single crossing should be read.

1. *The outer region is inherited from the seed.* A converged wind takes its $20\text{--}30R_p$ state from the solution it was started from, the JFNK finish does not re-solve it, and $\|R\|$ cannot separate two such states (Sect. 11.8; Update_EXHALE.md Sect. 83). That region lies outside the exobase, where the fluid solution is not valid in any case, so the absolute equivalent width of every rung of every ladder carries a systematic the closure tolerance does not reach.
2. *The closure crossing carries a seed systematic of order half a unit in reservoir.* Re-closing an arm from a neighbour instead of from the common parent moves its equivalent width by 0.2–1.9 per cent, which at the local slope is a large fraction of a unit in He/H, and no test the driver applies sees it (Sect. 11.7).
3. *The closure tolerance itself is visible in the answer.* At $\text{tol} = 0.05$ the ladder steps by $0.0122 \text{ \% \AA}$ between reservoirs 8.45 and 8.50 and by 0.0048 over the three times wider interval 8.50–8.65. Tightening it to 0.008 is refused by the driver, because the radial spread of the flux over the steady window would then exceed the residual being decided on it.
4. *Which chemistry produced the column.* The earlier results were made with Photochem 0.8.4; the closure results here are Photochem 0.9.0 with the local corrections of docs/photochem_solver_modification_*.md. On the

seven arms that can be compared like for like the equivalent width moves by at most 1.2×10^{-6} in relative terms and $\log_{10} \dot{M}$ does not move in its fourth decimal, *even though the correction takes the deepest-level elemental residual from 2×10^{-5} to 10^{-14}* — nine orders of magnitude in the equilibrium residual, and nothing in the observable (Sect. 11.7).

Everything below uses inputs verified against the authors' own released material (Zenodo 20723095): their p-winds driver script, their Mega-MUSCLES v23 spectra, their reduced transmission spectrum and their likelihood grid. Products live in LHS1140b/; the notebook LHS1140b/LHS1140b_analysis.ipynb regenerates every number and figure here.

2. Inputs, and how each was pinned

2.1 System

Planet and star from Cadieux et al. (2024) as quoted by C26: $R_p = 1.730 \pm 0.025 R_{\oplus}$, $M_p = 5.60 \pm 0.19 M_{\oplus}$, $T_{\text{eq}} = 226 \text{ K}$, $a = 0.0946 \text{ AU}$, $M_{\star} = 0.1844 M_{\odot}$, $R_{\star} = 0.2159 R_{\odot}$. One trap: EXHALE converts R_p with its own internal constants (mean radii, $R_J = 6.9911 \times 10^9 \text{ cm}$, $R_{\oplus} = 6.3725 \times 10^8 \text{ cm}$), so the input must be built with those and not with IAU equatorial values — a 2.2% error in radius, 4.4% in transit depth, if taken from the wrong pair.

2.2 Lower boundary condition

EXHALE anchors the lower boundary at the planetary radius R_p using the standard CETIMB pressure-anchored condition (Base BC: pressure 1.0), fixing the base pressure at $P_0 = 1.0 \mu\text{bar} = 1.0 \text{ dyn cm}^{-2} = 1.0 \text{ erg cm}^{-3}$ (0.1 Pa). The corresponding base total number density n_0 follows from the ideal gas law at the equilibrium temperature $T_0 = T_{\text{eq}} = 226.0 \text{ K}$:

$$n_0 = \frac{P_0}{k_B T_0 n_{\text{tot},\text{bc}}} = \frac{1.0 \text{ dyn cm}^{-2}}{(1.380649 \times 10^{-16} \text{ erg K}^{-1}) \times 226.0 \text{ K}} = 3.20485 \times 10^{13} \text{ cm}^{-3}, \quad (1)$$

where $n_{\text{tot},\text{bc}} = 1.0$ for neutral atomic gas. In logarithmic form, $\log_{10}(n_0/\text{cm}^{-3}) = 13.506$, which is entered in `input.inp` as `Log10 lower boundary number density [cm^-3]: 13.506`. When Base BC: pressure is specified, EXHALE dynamically re-evaluates and enforces this relation internally (`input_read.f90`).

2.3 Stellar spectrum

The paper describes its normalization two ways that cannot both hold: the Supplement's prose says the proxy SED is scaled so its integrated X-ray flux matches that measured for LHS 1140, while the quoted constants ($A_{1132} = 0.59$, $A_{699} = 0.056$) are the catalog X-ray flux *ratios*, a factor 1.8 apart. The authors' script settles it:

```
Lx_ratio_1132 = 0.59
```

```
norm_constant_1132 = Lx_ratio_1132*(d_1140/(smax_1140*au2m))**2
```

applied to the plain const-res v23 product. That combination also reproduces the fiducial XUV flux the main text quotes, 0.033 W m^{-2} (we measure $33.6 \text{ erg s}^{-1} \text{ cm}^{-2}$ over 10–1300 Å). All results here use it. Note the proxy SED and

the paper’s own APEC fit disagree in the X-ray band (4.4 against $2.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ at the orbit); that inconsistency is inherited from the paper’s procedure, not introduced here.

2.4 Measurement

The comparison target is the authors’ released Fig. 3B spectrum (129 samples, vacuum, planetary rest frame, correlated-noise corrected), not a reading of a figure. Fitting it with our own three-Gaussian extractor (`he_line_metrics.py`, the same one applied to every model spectrum here) reproduces the paper’s MCMC to within its published uncertainties:

	this extractor	paper
blended red depth	1.254%	$1.24^{+0.22}_{-0.23}\%$
blue depth	0.198%	$0.25^{+0.14}_{-0.12}\%$
red/blue	6.35	$6.7^{+12.7}_{-3.1}$
FWHM	0.841 Å	$0.86^{+0.15}_{-0.27} \text{ Å}$

so models and measurement can be compared through one instrument.

2.5 The atomic input that sets the line: $\text{He}(2^3S)$ destruction on atomic hydrogen

Every composition this memo quotes is read off the metastable population, and in this wind that population is drained almost entirely by one reaction, $\text{He}(2^3S) + \text{H}$. Its rate coefficient is therefore an input of the same standing as the spectrum, and it has been changed since the earlier draft. The record is `docs/Update_EXHALE.md`, Sect. 87; the grounds are stated here because every number below moved with it.

What was used, and why it cannot be. The code carried Taylor et al. (2025), Table 2, transcribed correctly: $1.9 \times 10^{-9}(300/T)^{0.07}$ below 4000 K and $9.1 \times 10^{-9}(300/T)^{0.50}$ above it. Those two branches do not meet: at 4000 K they give 1.5849 and 2.4921×10^{-9} , a jump of 1.5724. A rate coefficient is a continuous function of temperature, and “it is the published value” is not a defence of an expression that is discontinuous in a state variable. In these solutions the seam falls at $1.11\text{--}1.18R_p$, inside the line-forming region, and it was visible as a one-cell step of 33 per cent in $n(2^3S)$ in the figures of the earlier draft.

The fit also disagrees with its own paper. Taylor et al.’s Fig. 19 (bottom right) plots the quantity Table 2 is meant to represent — their Maxwell–Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections — and that curve has a maximum near 1400 K. A monotonically decreasing power law cannot produce a maximum, so the tabulated low branch cannot be the same quantity; digitized, the two differ by 1.13–1.62 over 500–4000 K. Above the seam the high branch is worse: 2.23×10^{-9} at 5000 K, above *every* cross-section determination collected in García Muñoz (2025), Fig. 5, which lie between about 1.3 and 1.7×10^{-9} there. Their appendix records how this happened — an equation already implemented in their KPP routine, fitted in two pieces with `curve_fit` and nothing tying the pieces together at the join.

What is used now. García Muñoz (2025), A&A 698, A199, publishes in the caption of its Fig. 5 a closed form for the

Maxwell–Boltzmann average of the Movre & Meyer (1997) total ionization cross sections,

$$k = 10^{-9} \exp\left(\frac{c}{T} + d_1 \ln T + d_2 (\ln T)^2 + d_3 (\ln T)^3\right) \text{ cm}^3 \text{ s}^{-1}, \quad (2)$$

with $c = -8.64804 \times 10^1$, $d_1 = -2.86766 \times 10^{-1}$, $d_2 = +8.68445 \times 10^{-2}$ and $d_3 = -5.73001 \times 10^{-3}$, stated there as accurate to better than 2% over 200–10⁴ K. Measured here, it reproduces that paper’s own Table A.5 to 0.4%, and it is continuous at 4000 K to the eighth digit ($1.35630953 \times 10^{-9}$ at 3999.9 K against 1.35630905 at 4000.1).

This is not one paper’s fit traded for another’s. Taylor’s own Fig. 19 and the García Muñoz curve agree to 10–20%; it is the tabulated fit that is the outlier, and the tabulated fit is what the code was using.

The 0.9:0.1 branching. Equation (2) is the *total* ionization rate: Penning, $\text{He}(2^3S) + \text{H} \rightarrow \text{He}(1^1S) + \text{H}^+ + e^-$, and the associative channel that leaves HeH^+ , sharing one cross section. García Muñoz splits them “an average 0.9:0.1”, and the machine-readable network file deposited with that paper (references/garcia_munoz_2025_network/, Zenodo 15453858) applies exactly that split through two amplitudes per channel. EXHALE now does the same, and the distinction is not a scale factor: the total removes the metastable, so every sink keeps the sum — using 0.9 there would make the observable, $n(2^3S)$, 10 per cent too high — while the 0.9 scales only the terms that leave a lasting proton behind and deposit the Penning exothermicity, and the remaining 0.1 becomes an HeH^+ source in the molecular system.

What it does to this memo. The new coefficient is lower than the old one everywhere in this wind (new/old 0.69 at 1000 K, 0.86 just below the seam, 0.54 just above, 0.81 at 10⁴ K), so the metastable level lives longer and the line is stronger at fixed composition: re-solved, `heh0p55` goes from $EW = 1.109$ to $1.310 \text{ \% } \text{\AA}$ and `heh2p13_diff_kzz1e9` from 1.129 to 1.338, with $\log_{10} \dot{M}$ unchanged in both. Every crossing therefore moves down, and the composition ladders were re-solved rather than scaled by a local slope, because a +18 per cent shift is well outside the range over which such a slope is valid. The step detector of LHS1140b/exhale/metastable_step_diagnosis flags no cell at the seam in any of the runs quoted here.

3. Reproducing the paper’s p-winds model

Run at the authors’ settings — their SED normalization, $R_p = 0.154 R_J$, $b = 0.23$, `turbulence_broadening=True`, five-phase transit averaging, and their fitted line-of-sight bulk velocity $v_{\text{wind}} = 2.26 \text{ km s}^{-1}$ — and at the MCMC medians ($\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$, $T = 5160 \text{ K}$, $\text{H}:\text{He} = 1.01 \times 10^{-3}$), the vector their Fig. 4 legend states for the plotted curve, our reproduction gives a red depth of 1.376% and FWHM $0.575 \text{ \AA}$ against the 1.32% and $\sim 0.71 \text{ \AA}$ read off their Fig. 4: a 3% match in depth, but a line about 20% too narrow.

The width residual is not a consequence of which point in parameter space is drawn. Running the same reproduction at the grid maximum-likelihood point ($1.83 \times 10^8 \text{ g s}^{-1}$, 5131.6 K , 8.338×10^{-4}) gives red 1.366% and FWHM $0.581 \text{ \AA}$, i.e. the two vectors differ by less than $0.01 \text{ \AA}$ in width. What produces the extra width in the published curve has not been identified.

Two points are worth recording because both cost time to find.

First, the curves drawn over the spectrum in the paper’s Fig. 3 are the *three-Gaussian empirical fits*, not p-winds; the physical model is overlaid only in Fig. 4. Fig. 3’s apparently perfect agreement is a fit to the data, and says nothing about the model’s line shape.

Second, the paper’s plotted model is visibly broader than a plain p-winds run, which we initially modelled as an undocumented instrumental kernel. It is not: their script sets `turbulence_broadening=True` and averages over five transit phases. Both widen the line, and together they account for the difference.

4. Turbulent broadening: what the literature does

Since that switch turned out to matter, all 43 papers under references/ were checked: every PDF converted to text, searched for `turbulen`, `microturbulen` and `non-thermal broadening`, and each hit read in context. Fifteen matched; most use “turbulent” for eddy diffusion K_{zz} or for flow structure in 3D simulations, not for line broadening. Section 4.1 tabulates the ones that bear on broadening.

The convention, stated most explicitly by Yan et al. (2024), replaces the thermal Doppler width by

$$\Delta\nu_D = \frac{\nu_0}{c} \left(\frac{2kT}{m} + v_{\text{turb}}^2 \right)^{1/2}, \quad v_{\text{turb}} = \sqrt{\frac{5kT}{3m}} = c_s, \quad (3)$$

the monatomic sound speed, as assumed by Salz et al. (2018), Lampón et al. (2020) and Czesla et al. (2022). The p-winds implementation uses $\sqrt{5kT/6m}$ in a σ -convention, which is the same physics expressed against a Doppler width that lacks the factor 2; in EXHALE’s b -parameter convention the same term is $b \rightarrow \sqrt{11/6} v_{\text{th}}$, a factor 1.354. That is what `EXHALE_TRANSIT_TURB=1` applies. It is opt-in and off by default, because the field does not agree that it belongs.

4.1 Paper by paper

Adopt it as an assumed value. Salz et al. (2016), Eq. (5): “ b is the Doppler parameter, v_{th} the thermal Doppler velocity, and for the microturbulence v_{micro} we use the sound speed”, applied when synthesizing the transit profile. (Their atmosphere solver separately carries a constant 1 km s^{-1} microturbulence at the lower boundary, a different quantity, which they note “has little impact on the overall atmospheric structure”.) Lampón et al. (2020, 2021) is the reference implementation, with turbulent broadening as a free or assumed parameter of 1D isothermal Parker fits; Czesla et al. (2022) follows it. Cherubim et al. (2026) inherits the choice through p-winds’ `turbulence_broadening=True`, so the LHS 1140 b retrieval carries it — a fact stated nowhere in the paper text.

Test it, then leave it out. Yan et al. (2024) give the cleanest statement of the convention and of the doubt. They record that “these authors assumed that the turbulence velocity $v_{\text{turb}} = \sqrt{5kT/3m}$, which corresponds to the speed of sound of a monatomic gas and, thus, to the transonic case with a Mach number of 1”, then keep it out of their nominal models and test $v_{\text{turb}} = 0.5, 1.0$ and $1.5 c_s$ as a sensitivity. Their finding for He 10830 is that turbulence “decreases the absorption depth slightly at the line center, but doesn’t significantly alter the absorption at the line wings”. They argue Mach 1 is likely an overestimate, since in the diffuse ISM turbulent velocity anti-correlates with temperature and the Mach number is below unity at a few thousand K, and conclude “the turbulence effect is likely less significant than the one explored in this paper”. Yan et al. (2019) computes the Doppler width explicitly “assuming no turbulence”.

Argue against it. Murray-Clay et al. (2009), on García Muñoz’s (2007) proposal to broaden $\text{Ly}\alpha$ by wind turbulence: “to generate the large velocities observed, the energy in such turbulence would need to exceed the thermal and bulk kinetic energies in the mean flow by a factor of ~ 100 . Such energy requirements seem insurmountable.” Koskinen et al. (2013b) find that “non-thermal broadening such as that proposed by Ben-Jaffel and Hosseini (2010) does not appear to be necessary to explain the current observations”, while allowing that stellar-wind interaction may produce broadening turbulence. Schulik & Owen (2025) offer a physical alternative — adiabatic cooling produces a metastable-helium population bump at high velocity, widening the line with no free parameter — and are pointed about the alternative: the isothermal model “can only explain the broadness of the data under physically inconsistent assumptions, i.e. those of even larger mass-loss rates, not supported by the energy input into this planet, and an empirical free parameter, turbulent broadening (Lampón et al. 2021)”. They add that $T = 10^4$ K already maximizes what thermal broadening alone can give.

Need it even so. Taylor et al. (2026) is the closest analogue to the present work: a self-consistent model (energy balance, multispecies diffusion, non-LTE $\text{H}(n=2)$) that reproduces the HD 189733 b line core but not its width. They write that “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, and quantify it as $\sim 12 \text{ km s}^{-1}$ of additional non-thermal broadening. They speculate — explicitly calling it speculative — that it is plasma turbulence from ions trapped on low-latitude planetary field lines, strengthening with stellar activity. Rumenskikh et al. (2022), as reported there, instead imposed reduced hydrogen line cooling and prescribed zonal jets of $10\text{--}20 \text{ km s}^{-1}$ in a 3D multifluid model.

Matches that are not about broadening. Recorded so the sweep need not be repeated: Wogan (2025), Lavvas (2014, 2019), Robeling (2026) and the second half of Koskinen et al. (2013b) use “turbulent” for eddy diffusion K_{zz} ; 2607.18193v1 for flow–flow interaction in a 3D stellar-wind simulation; Jensen (2012) for photospheric turbulence driving stellar activity; Taylor et al. (2025) for gravity-wave breaking. Vidal_2003_Nature.pdf matches only on its first page, which carries the tail of the preceding *Nature* letter (on Crab pulsar radio emission); Vidal-Madjar et al. themselves do not discuss turbulence.

4.2 Empirical baseline: turbulence in Solar System giant planets

To evaluate whether sonic or supersonic ($v_{\text{turb}} \sim c_s$) turbulence is physically plausible in a planetary atmosphere, it is instructive to compare with directly measured turbulence and non-thermal velocity fields across the stratified layers of Jupiter and Saturn:

- *Troposphere and cloud deck* ($\sim 0.1\text{--}1 \text{ bar}$). Cloud-tracking measurements from *Voyager*, *Galileo*, *Cassini*, and *Juno* (e.g., Salyk et al. 2006; Del Genio et al. 2007; Ingersoll et al.) isolate the fluctuating eddy velocities $v' = \sqrt{u'^2 + v'^2}$ from the mean zonal jets ($100\text{--}500 \text{ m s}^{-1}$). The resulting RMS turbulent eddy velocity is $v' \approx 3\text{--}10 \text{ m s}^{-1}$ on Jupiter (locally $20\text{--}30 \text{ m s}^{-1}$ in the Great Red Spot shear zone) and $5\text{--}15 \text{ m s}^{-1}$ on Saturn. Compared to the tropospheric sound speed ($c_s \approx 800\text{--}900 \text{ m s}^{-1}$), this is deeply subsonic turbulence, with Mach numbers $\mathcal{M} \sim 0.005\text{--}0.02$.

- *Stratosphere* ($\sim 10\ \mu\text{bar}$ – $10\ \text{mbar}$). High-resolution submillimeter spectroscopy of impact-delivered HCN/CO molecules with ALMA (e.g., Cavalié et al. 2021) and Herschel reveals stratospheric jet velocities up to 300 – $400\ \text{m s}^{-1}$ in the auroral zone, but the associated non-thermal velocity dispersion (turbulence and wave motions) remains limited to tens of m s^{-1} up to ~ 100 – $200\ \text{m s}^{-1}$ ($\mathcal{M} \lesssim 0.2$).
- *Thermosphere and auroral ionosphere* ($\lesssim 1\ \mu\text{bar}$). Ground-based high-resolution ($R \sim 10^5$) infrared spectroscopy of H_3^+ line profiles (e.g., with VLT/CRIRES, Keck, and CFHT) disentangles thermal Doppler broadening from non-thermal velocity fields. In non-auroral mid-latitudes, the non-thermal velocity is $v_{\text{nth}} \approx 0.2$ – $0.5\ \text{km s}^{-1}$. In the auroral oval, where intense magnetosphere–ionosphere coupling, ion drag, and Joule heating inject substantial kinetic energy, non-thermal broadening reaches $v_{\text{nth}} \approx 0.8$ – $2.0\ \text{km s}^{-1}$.

These empirical benchmarks show that pure hydrodynamic planetary turbulence is strictly subsonic ($\mathcal{M} \ll 1$), and even violent magnetospheric auroral driving in upper thermospheres caps non-thermal broadening at $\lesssim 2\ \text{km s}^{-1}$.

4.3 Turbulence and 3D hydrodynamic simulations of exoplanets

Beyond 1D parameterizations, 3D numerical simulations explore how turbulent mixing, non-thermal velocity structures, and line broadening emerge across diverse exoplanetary regimes:

1. 3D hot Jupiter escape and circulation.

- *3D atmospheric escape and kinematic line broadening*. Hydrodynamic 3D escape simulations (e.g., McCann et al. 2019; Debrecht et al. 2020; Carolan et al. 2020, 2021; Wang & Dai 2021; MacLeod & Oklopčić 2022) demonstrate that apparent transmission line broadening arises primarily from large-scale 3D kinematic velocity gradients (10 – $50\ \text{km s}^{-1}$) across the day-to-night outflow, Coriolis-deflected tails, and tidal distortion, rather than unresolved isotropic microscale turbulence. Furthermore, the collision between the expanding planetary wind and the incoming stellar wind creates a shear layer prone to Kelvin–Helmholtz instabilities and turbulent mixing at the contact discontinuity. In 3D multifluid models of HD 189733 b, Rumenskikh et al. (2022, 2023) show that upper-atmospheric zonal jets of 10 – $20\ \text{km s}^{-1}$ combined with 3D outflow geometry reproduce the observed broad transmission profiles without invoking ad hoc microturbulence.
- *3D general circulation models (GCMs) and macroturbulent mixing*. In the observable lower-to-middle atmosphere ($1\ \text{bar}$ to $\sim 1\ \mu\text{bar}$), 3D GCMs (e.g., Showman & Polvani 2011; Showman et al. 2013, 2015; Parmentier et al. 2013; Charnay et al. 2015; Zhang & Showman 2018) demonstrate that equatorial superrotation (1 – $5\ \text{km s}^{-1}$) and vertical mixing ($K_{zz} \sim 10^7$ – $10^{11}\ \text{cm}^2\ \text{s}^{-1}$) are driven by planetary-scale macroturbulence (non-linear interactions between planetary-scale Kelvin and Rossby waves and the mean flow) and overturning circulation. The sub-grid parameterization of small-scale turbulent dissipation (hyperdiffusion, Smagorinsky closures, or Shapiro filters) remains a known source of numerical sensitivity in GCM wind speeds and jet stability (Heng et al. 2011; Polichtchouk et al. 2014; Skinner & Cho 2021).
- *Magnetohydrodynamic (MHD) turbulence in ultra-hot atmospheres*. For planets with $T \gtrsim 1500\ \text{K}$, thermal ionization of alkali metals (Na, K) couples the gas to the planetary magnetic field. 3D MHD simulations (e.g., Perna

et al. 2010, 2012; Rauscher & Menou 2013; Rogers & Showman 2014; Rogers & Komacek 2014; Beltz et al. 2022) show that magnetic winding of the field by supersonic zonal winds generates Lorentz forces (magnetic drag) that brake the flow, generating 3D MHD turbulence, Ohmic dissipation, and time-dependent variability in wind patterns and hotspot offsets.

2. 3D simulations across other planetary classes. The reach of 3D fluid simulations extends far beyond hot Jupiters to other major categories of exoplanets and substellar objects:

- *Warm Neptunes and sub-Neptunes.* For warm Neptunes such as GJ 436 b and GJ 3470 b, 3D particle-mesh and hydrodynamic escape simulations (Bourrier et al. 2015, 2016; Villarreal D’Angelo et al. 2021; Carolan et al. 2021) reveal gigantic, asymmetric cometary hydrogen tails spanning tens of planetary radii, directly explaining the 56% Ly α transit obscuration and the He 10830 Å absorption through 3D stellar radiation pressure and charge-exchange interactions. In metal- and water-rich sub-Neptune atmospheres (e.g., GJ 1214 b, K2-18 b), 3D GCMs model cloud/haze circulation, photochemical transport, and vertical eddy mixing ($K_{zz} \propto P^{-0.4}$; Charnay et al. 2015; Innes & Pierrehumbert 2022).
- *M-dwarf habitable zone terrestrial planets.* Tidally locked rocky worlds (such as TRAPPIST-1 e/f/g, Proxima b, and LHS 1140 b) have been extensively modelled with 3D climate GCMs (Turbet et al. 2016, 2018; Wolf et al. 2017; Sergeev et al. 2020, 2022; the THAI intercomparison, Fauchez et al. 2020, 2022). These simulations demonstrate how thick dayside convective cloud decks elevate planetary albedo to prevent runaway greenhouse states, while stationary Rossby–Kelvin wave patterns redistribute heat to the nightside to avoid atmospheric freeze-out. Complementary 3D multi-species MHD simulations (Dong et al. 2017, 2018; Cohen et al. 2014) assess O $^+$ and H $^+$ ion escape driven by extreme stellar coronal mass ejections and flaring winds.
- *Lava worlds and ultra-short-period rocky planets.* For scorched rocky planets with dayside magma oceans (e.g., 55 Cancri e, K2-141 b, CoRoT-7 b), 3D hydrodynamic models of mineral-vapor atmospheres (Castan & Menou 2011; Nguyen et al. 2020; Zilinskas et al. 2022; Kang et al. 2021) reveal supersonic dayside-to-nightside winds ($\mathcal{M} > 1\text{--}3$, $v \sim 1\text{--}3 \text{ km s}^{-1}$), standing hydraulic jumps/shocks, and mineral vapor condensation leading to rock rain on the nightside.
- *Brown dwarfs and directly imaged giant planets.* In isolated brown dwarfs and young, directly imaged giants (e.g., HR 8799 b/c/d/e, VHS 1256 b), 3D convective-radiative GCMs (Tan & Showman 2019, 2021a,b; Showman & Kaspi 2013; Zhang & Showman 2014) couple rapid planetary rotation (strong Coriolis forces) with deep geothermal convection and silicate/iron cloud radiative feedbacks. The resulting turbulent vortices, patchy cloud bands, and storm dynamics provide the physical explanation for observed rotational photometric variability.

4.4 Where that leaves this comparison

There is no settled treatment. Turbulence at $v_{\text{turb}} = c_s$ is a common convention in 1D isothermal Parker-wind fits — and hence in the paper being reproduced — while Solar System observations and self-consistent or 3D models demonstrate that isotropic microscale turbulence cannot physically reach the required supersonic levels. 3D models instead supply

line width through kinematic flow gradients, zonal jets, and magnetospheric/plasma interactions. Enabling the $v_{\text{turb}} = c_s$ term in EXHALE is therefore right for a like-for-like comparison with C26 and wrong as a physical default, which is how it is implemented.

5. Searching for a match: the composition scan

Once the star, the planet and the spectrum are fixed, composition is the free parameter a match has to be sought in first. This section takes it in the well-mixed limit — element diffusion off, so the reservoir ratio is also the ratio everywhere in the wind. That is the simpler of the two mixing assumptions available, and Sect. 8.4 repeats the whole exercise with binary H/He diffusion active, where the answer depends on the eddy coefficient as well.

Eleven compositions were solved, from solar ($\text{He}/\text{H} = 0.083$) to $\text{He}/\text{H} = 10^4$, each a JFNK steady state ($\|R\| < 10^{-3}$, $\text{info} = 0$; one case, $\text{He}/\text{H} = 100$, stagnates at 1.18×10^{-3} and is recorded as such). Nothing is imposed: mass, momentum and energy are solved with photoionization heating and radiative cooling, so $\dot{M}$ and $T(r)$ are outputs.

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	red [%]	red (turb) [%]	FWHM [Å]	EW [% Å]
0.083	7.77	1.13	1.02	0.233	0.279
0.25	7.75	2.49	2.21	0.244	0.645
0.50	7.76	3.83	3.35	0.255	1.040
0.55	7.76	4.05	3.54	0.257	1.109
0.70	7.78	4.71	4.09	0.261	1.311
1	7.80	5.59	4.83	0.268	1.598
10	7.76	7.12	5.97	0.298	2.270
10 ³	7.74	5.46	4.55	0.315	1.844
10 ⁴	7.73	5.40	4.49	0.316	1.825
observed	—	1.254	—	0.841	1.108 ± 0.030

Nine of the eleven runs are listed; $\text{He}/\text{H} = 0.6$ falls between two tabulated entries and $\text{He}/\text{H} = 100$ is the case whose residual stagnates. The three helium-dominated rows are corrected values: their depths and equivalent widths were an artifact of the old advection gate, which pinned the metastable population to its equilibrium value across the hydrogen ionization front, and they fell by 89–93% when the gate was rebuilt on the slowest solved species (Update_EXHALE.md Sect. 72; at $\text{He}/\text{H} = 10$, $EW \ 21.53 \rightarrow 2.270 \text{ \% Å}$). The crossing of Sect. 7. is interpolated inside $\text{He}/\text{H} \leq 1$ and is untouched by that correction.

The escape rate is nearly composition-independent, $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$, about a third of the 2.03×10^8 the retrieval imposes.

Every equivalent width in this table predates the Penning coefficient replacement of Sect. 2.5 and is not the current value. The metastable level lives longer on the new coefficient, so each entry rises by about a fifth: at $\text{He}/\text{H} = 0.55$ the same run re-solved gives $EW = 1.310$ in place of 1.109 % Å and 0.2583 in place of 0.2573 Å, with $\log_{10} \dot{M}$

unchanged at 7.76. Only three of the eleven compositions here were re-solved, so the table is kept as the record of the scan that was run and the crossing of Sect. 7. is *not* read from it: it is read from a ladder of nine solutions over $\text{He}/\text{H} = 0.40\text{--}0.46$ solved on the current coefficient (LHS1140b/exhale/crossings_gm25), which brackets the measurement with arms that were run.

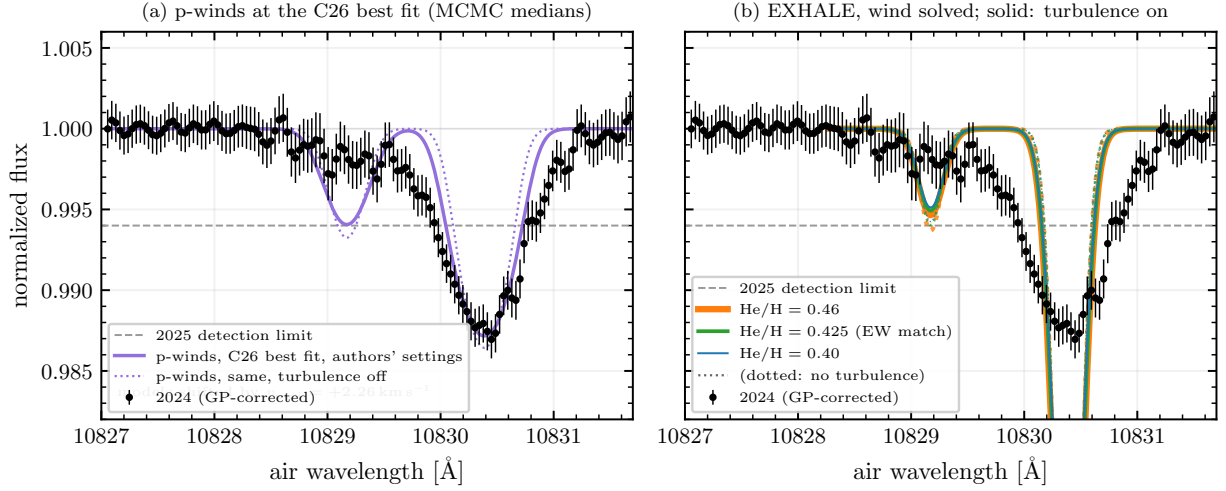


Figure 1: The measurement (black) with, left, the paper’s own p-winds model at its best fit — the MCMC medians its Fig. 4 legend states — computed with the authors’ released settings (solid; dotted is the same run with turbulence off), and right, EXHALE at the three arms of the well-mixed ladder of record that bracket the equivalent-width crossing, $\text{He}/\text{H} = 0.40, 0.425$ and 0.46 (crossings_gm25 lineage W2, on the current Penning coefficient). Solid EXHALE curves carry the same turbulence term p-winds uses; dotted are without it. Over that range of composition the profile barely moves, so the three nearly coincide; the composition dependence of the absorbing power is Fig. 4. Neither code applies a line-of-sight bulk velocity, so both sets of model curves are shifted by the $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ that C26 fit; the shift is rigid and leaves equivalent widths unchanged. Drawn in the presentation of the paper’s Fig. 4.

6. Why the two models disagree

At the same composition the two models are not small perturbations of each other. The *total* metastable column is comparable — at $\text{H}:\text{He} = 10^{-3}$ EXHALE’s radial column over $1\text{--}20R_p$ is 0.96 of p-winds’ — but its distribution is not, and transit depth is absorbing *area*:

	p-winds	EXHALE
50% of the 2^3S column inside	$1.33 R_p$	$5.7 R_p$
90% inside	$1.59 R_p$	$10.2 R_p$

Both radii are read from the local-equilibrium metastable profile, which is what Fig. 2 draws. The advection-corrected profile that the transit integral consumes is more concentrated, and for this helium-dominated composition it collapses onto the base once the advection gate is formed on the metastable’s own clock rather than on hydrogen’s

(Update_EXHALE.md Sect. 72). This section is therefore a statement about the two wind solutions; every transit number quoted elsewhere in the memo is measured on the advection-corrected profiles.

The star is $13.6R_p$ across, so metastable helium spread to $10R_p$ covers most of the disk while a column confined to $1.6R_p$ can darken only a narrow ring. The distribution difference is the wind structure — the one thing p-winds imposes and EXHALE solves. At $r = 10R_p$, same composition and spectrum: p-winds is isothermal at 6100 K by construction, EXHALE's energy balance gives 1392 K; p-winds flows at 4.3 km s^{-1} , EXHALE at 0.53 ; and the cold, slow, denser gas keeps helium neutral and the metastable level populated far out, $n(2^3S) = 40 \text{ cm}^{-3}$ against 2×10^{-4} (Fig. 2).

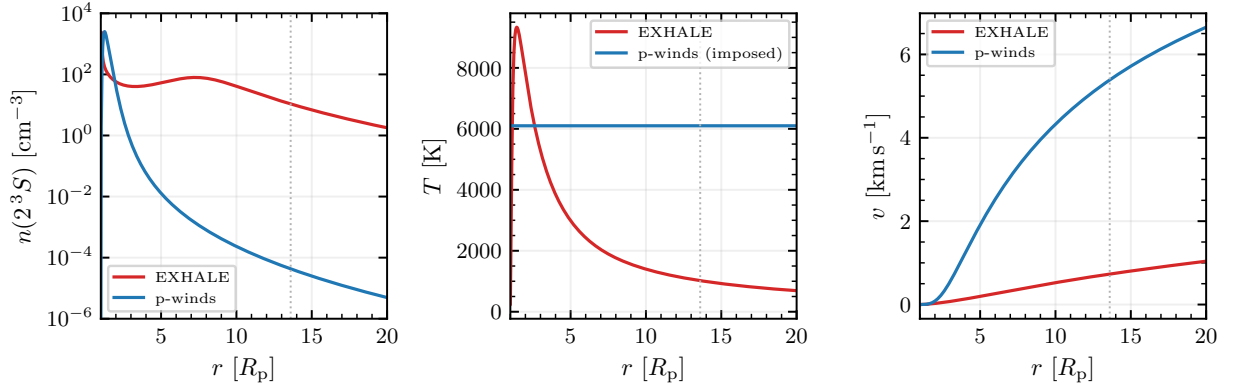


Figure 2: Same composition ($\text{H}:\text{He} = 10^{-3}$), same spectrum, two wind structures. Dotted vertical line: the stellar radius, $13.6R_p$.

6.1 Density and ionization, the same pair

Figure 2 draws three of the five profiles the comparison was set up to make; the other two, mass density and hydrogen ionization, are in Fig. 3, from the same pair of solutions.

The densities are the part that agrees. Outside the base the ratio EXHALE/p-winds runs from 0.15 at $1.2R_p$ through unity near $2R_p$ to 2.7 at $5R_p$, and stays inside a factor 2.3 from $3R_p$ out. At the base cell itself EXHALE sits $176\times$ higher, which is the two lower boundary conditions and not a profile difference: p-winds sets its base density from the imposed $\dot{M}$ through the Parker solution, EXHALE from its own base state.

The ionization does not agree at all. EXHALE has hydrogen completely ionized above $1.03R_p$ — $n(\text{HI})$ is an exact zero in the output — while p-winds keeps it 91–94% neutral throughout, $x(\text{HII})$ rising only from 0.056 at $1.2R_p$ to 0.086 at $20R_p$. Neither end of that is the other's error, and both are worth stating.

p-winds' hydrogen is frozen far below its own photoionization equilibrium. Integrating the same SED against the Verner et al. (1996) ground-state cross section gives an unattenuated rate $\phi = 5.17 \times 10^{-8} \text{ s}^{-1}$ (p-winds' internal value for this file agrees to 0.1%), an ionization time of 224 d, while the flow time r/v through the imposed $\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$ falls to $2.6 \times 10^5 \text{ s}$ at $2R_p$ and $2.6 \times 10^4 \text{ s}$ at $10R_p$. Equilibrium on p-winds' own density and temperature would be $x(\text{HII}) = 0.76$ at $2R_p$ and 0.997 at $5R_p$ (dotted in Fig. 3b); the wind carries the gas out before it gets there. Shielding is not what holds it down — hydrogen is a trace species here and the neutral-H column above $1.2R_p$ is optically thin at every energy.

EXHALE's hydrogen goes the other way: it is more ionized than any photoionization balance on its own structure allows. With the same ϕ , its own electron density at $3R_p$ ($2.0 \times 10^6 \text{ cm}^{-3}$, set by helium) and its own temperature there (5247 K), primary photoionization alone gives $x(\text{HII}) = 0.059$ — close to what p-winds carries, and nowhere near 1. An exact zero can only come from a hydrogen ionization source that does not vanish with $n(\text{HI})$, and the code has one: the Shull & van Steenberg (1985) secondary channel, added in `util_ion_eq.f90` as $P_{\text{HI}} \rightarrow P_{\text{HI}} + R_{\text{sec,HI}}/n(\text{HI})$, where $R_{\text{sec,HI}}$ is accumulated over *every* absorber's photoelectrons. In a helium-dominated gas that numerator is helium's. Evaluating it at $3R_p$ on this solution's own radiation field — the He I photoelectrons above the code's 40 eV secondary threshold, with $f_{\text{ion,HI}} = 0.091$ at the local ionized fraction $x = 0.246$ — gives $R_{\text{sec,HI}} = 4.9 \times 10^{-2} \text{ cm}^{-3} \text{ s}^{-1}$ against a hydrogen recombination rate of 6.6×10^{-3} , a factor 7.5. Complete ionization follows.

That is an approximation used outside the composition it was made at. The SvS85 fits are functions of the ionized fraction alone; the helium abundance of the gas their Monte Carlo followed is fixed inside them and is not the $\text{He}/\text{H} = 10^3$ of this run, and their two branches hand hydrogen $7 \times$ helium's share of the secondary ionizations ($f_{\text{ion,HI}}/f_{\text{ion,HeI}} = 0.3908/0.0554$ at $x \rightarrow 0$) where the abundances say the opposite by three orders of magnitude. It is recorded here rather than changed, because the reach was measured and is bounded: $n(\text{HI})$ reaches an exact zero only at $\text{He}/\text{H} \geq 10$ over a prescribed metal-free base (`exhale/heh10` above $6.6R_p$, `heh100` above 1.06, `heh1000` above 1.03), and in none of the solutions this memo quotes science from — `heh0p55`, `heh2p13_diff_kzz1e9`, and every rung of the flux-closed ladder out to $\text{He}/\text{H} = 12$, where $x(\text{HII})$ at $2R_p$ runs 0.060 to 0.178 and no cell is fully ionized. The $\text{H}:\text{He} = 10^{-3}$ comparison of this section is the one place in the memo that sits inside it, and the quantity it distorts is hydrogen, not the metastable helium the line is made of.

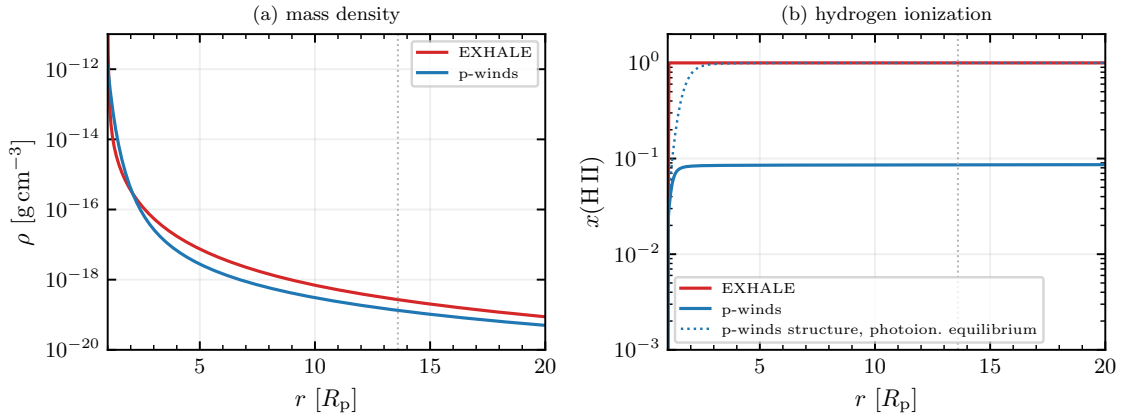


Figure 3: The other two profiles of the same comparison ($\text{H}:\text{He} = 10^{-3}$, same spectrum): (a) mass density, (b) hydrogen ionization. The dotted curve in (b) is photoionization equilibrium evaluated on p-winds' own density and temperature, so its distance from the solid p-winds curve is how far the imposed flow holds hydrogen out of equilibrium. Dotted vertical line: the stellar radius.

7. What can be matched: the equivalent width

Depth and width trade against each other under any broadening, but the equivalent width does not: it is invariant under the instrument profile, the turbulence term and the phase averaging. It is therefore the quantity on which a self-consistent model can be matched without first settling the broadening question.

Over the blended red pair (vacuum 10832.6–10834.2 Å), the released spectrum gives $EW = 1.108 \pm 0.030 \%$ Å. EXHALE crosses it at

$$\text{He}/\text{H} = 0.4245 \quad (0.410\text{--}0.443 \text{ on the } EW \text{ } 1\sigma), \quad \text{H}:\text{He} = 2.4, \quad (4)$$

that is $5.1 \times$ solar, a helium mass fraction of 0.63. The crossing is read inside a bracket of two solved arms, 0.42 at $EW = 1.1008$ and 0.425 at 1.1088% Å against the observed 1.1084 ± 0.0295 , and a chord and a quadratic through the ladder agree on it to 0.02 per cent (0.4245 against 0.4244). A run at 0.4245 itself gives $EW = 1.1068 \%$ Å, 0.04σ from the measurement, on the lineage the crossing was read on (LHS1140b/exhale/confirm_gm25); re-seeded from the nearest neighbouring arm the same composition gives 1.1167, still inside the band and about 1 per cent away in He/H, which is the path spread on this axis. Two lineages were solved, the second re-seeded from a solution of the current binary; the seed moves the crossing from 0.4268 to 0.4245, which is the size of the path spread on this axis. Enabling the turbulence term was not re-measured here — the transit synthesis was run in its nominal configuration only — but on the earlier scan it moved the crossing by 1.5 per cent, a direct demonstration that the estimate does not depend on the broadening treatment. The value sits three orders of magnitude away from the paper’s $\text{H}:\text{He} \lesssim 10^{-3}$, and the reason is the structural difference of the previous section: a wind that spreads metastable helium over most of the stellar disk needs far less helium to absorb as much.

Two assumptions are carried inside that number and are taken up later. It is the well-mixed crossing, and it is the crossing on the GJ 1132 proxy spectrum. Sect. 8.4 rescans it with element diffusion active at every decade of the eddy coefficient, and Sect. 10. rescans it on the brighter proxy; both move it, the spectrum by much the larger factor.

8. What cannot be matched: the profile

Matching the equivalent width is not matching the line. At the crossing $\text{He}/\text{H} = 0.4245$ the profile is 3.3 times too deep (4.12% against 1.254%) and 3.3 times too narrow (0.253 against 0.841 Å). The equivalent width states how much absorbing power the model has; the profile states how that power is spread in velocity, and composition does not act on the second. Every solved composition gives 0.23–0.32 Å, rising only to 0.26–0.39 Å with the turbulence term.

8.1 How much is missing, in velocity

The measured FWHM is 0.841 Å, that is 23.3 km s^{-1} at 10833 Å. Three ways to supply it, and what each would require:

- *Thermal alone.* A Doppler width of 23.3 km s^{-1} FWHM in helium needs $T = 4.7 \times 10^4 \text{ K}$. Helium is fully ionized far below that, so there would be no metastable population left to absorb. This is excluded.

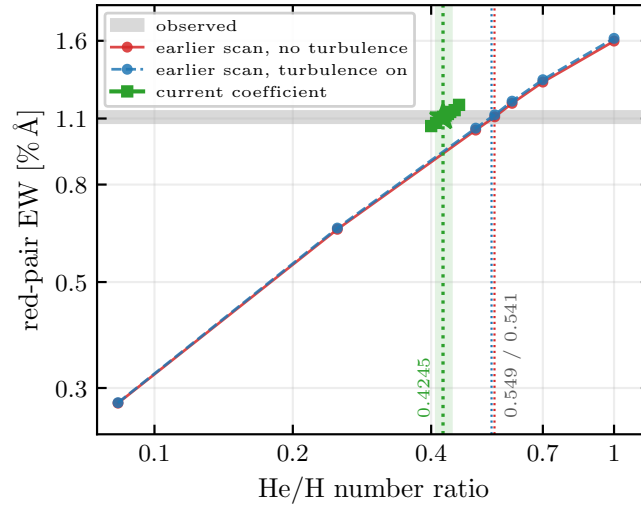


Figure 4: Red-pair equivalent width against composition, with and without the turbulence term, against the measured band. The crossing is insensitive to the broadening treatment. *All eight arms of this scan are on the retired Penning coefficient* (Sect. 2.5) and are drawn that way on purpose, so the figure is internally consistent; two of them have resolved counterparts but the other six do not, and swapping in only those two would mix coefficients along one curve. The crossing this memo quotes for this axis, 0.4245, is the one of Sect. 7., read off a ladder solved on the current coefficient, and is not read from this figure.

- *Bulk outflow.* A line-of-sight velocity spread of order $\pm 12 \text{ km s}^{-1}$ across the line-forming region would do it. The EW-matched EXHALE solution reaches 1.2 km s^{-1} at $20R_p$; the paper's p-winds model, with its imposed temperature, reaches 4–7. Both are an order of magnitude short.
- *Non-thermal (turbulent) broadening.* Added in quadrature to what EXHALE already has, closing the gap takes 22.2 km s^{-1} FWHM — 9.4 km s^{-1} of turbulent velocity (σ), against the 2.9 km s^{-1} that the sound-speed convention of Sect. 4.1 supplies for helium at 5000 K. This is the one of the three that can be measured directly, and Sect. 8.2 measures it.

So the model is not short by a detail. It is short by an amount of velocity comparable to the escape speed, and no choice of composition supplies it.

8.2 The broadening the data demand, measured

The three routes above can be replaced by one measurement. Take the EW-matched EXHALE profile as solved, convolve it with a Gaussian of FWHM f in velocity, and ask which f reproduces the measured line. The convolution is applied to the excess absorption $(\max T - T) / \max T$ built from the instrument-convolved column of `tpm_He10830.txt`; depth, width and the red/blue ratio are then read off with the same three-Gaussian extractor applied to the data, and the equivalent width by the same integral over the vacuum window 10832.60–10834.20 Å. So f is the only quantity varied, and every number in Table 1 is measured the way the corresponding observed number was.

Table 1: The EW-matched EXHALE profile (crossings_gm25/rs_heh0p425, He/H = 0.425, the arm that sits on the crossing 0.4245; no turbulence term) convolved with an added Gaussian of FWHM f . Observed values on the last row: red and blue depths from the C26 MCMC posterior, width and equivalent width measured here on their spectrum.

f [km s ⁻¹]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.117	0.595	6.91	0.253	1.109
10	2.363	0.318	7.43	0.440	1.109
15	1.742	0.231	7.54	0.597	1.106
20	1.361	0.179	7.58	0.765	1.089
22.2	1.238	0.163	7.59	0.841	1.077
25	1.110	0.146	7.61	0.938	1.056
observed	1.24 ^{+0.22} _{-0.23}	0.25 ^{+0.14} _{-0.12}	6.7 ^{+12.7} _{-3.1}	0.841	1.108 ± 0.030

The width is reached at $f = 22.22$ km s⁻¹ FWHM, that is $\sigma = 9.44$ km s⁻¹. At that one value the depth is 1.238% against the measured 1.24^{+0.22}_{-0.23}, the blue component 0.163% against 0.25^{+0.14}_{-0.12}, and the ratio 7.59 against 6.7^{+12.7}_{-3.1}: depth, blue depth and ratio all land inside the observed 1 σ interval at the value fixed by the width alone. Nothing was tuned to them. Figure 5 shows the profile and the two crossings.

The number does not depend on which solution is broadened. The neighbouring arm of the same ladder, He/H = 0.42, asks for 22.22 km s⁻¹ and the $K_{zz} = 0$ arm at He/H = 3.79 for 22.24; the He/H = 0.55 run with the turbulence term already on asks for 21.75 and solar composition for 22.40, although the solar run is a third of the observed absorption (0.424% red, EW 0.380) and so matches the shape while missing the amount. Every arm of this comparison is on the current Penning coefficient (LHS1140b/exhale/misc_gm25/), and the replacement is what makes the point measured rather than assumed: it raises the metastable population of the 0.55 arm by 18 per cent and moves the kernel it asks for by 0.02 km s⁻¹, one part in 1100. The required broadening is therefore set by the velocity field, not by composition or by the broadening treatment already in place: the direct measurement of what Sect. 8. inferred. The equivalent width falls by 3% between $f = 0$ and $f = 22.2$ because the Gaussian wings carry absorption out of the red-pair integration window, not because absorbing power is lost.

This is descriptive, not explanatory. It states that one free parameter, applied to a solved profile, reproduces the measured line; it does not say what produces 22.2 km s⁻¹. The candidates are the ones listed below, and the measurement sharpens them by fixing the number each has to deliver.

One conversion narrows the list before it is read. If the missing 22.2 km s⁻¹ is called turbulence, it has to be turbulence in gas whose sound speed the solution itself fixes. Evaluating $c_s = (\gamma kT / \mu m_H)^{1/2}$ with $\gamma = 5/3$ on the well-mixed heh0p55 output, taking μ from the solved H I/H II/He I/He II/He III and electron densities, gives $c_s = 5.5$ km s⁻¹ at $2R_p$ ($T = 4311$ K, $\mu = 1.94$), 3.8 at the metastable peak $5.4R_p$ (1609 K, 1.57) and 3.0 at $10R_p$ (845 K, 1.32); weighted over the line-forming region by $n(2^3S)r$, $T = 2022$ K, $\mu = 1.59$ and $c_s = 3.9$ km s⁻¹. Note this is the bulk-gas sound speed, the quantity a Mach number is defined against, and not the helium thermal speed that enters the line profile.

Against that, the kernel is supersonic on any convention. Read as a line-of-sight velocity dispersion, $\sigma = 9.44$ km s⁻¹ is $\mathcal{M} = 2.4$. Read in the b -parameter convention of Lampón et al. and p-winds, where v_{turb} is the quantity added

to $b^2 = 2kT/m + v_{\text{turb}}^2$, it is $v_{\text{turb}} = \sqrt{2}\sigma = 13.4 \text{ km s}^{-1}$, $\mathcal{M} = 3.4$; as an isotropic three-dimensional turbulent speed, $\sqrt{3}\sigma = 16.3 \text{ km s}^{-1}$, $\mathcal{M} = 4.2$. The $v_{\text{turb}} = c_s$ convention of Sect. 4.1 supplies only $3\text{--}4 \text{ km s}^{-1}$ in this gas, about a third of what the data demand. That is the opposite corner of the literature reviewed there: Yan et al. (2024) argue $\mathcal{M} = 1$ is already an overestimate at a few thousand K, and Murray-Clay et al. (2009) call the energy requirement of supersonic turbulence “insurmountable”. Supplying the shortfall under the name of turbulence therefore looks physically hard to sustain, and the measurement points to the candidates of the next section — an ordered velocity field, three-dimensional flow, or heating the 1D energy equation does not carry — rather than to a missing thermal width.

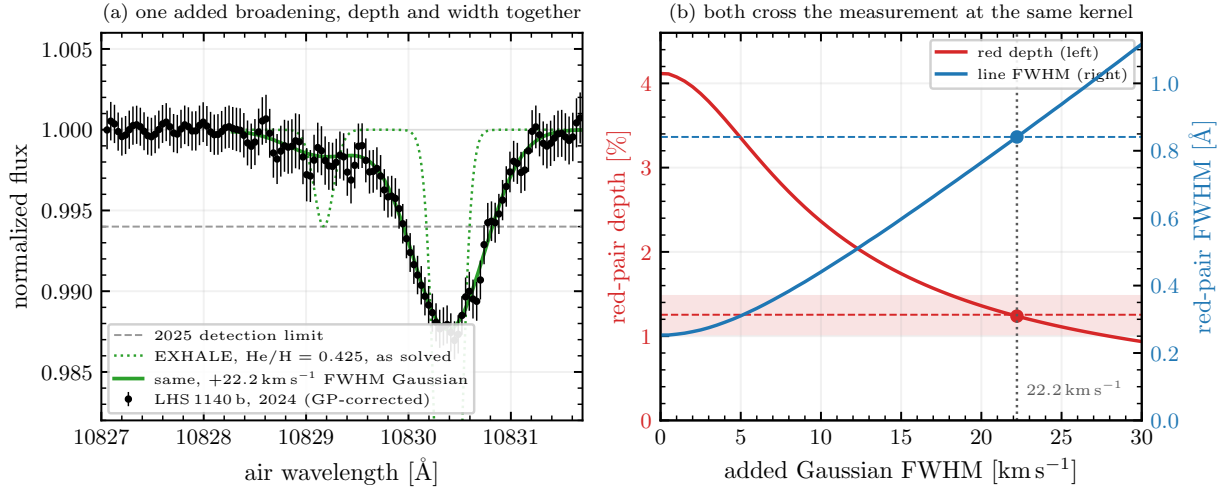


Figure 5: Left: the EW-matched EXHALE profile as solved (dotted, its core running below the plotted range) and convolved with a 22.22 km s^{-1} FWHM Gaussian (solid), against the measurement. Both are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ used in the earlier figures. Right: red-pair depth (red, left axis) and red-pair FWHM (blue, right axis) as the added kernel is scanned. Dashed lines are the observed values, the shaded band the C26 1σ interval on the depth; the dotted vertical marks where the width is matched. Both curves meet the measurement at the same kernel.

8.3 Is the demand an artifact of the measurement?

The number above is a difference between two widths, so it is worth no more than the two definitions behind it. Before the missing velocity is treated as physics, the observing and fitting side was audited on its own; the record, with every number recomputed there from the released spectrum, the released MCMC chain and the code in this repository, is `docs/lhs1140b_width_measurement_audit.md`.

Nothing on the measurement side supplies the width. Term by term, in the units the requirement is stated in:

- *The instrument profile.* The WINERED HIRES-Y kernel the Supplement states, and validates against nearby telluric features, is $10833 \text{ Å} / 68,000 = 0.1593 \text{ Å} = 4.41 \text{ km s}^{-1}$ FWHM. The observed line is 5.4 times wider, so it is resolved, and removing the kernel in quadrature moves the intrinsic width from 23.9 to 23.4 km s^{-1} .
- *The triplet splitting.* The 0.0897 Å separation of the red pair is present on both sides of the comparison. It is 0.7 per cent of the observed blend width but 8 per cent of the model’s, because at $\sigma = 0.35 \text{ Å}$ the pair is a single

hump; the asymmetry nearly cancels, moving the requirement by less than 0.1 km s^{-1} and in the direction that makes the gap larger, since the extractor already credits the model with width it does not get from gas motion.

- *Stellar rotation.* With $P_{\text{rot}} = 131 \text{ d}$ and $R_* = 0.216 R_{\odot}$, $v_{\text{eq}} = 0.083 \text{ km s}^{-1}$ and the rotational broadening is 0.14 km s^{-1} . The Rossiter–McLaughlin distortion and the center-to-limb residual scale with the same $v \sin i$ and are correspondingly irrelevant; neither is mentioned anywhere in the paper or its Supplement, and on this star there is nothing to mention.
- *Planetary rotation.* Tidally locked: 0.032 km s^{-1} at R_p and 0.097 at $3 R_p$. It is already inside the model curves, which carry the solid-body spin.
- *Orbital motion.* 0.93 km s^{-1} across the full transit if left uncorrected, but the spectra are shifted into the planetary rest frame before averaging, which leaves the 0.022 km s^{-1} of the ephemeris uncertainty; a 300 s exposure smears 0.037 km s^{-1} .

Together these come to at most about 0.4 km s^{-1} , and most of that is removed as soon as each side is deconvolved from its own kernel. Done that way — component width against component width, both intrinsic — the requirement is 22.00 km s^{-1} FWHM ($\sigma = 9.34 \text{ km s}^{-1}$), against the 22.22 the convolution scan above returns without deconvolving either side. The audit moves the number by 0.2 km s^{-1} , not by a factor. (The audit itself was run on the earlier crossing; what is carried here is the differential it measured, -0.22 km s^{-1} , applied to the present scan.)

One defect was on our side, and is corrected here. The transit synthesis convolves the He 10830 line with a Gaussian of FWHM λ/R , and until this revision every LHS 1140 b run in this memo took R from the built-in default of EXHALE_transit.py, which is the CARMENES 8×10^4 , where this observation was made at WINERED’s 68,000. Every model curve quoted here has been re-synthesized at the WINERED value; the runs now read it from LHS1140b/winered_hires_y.sh, and the built-in default is left alone because the other planets in this repository were observed through other instruments. The correction widens the modelled blend by about $0.013 \text{ \AA}$ — at $\text{He}/\text{H} = 0.55$, $0.244 \rightarrow 0.257 \text{ \AA}$ — lowers the modelled peak depth by about 5 per cent, and leaves every equivalent width in this memo unchanged to $10^{-4} \% \text{ \AA}$, because a convolution conserves absorbing power. So it moves no crossing and no composition, only the widths and the broadening they demand, the latter by 0.12 km s^{-1} . The p-winds curves were already computed at $R = 68,000$, so the comparison of Sect. 8.5 is now through one instrument on both sides.

The paper’s own model does not reach the width either, and that changes what the shortfall means. Reproducing the authors’ p-winds setting at their retrieved medians and measuring it with the same extractor gives FWHM $0.458 \text{ \AA}$ with turbulence off and $0.575 \text{ \AA}$ with it on, against the measured 0.841. The retrieval that produced the published parameters therefore lands 1.5 times short of the width printed in the same paper. Taken with the accounting above, the reading this supports is that *the observation is not described by a one-dimensional steady-state wind of either code* — rather than that the wind solved here carries a defect the comparison model does not. That is an interpretation, not a measurement; what is measured is that both models are too narrow and that the measurement side does not account for the difference.

A second difference runs the same way. The p-winds curve reaches part of its width through saturation: its red-to-blue amplitude ratio is 2.36 with turbulence off and 2.57 with it on, against the optically thin limit of 8, whereas the

measurement sits at 6.35 and the solutions here at 6.9–7.0. The premise that this line is optically thin, so that column density cannot be traded against width, holds for the data and for these solutions but not for the comparison model.

Two remarks on the published width, offered as observations rather than conclusions. First, the three widths the Supplement’s stellar-variability test reports as FWHM — 0.22176 Å for the stellar helium line, 0.323 for the in-transit feature and 0.392 for the pre-ingress feature — appear to be Gaussian standard deviations rather than full widths. Fitting the same three-Gaussian model to the same pre-GP spectrum with our extractor returns $\sigma = 0.3241$ Å against their 0.323, a 0.3 per cent match, where the corresponding FWHM is 0.763 Å and matches nothing on that list. The $\sim 9\sigma$ separation the paragraph concludes with is a difference between two widths and is unaffected; the absolute values are. Second, the asymmetric uncertainties on the headline FWHM appear to be transposed. The deposited MCMC chain — identified by its shift column reproducing the published $0.072^{+0.080}_{-0.073}$ Å exactly — gives $2.3548\sigma = 0.865^{+0.261}_{-0.155}$ Å ($23.93^{+7.24}_{-4.30}$ km s⁻¹), where the paper prints $0.86^{+0.15}_{-0.27}$ Å and $23.9^{+4.2}_{-7.5}$ km s⁻¹: the magnitudes agree once the two are swapped. Neither remark can be settled from outside the collaboration, since the printed bars may come from a chain other than the deposited one, and neither changes the comparison except in one direction — on the deposited posterior the 1σ lower bound on the measured width is 0.709 Å rather than 0.59, so narrow models are excluded somewhat more firmly than the printed interval suggests.

What the audit could not exclude. The star’s own metastable helium line is broad: taking the Supplement’s 0.22176 Å as a standard deviation, it is 0.522 Å = 14.5 km s⁻¹ FWHM as observed and 13.8 km s⁻¹ with the instrument kernel removed. An imperfectly divided stellar line would therefore inject a residual much wider than the modelled planetary line, though still short of the observed feature on its own. The released data do not include the stellar template, so a mixture of a narrow planetary line with a displaced stellar residual could not be tested here and is not excluded. Whether the published σ is itself deconvolved from the instrument profile was also left unsettled; it is worth 1.7 per cent in width either way.

8.4 The same demand, under diffusion and the adopted K_{zz}

Everything above was solved in the well-mixed limit. With binary H/He element diffusion active, helium settles out of the base unless an eddy term lifts it, and how much of it reaches the line-forming region is set by the eddy coefficient K_{zz} , which is not measured for this system. The diffusion operator is described in `binary_diffusion_design.md`; the literature range at the 1 μbar base of these runs, the ladder of solved cases, and the adoption of $K_{zz} = 1.0 \times 10^9$ cm² s⁻¹ are in `LHS1140b/kzz_decision.md`.

The composition, and how much of it K_{zz} carries

Turning the operator on moves the equivalent-width crossing of Fig. 4 away from He/H = 0.4245, and by how much depends on K_{zz} . The composition was therefore rescanned at every decade of the eddy coefficient from $K_{zz} = 0$ to 10^{11} cm² s⁻¹, by the same route in every case: the same control input, the same direct-steady JFNK solve at $\|R\|_{\text{tol}} = 5.0 \times 10^{-3}$, the same single post-processing pass and transit synthesis, and the same red-pair equivalent width on the vacuum window 10832.60–10834.20 Å. Table 2 is the result. Five arms were solved at every decade, placed so that both ends of the measurement’s 1σ interval as well as the crossing itself fall inside solved arms: *every entry of the table*

is interpolated inside a bracket of two runs, and nothing is extrapolated. Two of the entries, $K_{zz} = 0$ and $K_{zz} = 10^9$, were afterwards solved at the interpolated composition itself and return $EW = 1.1081$ and $1.1081 \text{ } \text{\AA}$ against the measured 1.108 (LHS1140b/exhale/confirm_gm25), each on the lineage its crossing was read on — re-seeded from the nearest neighbouring arm the plateau entry gives 1.1001 instead, still inside the band and about 1 per cent away in He/H. Two readings of each bracket agree — a chord and a quadratic in log–log differ by 0.06 per cent at worst and by 0.02 or better at five of the seven decades.

Table 2: Where the red-pair equivalent width crosses the measured $1.108 \pm 0.030 \text{ } \text{\AA}$, at every decade of the eddy coefficient, with binary H/He diffusion active throughout. Five arms per decade (six at 10^{10}); the crossing is the chord reading and the 1σ range propagates the measurement error alone, within one K_{zz} . The decades 10^1 – 10^4 were not scanned in composition: at a fixed He/H = 3.79 their equivalent width differs from the $K_{zz} = 0$ one by 0.00003 to 0.014 of the measurement error, which is the entry they carry. The last column is the ratio to the value the earlier draft of this memo quoted, on the retired Penning coefficient. Runs, seeds and residuals: LHS1140b/exhale/ladder_gm25/results.txt.

$K_{zz} [\text{cm}^2 \text{ s}^{-1}]$	He/H at the crossing	1σ range	bracketing runs [EW in % $\text{\AA}$]	new/old
0 (molecular only)	3.8103	3.7056–3.9214	3.79 [1.1025], 3.93 [1.1403]	0.814
10^1 – 10^4	not distinguishable from $K_{zz} = 0$ by this line			
10^5	3.7911	3.6874–3.9025	3.79 [1.1077], 3.93 [1.1454]	0.810
10^6	3.6167	3.5160–3.7242	3.60 [1.1033], 3.74 [1.1424]	0.813
10^7	3.0781	2.9856–3.1715	3.06 [1.1022], 3.18 [1.1407]	0.814
10^8	2.2821	2.2112–2.3567	2.23 [1.0857], 2.32 [1.1242]	0.795
10^9 (adopted)	1.6261	1.5692–1.6833	1.60 [1.0944], 1.65 [1.1204]	0.778
10^{10}	1.1432	1.1004–1.1859	1.10 [1.0777], 1.15 [1.1128]	0.742
10^{11}	0.8468	0.8138–0.8808	0.83 [1.0928], 0.865 [1.1244]	0.763
none (well mixed)	0.4245	0.410–0.443	Fig. 4	0.772

The full ladder is three regimes, not one power law (Fig. 6). Above 10^7 the degeneracy is shallow and very nearly a power law, $d \log(\text{He}/\text{H})/d \log K_{zz} = -0.144$ (-0.147 , -0.153 and -0.130 across the three decades 10^8 – 10^{11}), so $\text{He}/\text{H} \propto K_{zz}^{-0.14}$. Below 10^7 the curve rolls over (-0.070 across 10^6 – 10^7 , -0.020 across 10^5 – 10^6) onto a plateau at $\text{He}/\text{H} = 3.81$, the value the line implies when the eddy term does nothing.

This shape is what survived the coefficient replacement of Sect. 2.5 unchanged. Decade by decade the exponent was -0.022 , -0.071 , -0.120 , -0.138 , -0.133 and -0.142 before and is -0.020 , -0.070 , -0.130 , -0.147 , -0.153 and -0.130 now, and the level fell by a nearly uniform factor: the ratio of the new crossing to the old is 0.81 at every decade below 10^8 and 0.74–0.78 above (last column of Table 2). What a rate coefficient this wrong moved was the absolute composition, by about a fifth; the conditionality the ladder is drawn to show is a property of the diffusion operator and the line, and it is untouched.

That plateau is where it is for a reason that is measured rather than assumed: the operator adds K_{zz} to the molecular coefficient, and the base coefficient of the $K_{zz} = 0$ arm at the crossing composition is $D_{\text{eff}} = 1.2370 \times 10^6 \text{ cm}^2 \text{ s}^{-1}$ (measured at $r = 1.00029 R_p$ in output/element_flux_profile.txt, dominant pair He I–H I), so $K_{zz} = 10^6$ raises it

by 81%, 10^5 by 8% and 10^4 by 0.8%. The sharper measurement holds the composition fixed at $\text{He}/\text{H} = 3.79$ and solves one common seed at every decade: the equivalent width departs from the $K_{zz} = 0$ one by 0.00003, 0.0003, 0.0017 and 0.014 of the measurement error at 10^1 , 10^2 , 10^3 and 10^4 , by 0.22 at 10^5 and by 1.75 at 10^6 : *no eddy coefficient at or below $10^4 \text{ cm}^2 \text{ s}^{-1}$ is distinguishable from none at all by this line, and 10^6 is the first decade that moves it by more than the measurement error.* At the other end the operator never reaches the well-mixed limit either — even at 10^{11} the inferred reservoir, 0.847, is twice the well-mixed 0.4245.

Over the span the literature leaves open at the base pressure of these runs the inferred reservoir ratio moves only from 1.14 to 2.28, a factor 2.0, against a factor 9 in equivalent width across the same two decades at fixed composition: the line responds to composition much faster than it responds to K_{zz} , so a large change in the eddy coefficient is bought back by a small change in the reservoir. *The composition inferred from this line under diffusion is therefore only weakly conditional on K_{zz} , as $K_{zz}^{-0.14}$, and strongly conditional on the assumed spectrum* — the SED moves the well-mixed crossing by a factor 8.9, from 0.4245 to 0.0479 (Sect. 10.), which is four times the entire K_{zz} lever.

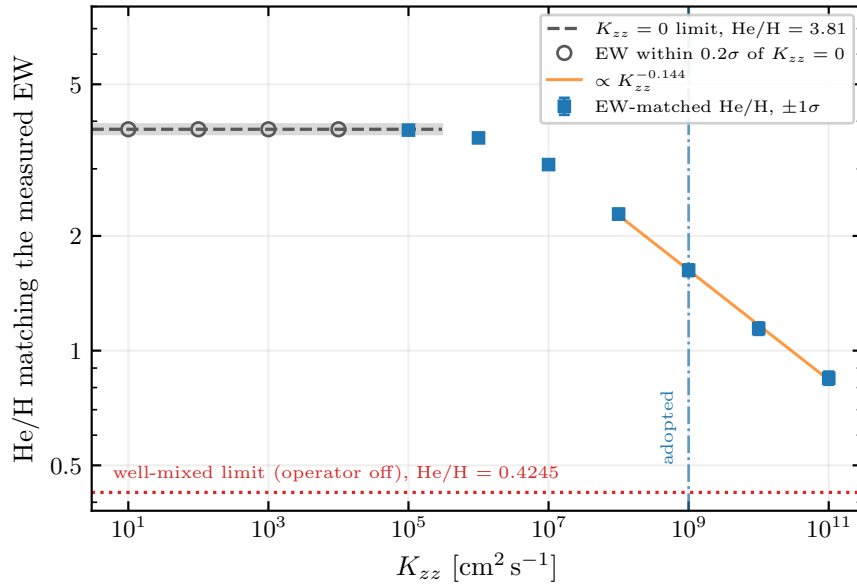


Figure 6: **The composition–mixing degeneracy in full.** Each square is the He/H reservoir ratio whose red-pair equivalent width equals the measured one, solved at one decade of K_{zz} with binary H/He diffusion active (Table 2); the error bars propagate the measurement error alone and are smaller than the symbols over most of the range. The landscape has three parts. Below $K_{zz} \simeq 10^5$ the eddy term is a small correction to the molecular coefficient the base already carries, the inference sits on a plateau at $\text{He}/\text{H} = 3.81$ (dashed, with its 1σ band), and the open circles mark the decades whose equivalent width differs from the $K_{zz} = 0$ one by less than a fifth of the measurement error. Between 10^6 and 10^7 the curve rolls off. From 10^8 up it is the shallow power law $K_{zz}^{-0.144}$ (orange). Neither end of the axis reaches the well-mixed crossing of 0.4245 (dotted): with the operator active, helium settles at every eddy coefficient tried, and He/H inferred from this line is a statement about the assumed mixing as much as about the planet. The adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ is marked.

What K_{zz} does act on strongly is the helium profile itself, and Fig. 7 separates that from the wind. At a fixed reservoir

$\text{He}/\text{H} = 0.55$, raising K_{zz} from 0 to 10^{10} lifts the homopause from below the base to $1.13 R_p$ and the metastable population by a factor 160 — read against the $K_{zz} = 0$ arm, which carries the defect of Update_EXHALE.md Sect. 88.4(a) and whose advection-corrected metastable column is therefore not to be trusted; between 10^8 and 10^{10} , where no defective arm enters, the factor is 4.1 — while the velocity and density profiles move by less than 16% and $\dot{M}$ by 0.19 dex. All four runs are re-solved on the current coefficient (LHS1140b/exhale/misc_gm25/). The replacement does *not* act equally on them, so this is not a comparison the coefficient cancels out of: the three arms with eddy mixing move by 0.02 dex and 1–3% in density while the settled $K_{zz} = 0$ arm moves by 0.06 dex and 20%. It sharpens the reading rather than weakening it — the spread of the wind panels narrows from 28 to 7 per cent in density while the composition panels still span two to four orders of magnitude. Temperature is the one bulk variable that follows, because the helium the eddy term lifts is itself a heating agent. Along the equivalent-width-matched sequence of Table 2, where each K_{zz} carries its own crossing composition, $\log_{10} \dot{M} = 7.80\text{--}7.81$ in every run: the escape rate does not carry the K_{zz} dependence at all.

The width the data demand, again

The question that remains is narrower: whether the width the data demand moves with the composition.

It does not. Repeating the measurement of Sect. 8.2 on crossings_gm25/kz_heh1p65, the arm that straddles the crossing $\text{He}/\text{H} = 1.6261$ from above (red-pair EW 1.120 % Å, red depth 4.186%, FWHM 0.251 Å), with the same convolution, the same three-Gaussian extractor and the same integration window, gives Table 3.

Table 3: The EW-matched diffusive solution (crossings_gm25/kz_heh1p65: $\text{He}/\text{H} = 1.65$, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$) convolved with an added Gaussian of FWHM f , measured exactly as Table 1.

f [km s ⁻¹]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.186	0.606	6.91	0.251	1.120
10	2.391	0.321	7.44	0.439	1.120
15	1.761	0.234	7.54	0.597	1.117
20	1.375	0.181	7.59	0.765	1.099
22.2	1.250	0.165	7.60	0.841	1.086
25	1.122	0.147	7.61	0.938	1.064
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

The width is reached at $f = 22.23 \text{ km s}^{-1}$ FWHM, against 22.22 km s^{-1} for the diffusion-off well-mixed solution of Sect. 8.2: the two differ by 0.01 km s^{-1} , more than three orders of magnitude below the number itself. At that kernel the depth is 1.250% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.165% against $0.25^{+0.14}_{-0.12}$, the ratio 7.60 against $6.7^{+12.7}_{-3.1}$ and the equivalent width 1.086 against 1.108 ± 0.030 — all four inside the observed 1σ interval at the value fixed by the width alone, as in the well-mixed case. So the velocity the profile demands is unchanged by turning diffusion on and by the adopted eddy coefficient: those change how much helium is there, not how fast it moves. Figure 8(a) displays the radial number density profiles of the hydrogen and helium species in this diffusive steady state,

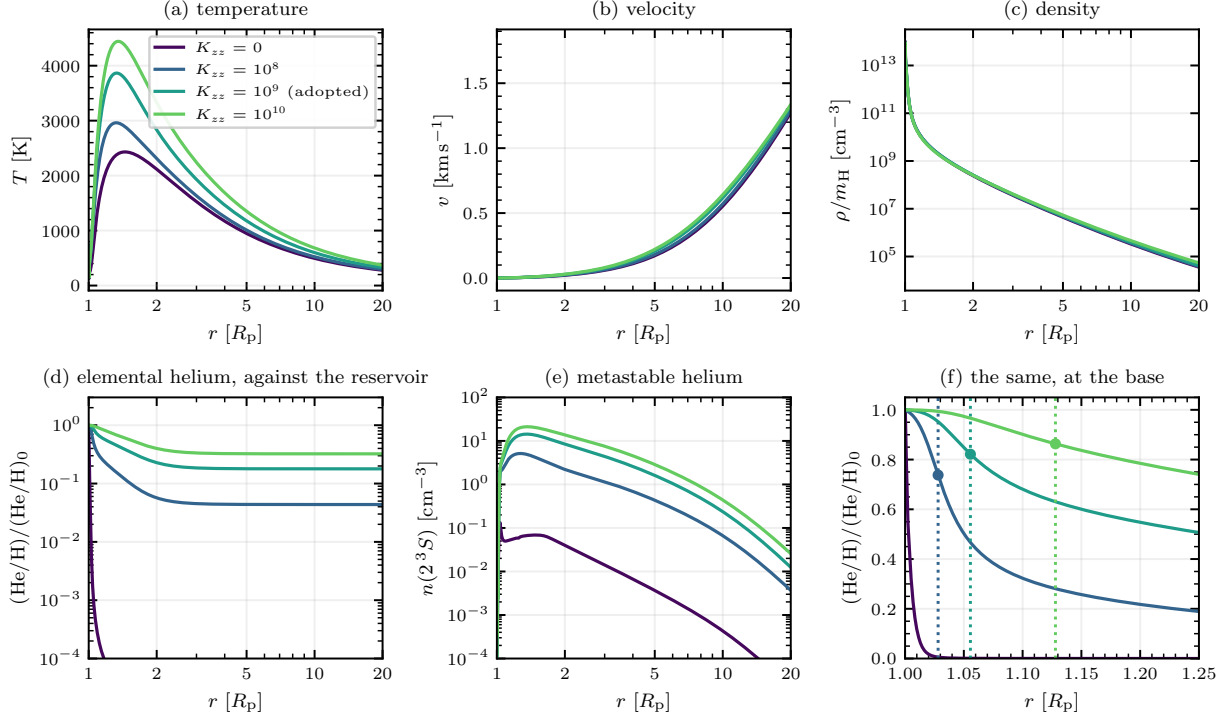


Figure 7: Four converged solutions differing only in the eddy coefficient, at a fixed reservoir $\text{He}/\text{H} = 0.55$ with binary H/He diffusion active: $K_{zz} = 0, 10^8, 10^9$ (adopted) and $10^{10} \text{ cm}^2 \text{ s}^{-1}$, dark to light. **The eddy coefficient is a composition knob, not a wind knob.** Panels (d)–(f), the elemental helium against its base reservoir value and the metastable population, span two to four orders of magnitude across the four runs — at $1.05 R_p$ the elemental ratio is 0.001, 0.503, 0.848, 0.974, and the peak $n(2^3S)$ is 0.134, 5.1, 14.3, 21.2 cm^{-3} — while in panels (b) and (c) the velocity at $10 R_p$ moves by 16% and the density at $2 R_p$ by 7%, and $\log_{10} \dot{M}$ by 0.19 dex. Panel (a) is the exception: T at $1.5 R_p$ rises from 2425 to 4287 K, because the helium the eddy term lifts is itself absorbing. Panel (f) repeats (d) on a linear axis over the base, where the dots and dotted lines mark each run’s homopause — the radius at which its own molecular coefficient equals its K_{zz} — at 1.028, 1.056 and 1.128 R_p , with the $K_{zz} = 0$ homopause below the base. All four are re-solved on the current Penning coefficient; the $K_{zz} = 0$ metastable curve of panel (e) carries the defect of `Update_EXHALE.md` Sect. 88.4(a) and is drawn for the trend, not for its value.

and Figure 8(b) shows the resulting broadened transit spectrum overlay.

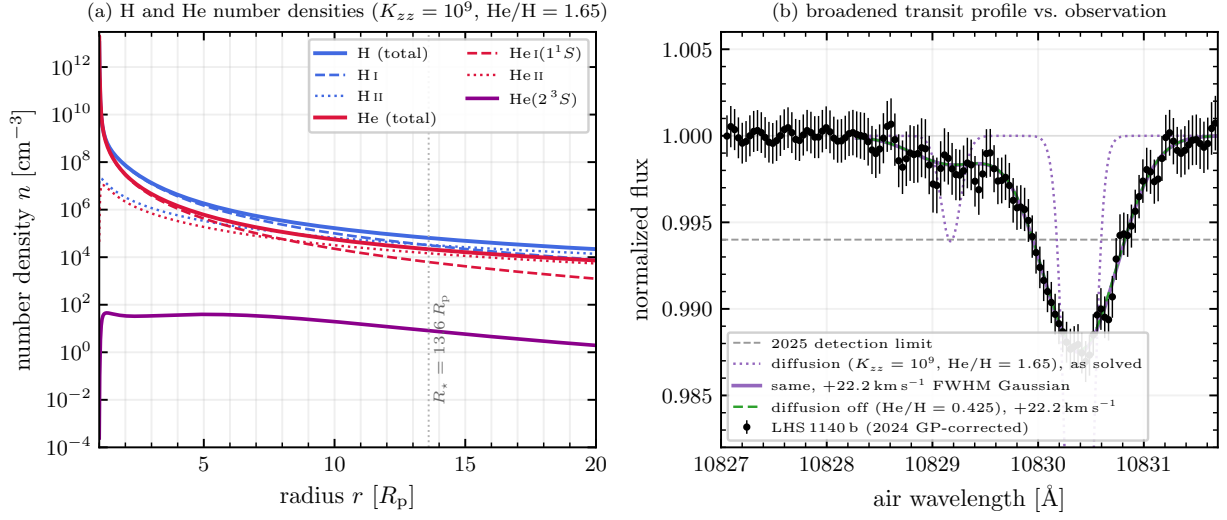


Figure 8: **The EW-matched diffusive solution** (crossings_gm25/kz_heh1p65: $\text{He}/\text{H} = 1.65$, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$). **(a)** Absolute number densities of hydrogen species (total H, H I, H II) and helium species (total He, He I(1^1S), He II, and metastable He(2^3S)) as a function of planetary radius r [R_p]. At the lower boundary ($r = 1.0 R_p$), $\text{He}/\text{H} = 1.65$; binary element diffusion causes helium to settle outward relative to hydrogen, with the local abundance falling to $\text{He}/\text{H} = 0.41$ at $2 R_p$ and 0.33 at $10 R_p$. Dotted vertical line marks the stellar disk radius ($13.6 R_p$). **(b)** The transmission spectrum as solved (dotted purple, its core running below the plotted range) and convolved with the 22.23 km s^{-1} FWHM Gaussian that reproduces the measured width (solid purple), against the measurement; the diffusion-off well-mixed curve at its own matched kernel (dashed green) is repeated from Fig. 5. The two broadened curves are indistinguishable — their kernels differ by 0.01 km s^{-1} . All model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

8.5 Broadening cannot repair a deficit in equivalent width

The same scan was applied to the p-winds curves, in their own vacuum frame and with the same three-Gaussian extractor and the same integration window (Table 4). The convolution conserves absorbing power: it redistributes the excess in velocity and leaves the equivalent width unchanged, apart from the few percent that the wings carry out of the integration window. Whether one kernel can reproduce depth and width together is therefore decided before the scan starts, by the equivalent width the curve already has at $f = 0$.

The p-winds best fit carries 0.837% $\text{\AA}$ against the measured 1.108 ± 0.030 , 24% short. At the kernel that reproduces the width, $f = 17.8 \text{ km s}^{-1}$, the red depth is 0.926% against $1.24^{+0.22}_{-0.23}$, about 1.4σ low. What the broadening does is move the disagreement rather than remove it: unbroadened, the curve is too deep in the blue component (0.534% against $0.25^{+0.14}_{-0.12}$, some 2σ high) and too narrow; broadened to the measured width, the blue component falls to 0.339% , inside 1σ , and the red component drops below the measurement. The blue side is genuinely improved; the red side is not, and the total is fixed.

The intensity ratio barely moves, 2.57 at $f = 0$ against 2.73 at the matched kernel. It is the amplitude ratio of a saturated

Table 4: The same added-Gaussian scan applied to the p-winds curves, with the EXHALE row of Table 1 repeated for comparison. The kernel is the one that reproduces the measured width, $0.841 \text{ \AA}$; the depths and the ratio are read at that kernel. Equivalent widths are for the unbroadened curve, over the vacuum window $10832.60\text{--}10834.20 \text{ \AA}$.

curve	EW($f=0$) [% $\text{\AA}$]	f [km s^{-1}]	red [%]	blue [%]	red/blue
p-winds best fit, turbulence on	0.837	17.8	0.926	0.339	2.73
p-winds best fit, turbulence off	0.701	20.1	0.777	0.301	2.58
p-winds depth-matched, 6100 K	0.480	20.2	0.535	0.140	3.82
EXHALE, He/H = 0.425	1.109	22.2	1.238	0.163	7.59
observed	1.108 ± 0.030	—	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$

pair, set by how the two components share the optical depth, and a common convolution acts on both alike. It stays well below the measured $6.7^{+12.7}_{-3.1}$, though the observed interval is wide enough that this alone is not a discriminator.

Matching the depth as well would require 24% more absorbing power, and that is not something a kernel supplies: it is a change in $\dot{M}$, temperature or composition. Fitting the broadening as a free parameter alongside those — which is what the measurement asks for — would presumably shift the C26 parameter estimates, since the composition is constrained through the same absorbing power that would have to rise.

The asymmetry with EXHALE should not be read as one code outperforming the other. The EXHALE curve entered the scan with its equivalent width already set to the observed value by the choice of He/H = 0.425 (Sect. 8.2), so only the shape remained free, and one kernel settled it. The p-winds curve entered at the composition C26 report, whose equivalent width falls short, so no kernel could settle both. The two scans test the same physical statement — broadening redistributes, it does not create — from two different starting points.

8.6 The adiabatic-cooling population bump, looked for

Schulik & Owen (2025) propose that the missing width is already contained in the metastable population itself. Writing the steady state of their Eq. (1) as

$$n_{2^3S}(T_e) = \frac{\alpha(T_e)}{q(T_e) + t_{\text{ion}}/n_e} n_{\text{He}^+}, \quad (5)$$

they note that as adiabatic expansion drops the temperature “from $\simeq 10^4 \text{ K}$ to $\simeq 10^3 \text{ K}$ ”, “a change in slope in the population occurs as the dominant triplet loss rate $q(T_e)$ drops significantly at lower temperatures”—the metastable fraction climbs above the curve computed with the rates frozen at the peak temperature. The rise is “quenched” where the advection timescale overtakes the excitation to 2^1S , after which $\dot{m}_{2^3S} = 4\pi r^2 v n_{2^3S} m_{\text{He}}$ falls through ionization and the $2^3S \rightarrow 1^1S$ decay. They credit Yan et al. (2022) and Biassoni et al. (2024) with first identifying the effect, and their conclusion states the condition it must satisfy to be observable: “The helium triplet population bump must be situated at high velocities to help explain anomalous line widths.”

EXHALE carries the temperature dependence the mechanism needs, and carries it in exactly the balance above. The

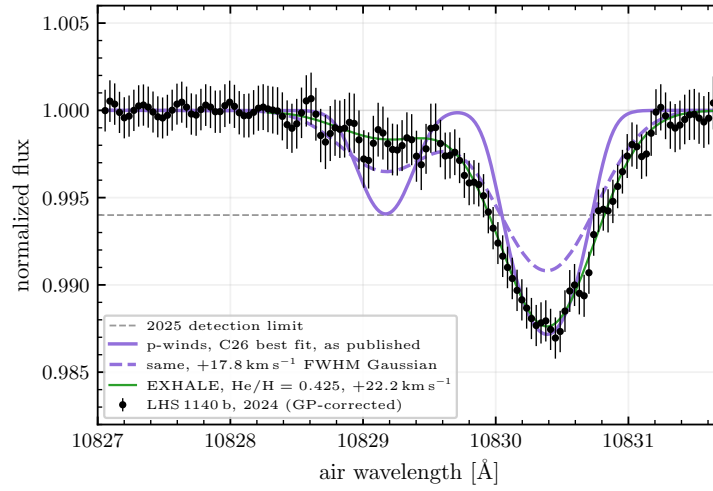


Figure 9: The C26 p-winds best fit as published (solid purple) and convolved with the 17.8 km s^{-1} FWHM Gaussian that reproduces the measured width (dashed purple), against the measurement; the EXHALE $\text{He}/\text{H} = 0.425$ curve at its own matched kernel (22.2 km s^{-1} , thin green) is repeated from Fig. 5. Broadened to the observed width, the p-winds curve passes above the red core: the width is matched and the depth is not. All curves are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

metastable row (`tr_triplet_row`, `ion_residual_core.f90:81`) is

$$n_e (n_{\text{He}^+} \alpha_{2^3S} + n_{\text{HeI}(1^1S)} q_{13}) = n_{2^3S} [\Gamma_{2^3S} + A_{31} + n_{\text{HI}} Q_{31} + n_e (q_{31a} + q_{31b} + b_{2^3S})], \quad (6)$$

with every coefficient a function of T : $\alpha_{2^3S} = 2.10 \times 10^{-13} (T/10^4)^{-0.778}$ (`Cool_coeff.f90:985`), the collisional losses q_{31a} ($2^3S \rightarrow 2^1S$, 0.80 eV) and q_{31b} ($2^3S \rightarrow 2^1P$, 1.40 eV) as Bray (2000) collision strengths folded with $\exp(-\Delta E/kT)$ (`Cool_coeff.f90:920` and `943`), the destruction term Q_{31} on atomic hydrogen as the García Muñoz (2025) closed form of Eq. (2) (`ioniz_HeI23S_H` in `Cool_coeff.f90`), collisional ionization of the metastable (`Cool_coeff.f90:1091`), and the radiative decay $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ (`util_ion_eq.f90:1235`). Between 5000 and 1000 K the loss $q_{31a} + q_{31b}$ falls by a factor 1.3×10^3 while α_{2^3S} rises by a factor 3.5, so the ingredients of the bump are present and are not an imposed parameterization.

LHS1140b/bump_analysis.py measures the result. It reconstructs the balance above in Python from the solved profiles, recovering from the row itself the one term not written to file—the 2^3S photoionization rate Γ_{2^3S} —and then recomputes the reference curve with α , q_{13} , q_{31a} , q_{31b} and Q_{31} frozen at the peak temperature. Each transcribed coefficient was checked against the compiled Fortran over 300– 10^4 K: Q_{31} agrees to 8×10^{-16} and the other five to 1.2×10^{-5} or better, the residue being the single-precision literals the Fortran carries.

The bump is there. In every case the metastable fraction $f_3 = n_{2^3S}/n_{\text{He}}$ rises monotonically outward across the whole cooling leg, from the temperature maximum to a maximum of f_3 at $1.1\text{--}14.5 R_p$ —a rise of $1.3\text{--}4.3 \times 10^3$ in the equilibrium profiles and $22\text{--}174$ in the advection-corrected ones—and the rise is attributable to the temperature dependence in the way Schulik & Owen describe: against the same balance with the rates frozen at the peak temperature, the solved population stands a factor 2.4–31 high at the density maximum and up to 9.7–156 high further out. On the GJ 1132

Table 5: The metastable bump in the converged solutions. Columns 2–4 locate the n_{2^3S} maximum in the equilibrium profile and give the temperature and outflow speed there; the last three characterize the absorber-weighted ($w = n_{2^3S} r^2 dr$) distribution of $|v_r|$ over the advection-corrected 2^3S column that the transit tool integrates. The He/H = 1000 solutions have no outward density maximum—their n_{2^3S} peaks at the base—and are marked $\dagger$. All six arms were re-solved for this table, each restarted on its own stored solution with the composition unchanged, so every row carries the He $2^3S + H$ ionization coefficient of García Muñoz (2025) (Update_EXHALE.md Sect. 87) and the current advection post-process (Sect. 88). The runs are LHS1140b/exhale/bump_gm25/; all six reached info = 0 with a JFNK residual $\|R\| \leq 8.3 \times 10^{-4}$ against their own tolerance of 10^{-3} , and the last two of the four passes each arm ran accepted no step, so the quoted values are Newton-finished.

case	r_{peak} [R_p]	T [K]	v [km s $^{-1}$]	med. $ v_r $ [km s $^{-1}$]	> 2 [%]	> 5 [%]
<i>GJ 1132 proxy SED</i>						
He/H = 0.55	5.66	1537	0.28	0.36	0.0	0.0
solar He/H	3.91	1475	0.13	0.40	0.0	0.0
He/H = 1000	1.05 †	2549	0.00	0.00	0.0	0.0
<i>GJ 699 proxy SED</i>						
He/H = 0.06	5.91	942	1.54	2.46	62.6	0.0
solar He/H	5.91	981	1.52	2.42	61.8	0.0
He/H = 1000	1.05 †	2530	0.00	0.00	0.4	0.0

proxy at the equivalent-width-matched composition $\text{He}/\text{H} = 0.4245$ (crossings_gm25/rs_heh0p425) the maximum sits at $5.269 R_p$ where $T = 1553 \text{ K}$, against $T_{\text{max}} = 4623 \text{ K}$ at $1.449 R_p$: the temperature has fallen to 0.34 of its peak, which is the regime the mechanism asks for. The quench is visible too—in the advection-corrected profiles the density maximum has moved inward to $1.05\text{--}1.65 R_p$ and n_{2^3S} then falls monotonically, so the outflow appears to be sweeping the metastable out at least as fast as recombination resupplies it, and $\dot{m}_{2^3S}$ drops past the peak by 98–99.9% on the helium-dominated $\text{He}/\text{H} = 1000$ runs, while every other case, the equivalent-width-matched ones included, carries it outward without loss.

The velocity condition, however, is not met, and not narrowly. At the bump the outflow is moving at 0.25 km s^{-1} on the GJ 1132 proxy and 1.5 km s^{-1} on the GJ 699 proxy. Weighting by absorbers over the whole 2^3S column, the median $|v_r|$ is at most 0.4 km s^{-1} and 2.5 km s^{-1} on the two spectra, and the largest $|v_r|$ anywhere in the column is 1.6 and 3.9 km s^{-1} . No part of the column on either spectrum reaches 5 km s^{-1} , against the $\sigma = 9.4 \text{ km s}^{-1}$ of extra broadening the measurement demands (Sect. 8.2). Radial velocity is the generous choice here: the line-of-sight projection of a radial field over the transit chord is smaller still. Figure 10 shows the three statements together.

So the bump exists in these solutions and contributes nothing measurable to the width. That is consistent with what Schulik & Owen themselves report for the coolest hosts: for their M2 case they find that “the transit signal probes the same velocity region for both treatments of thermodynamics, making the line insensitive to velocity”, and they conclude that K stars, not M stars, are the favorable testbeds—G-type hosts suppress the bump by photoionizing the triplet, and M-type hosts have weaker tidal fields and so rarely drive fast enough outflows. LHS 1140 is an M4.5 dwarf, cooler than the M2 they model, and its planet is a temperate super-Earth rather than a hot giant. The mechanism appears to be present and correctly captured; the host is the wrong one for it to help.

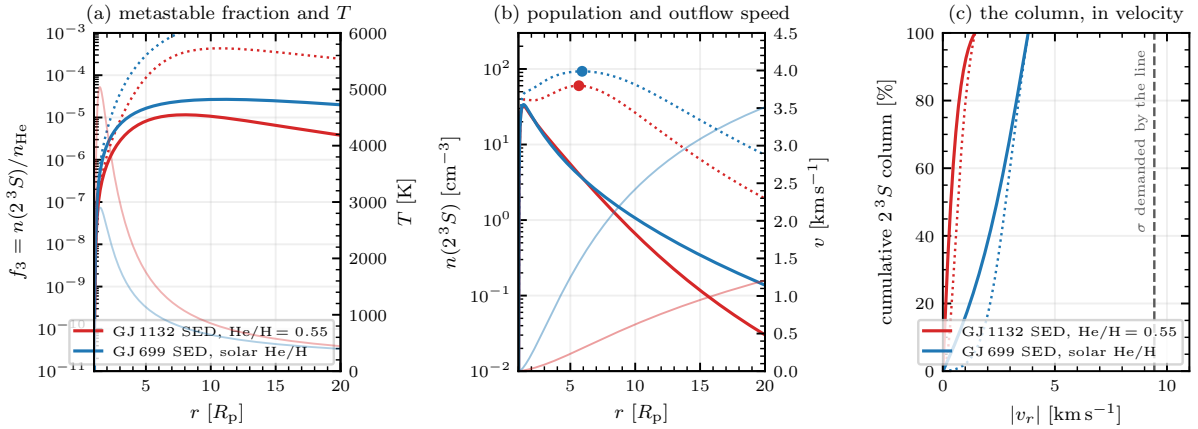


Figure 10: The adiabatic-cooling metastable bump in the EXHALE solutions (solid: advection-corrected, the profiles the transit integral uses; dotted: local equilibrium; faint curves on the right-hand axes). Two arms of Table 5 are drawn: $\text{He}/\text{H} = 0.55$ on the GJ 1132 proxy and solar He/H on the GJ 699 one. (a) The metastable fraction rises outward over the cooling leg, from the temperature maximum out to a maximum of f_3 at 10.7 and $12.0 R_p$ in equilibrium, which is the bump in the sense of Schulik & Owen (2025). (b) The equilibrium n_{2^3S} peaks at $5.7\text{--}5.9 R_p$ (markers); advection moves the maximum inward, to $1.4 R_p$, and quenches the outer rise. (c) The absorber-weighted distribution of $|v_r|$ over the 2^3S column: the whole column lies below 4 km s^{-1} , far short of the $\sigma = 9.4 \text{ km s}^{-1}$ the measured width demands.

8.7 The width in the wider He 10830 literature

The comparison so far has been between this model and this measurement. Put against the published population, the roles change: **the measured width is ordinary, and it is the model that is anomalous.**

The measured width is typical. Few He 10830 analyses fit the width freely rather than predicting it; where they do, the answer is 19–33 km s^{−1} FWHM. Table 6 collects them. LHS 1140 b’s 23.9^{+7.2}_{−4.3} km s^{−1} is an unremarkable member of that set. Guilluy et al. (2024) is the only survey that tabulates a free FWHM for several planets at once; their GJ 3470 b entry should not be taken at face value, since they write that “The larger FWHM for GJ 3470 b is believed to be attributed to residuals from OH line contamination”. The entries are quoted as published — different instruments, different extractors, no common deconvolution of the instrumental profile — so the clustering is a statement about published numbers rather than a uniformly reduced sample.

Table 6: Published He 10830 transit widths where the width was fitted freely. Conversions use 10833 Å. The last two rows are velocity dispersions of a different definition, given for scale, not as FWHM.

system	instrument (<i>R</i>)	FWHM [km s ^{−1}]	source
LHS 1140 b	WINERED (68,000)	23.9^{+7.2}_{−4.3}	this memo, on the released spectrum
WASP-107 b	GIANO-B (~ 50,000)	32.6 ^{+4.2} _{−3.6}	Guilluy et al. (2024), Table A.3
HAT-P-11 b (3 nights)	GIANO-B	29.3/19.0/22.9	Guilluy et al. (2024), Table A.3
WASP-69 b	GIANO-B	26.6	Guilluy et al. (2024), Table A.3
GJ 3470 b	GIANO-B	60.3	Guilluy et al. (2024); OH residuals
4 young sub-Neptunes	NIRSPEC (32,000)	27–33	Zhang et al. (2023)
SPIRou detections	SPIRou	~ 22	Masson et al. (2024)
HD 189733 b	CARMENES	$b = 15.1 \pm 0.6$	Salz et al. (2018), Table 3
HAT-P-32 b	CARMENES	$v_{t,a} = 15.2 \pm 3.1$	Czesla et al. (2022)

Zhang et al. (2023) treat that width as *evidence for* photoevaporation: “The width is the most consistent metric and is within 10% of 1.1 Å for all planets”, because “For a Parker outflow, the sound, bulk outflow, and thermal speeds are all close to $\sqrt{2k_B T/\mu m_H} = 9.4$ km s^{−1} for $T = 7000$ K and $\mu = 1.3$, leading to an FWHM of $2.355 \sigma \sim 22$ km s^{−1}.” That argument needs a 7000 K hydrogen-dominated transonic outflow. This solution has 5200 K at $2R_p$, a mean mass of 2.6–3.5 amu because of the helium enrichment, and an outflow of 0.024 km s^{−1} there; the same construction returns the model’s 2.4 km s^{−1}, not 9.4.

The one broadening term the field offers is Mach 1 by definition. There is a single turbulence formula in this literature, $v_{\text{turb}} = \sqrt{5kT/3m}$ with m the mass of a helium atom (Lampón et al. 2020, Eq. 16), adopted unchanged by p-winds as its Eq. (21) (Dos Santos et al. 2022). It is the adiabatic sound speed of a monatomic gas — a Mach number of exactly one — and it is a fixed function of T , not a free parameter. Evaluated on this solution it gives 4.2, 4.2, 3.6, 3.0 and 2.7 km s^{−1} at 1.2, 2, 3, 5 and $8R_p$, with a metastable-weighted mean of 3.8–4.0 km s^{−1} over $1\text{--}8R_p$. Against

the required $\sigma = 9.34 \text{ km s}^{-1}$ that is short by a factor 2.1–3.7 across the absorbing region and 2.3–2.5 on the absorber-weighted average. **The measurement asks for Mach 2–4 and the term on offer is Mach 1; it cannot absorb the requirement, whatever one concludes about its physical justification.** Three published analyses report the same failure in their own systems: Pallé et al. (2020) on GJ 3470 b (“The inclusion of the broadening of the lines due to turbulence ($v_{\text{turb}} = 5kT/3m$, where m is the mass of a helium atom), in addition to the standard Doppler broadening, was not enough to explain the measured broadening in the observations”), Lampón et al. (2021) on HD 189733 b (“Including a turbulence broadening component (see Sect. 3.3), the profile broadens, but it is still narrower than the measured absorption. . . . Hence, our hydrodynamic model alone is not able to explain the width of the absorption profile.”), and Czesla et al. (2022) on HAT-P-32 b (“Thermal broadening alone remains insufficient to explain the observed line width, even if the turbulence term is included.”).

Where a width is instead fitted freely, it comes back two to three times larger: Salz et al. (2018) obtain $b = 15.1 \pm 0.6 \text{ km s}^{-1}$ (extended model, $f = 20\%$) and 11.3 ± 0.5 (compact) at mid-transit for HD 189733 b, and Czesla et al. (2022) obtain $v_{t,a} = 15.2 \pm 3.1$, 11.2 ± 3.1 and $11.7 \pm 1.0 \text{ km s}^{-1}$ for HAT-P-32 b. Our own 9.34 km s^{-1} sits at the low end of that set.

An independent code reports the same shortfall, on the same axes. Taylor et al. (2026) apply a 1D self-consistent model of the same class — energy balance, lower-atmosphere coupling, multispecies diffusion, non-LTE $H(n = 2)$ — to HD 189733 b and find that “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, at “a factor of ~ 2 –3” in width. Decisively for the present work, the axes they varied are the axes varied here: “We tested variations in the free parameters (heating efficiency, K_{zz} , stellar activity level/XUV input, base composition, and lower-boundary placement): configurations that increase the width either deepen the core transit depth too much or violate the $H\alpha$ constraints, and none simultaneously reproduce both core depth and width.” Their proposed mechanism is stated by them as speculative — “Given that excited He is mostly produced by recombination of helium ions, this broadening can be driven by plasma turbulence . . . While these ideas are speculative, we encourage further exploration of plasma effects and magnetic field interactions” — and its first clause is the same physics Sect. 8.8 derives independently below. Rumenskikh et al. (2022), as reported there, reached the observed width in 3D only by prescribing zonal jets of 10 – 20 km s^{-1} together with reduced hydrogen line cooling; Schulik & Owen (2025) offer the adiabatic-cooling population bump as the one route with no free parameter, which Sect. 8.6 shows is present here and worth 0.25 – 1.5 km s^{-1} .

What follows. The width shortfall is not specific to EXHALE and not specific to LHS 1140 b: it is an unresolved problem of the 1D steady-state model class. The evidence for that statement is threefold — the measured width is a typical member of the published population (Table 6); the deficit is reproduced at the same factor of 2–3 by an independently written code, on a different planet, around a star at the opposite end of the activity range, after testing the same axes (Taylor et al. 2026); and the field’s own remedy is a term that is Mach 1 by construction and that its users report in print to be insufficient. Every published route to the observed width adds velocity by hand — a sonic turbulence term, a free Gaussian, or prescribed jets — and none of them follows from the 1D energy equation.

One citation is worth flagging, as an observation and not as a correction to anyone’s result. Lampón et al. (2023) and Yan et al. (2024) both cite Salz et al. (2018) as a source of the $v_{\text{turb}} = \sqrt{5kT/3m}$ convention, but “turbulen” does not occur anywhere in the published Salz et al. (2018), which fits a free Doppler parameter instead. The convention itself is stated in Salz et al. (2016), Eq. (5); the 2018 paper appears to be intended as the source of the measured *width*. As written, the attribution invites the reader to believe that a Mach-1 turbulence term reproduced the HD 189733 b width, when in that paper it was a free 15.1 km s^{-1} Doppler parameter.

8.8 The size of the non-thermal candidates

If the missing velocity is not thermal and not in the 1D energy equation, the next question is which non-thermal mechanism could supply $\sigma = 9.34 \text{ km s}^{-1}$. The estimates below are order-of-magnitude requirements computed on the reference solution (flux_closure/heh11p1/k01); the full set, including the stellar-wind standoff, the turbulent and kinetic power budgets, charge exchange and radiation pressure, is in docs/lhs1140b_width_nonthermal_candidates.md.

Magnetic. Setting the Alfvén speed equal to the required dispersion, $v_A = B/\sqrt{4\pi\rho} = 9.34 \text{ km s}^{-1}$, and reading ρ off the advection-corrected profile, the field needed in the line-forming region is $B = 0.423, 0.118, 0.043$ and 0.016 G at $1.2, 1.5, 2$ and $3R_p$. Continued inward as a dipole, $B_s = B(r/R_p)^3$, those are surface fields of $0.73, 0.40, 0.34$ and 0.43 G — a shallow minimum near $2R_p$, so a dipole of a few tenths of a gauss would put v_A near the required value over the whole region at once. *As a scale comparison* that is unremarkable: it lies between Earth’s surface dipole ($\approx 0.3 \text{ G}$) and Jupiter’s ($\approx 4 \text{ G}$). The difficulty is what such a field would do rather than whether a planet can have it: the plasma β at that strength is $0.28\text{--}0.38$ over $1.2\text{--}3R_p$, so the field would be dynamically dominant and would restructure the outflow rather than decorate it, and this solution does not model that. The wave energy flux required to sustain $\sigma = 9.34 \text{ km s}^{-1}$, $\rho(3\sigma^2)v_A$ through a sphere, is $12\text{--}870$ times the solution’s total XUV heating of $1.22 \times 10^{20} \text{ erg s}^{-1}$ over the same radii, which only a star–planet interaction driven by a wind of at least solar strength at this orbit could feed. **Not excluded by field strength; constrained by the power it has to be given, and by the fact that this comparison is being made for a 131 d rotator with $L_X/L_{\text{bol}} = 6.5 \times 10^{-6}$.**

Where a metastable atom can be made. This is the constraint that bears hardest on any mechanism that manufactures fast helium somewhere else — a stellar-wind interaction region, a charge-exchange or energetic-neutral population — and brings it into the beam. $\text{He } 2^3\text{S}$ is fed by He^+ recombination and destroyed by Penning ionization with atomic hydrogen (tr_triplet_row in ion_residual_core.f90). The collisional channel from the singlet ground state supplies a fraction 1×10^{-13} of the source at $1.2R_p$ and 2×10^{-14} at $2R_p$, falling to 10^{-33} by $5R_p$; the metastable population is a recombination product without qualification. Table 7 evaluates the two timescales on the solution, with $\alpha_{2^3\text{S}} = 2.10 \times 10^{-13}(T/10^4)^{-0.778}$ and the García Muñoz (2025) destruction rate of Eq. (2). Both columns are recomputed here from the solved profiles. What the coefficient replacement leaves alone is the wind the table stands on: between the retired-coefficient solution and the re-solved one T agrees to 1.3 per cent and t_{rec} to 1.4 at every radius, because the metastable is a trace population and not an energy-budget term. The last three columns are not preserved that way and should not be: they *are* the destruction rate, and it is the quantity Sect. 2.5 replaced, so they move by factors of 1.0–1.8 between the two fits. A reconstruction that reproduced them would be evidence that the replacement

had not been applied.

Table 7: Metastable helium source and sink timescales in the reference solution, and the distance an atom moving at the required 9.34 km s^{-1} covers within its total lifetime. The reference solution is re-solved on the current Penning coefficient (LHS1140b/exhale/misc_gm25/fc_heh11p1).

$r [R_p]$	$T [\text{K}]$	$t_{\text{rec}} [\text{s}]$	$t_{\text{Penning}} [\text{s}]$	$t_{\text{life}} [\text{s}]$	$L = 9.34 \text{ km s}^{-1} t_{\text{life}} [R_p]$
1.2	5124	8.4×10^4	1.75	1.75	0.0014
2.0	5298	8.8×10^5	60.3	59.8	0.050
3.0	3877	3.5×10^6	3.9×10^2	3.7×10^2	0.30
5.0	2669	1.6×10^7	2.9×10^3	2.1×10^3	1.7
8.0	2091	5.9×10^7	1.5×10^4	5.1×10^3	4.2
20	539	2.6×10^8	2.9×10^5	7.7×10^3	6.3

Two requirements follow, and they are severe. First, recombination is slower than destruction by 7.5×10^3 – 4.8×10^4 everywhere, so only a tiny standing fraction of the helium is ever metastable and the population is set locally and continuously. Second, and more restrictive, within its own lifetime an atom moving at the required speed travels $0.05 R_p$ at $2 R_p$ and $0.30 R_p$ at $3 R_p$ (Table 7, last column). **The inner metastable helium, which carries roughly a third of the absorption, cannot have been manufactured anywhere but where it is seen.** A stellar-wind or energetic-neutral population would therefore have to be re-ionized and recombine *inside* the absorbing region: the recombination time there is 10^6 – 10^7 s against a crossing time of $5 R_p$ at 9.34 km s^{-1} of 5.9×10^3 s, short by three to four orders of magnitude. The candidate is not excluded as a source of *some* broad outer emission, but it cannot be the source of the bulk of the line.

Three-dimensional flow. With $P = 24.737 \text{ d}$ and tidal locking, $\Omega = 2.94 \times 10^{-6} \text{ s}^{-1}$ and the corotation speed Ωr is 0.032 km s^{-1} at the surface, 0.098 at $3 R_p$ and 0.94 at the $29 R_p$ domain edge — already measured in Sect. 8.3 and two orders below the requirement. A pressure-driven day–night flow is bounded above by the thermal energy available, i.e. by a few times the 4 – 5 km s^{-1} sound speed in the complete-conversion limit, and independently by the measured line shift, whose posterior caps any coherent line-of-sight flow of the whole absorbing column at 4.7 km s^{-1} . Both are small here for the same reason: the driver of 3D circulation is the day–night temperature contrast, and LHS 1140 b is a temperate planet at $T_{\text{eq}} = 226 \text{ K}$ around an M4.5 dwarf, with an irradiation and a rotation rate one to two orders of magnitude below the hot Jupiters for which 10 – 20 km s^{-1} jets are prescribed. Taylor et al. (2026) reach the same conclusion for HD 189733 b, where the case is far more favorable: “typical GCM wind speeds (a few kilometers per second) are unlikely to supply the additional $\sim 12 \text{ km s}^{-1}$ required to match the observed width.”

What this does not settle. These are requirement estimates evaluated on *this* solution’s density, temperature and ionization structure. They say what each mechanism would have to be given in order to produce 9.34 km s^{-1} here, and on that basis none of them is comfortable: the magnetic route needs a power it has no obvious source for around this star, the fast-neutral route cannot make metastables where the absorption is, and the 3D route is short by two orders of

magnitude. They do not identify which mechanism is in fact responsible, and they do not exclude the possibility that the absorbing gas is not this solution's gas at all — the measured pre-ingress signal, at least as broad as the in-transit one at a phase when the planet occults nothing, is direct evidence that some of the signal comes from a structure no 1D radial model contains. **Which mechanism supplies the width is not determined by this work, and that is also the state of the literature.**

8.9 What would have to change

The measured line wants gas that is fast enough to be broad but not so extended that it absorbs too much. EXHALE at this XUV flux delivers neither. The candidates, in the order they should be tested:

1. *More energy* — tested (Sect. 10.), and it does not do it. The GJ 699 proxy carries five times the XUV of the GJ 1132 proxy used here, and the paper itself runs both. Solving the scan again on it raises $\dot{M}$ sevenfold and moves the composition estimate by an order of magnitude, but leaves the required extra broadening at 21.95 km s^{-1} against 22.18 : the width gap is not an energy-input problem.
2. *A different helium distribution* — tested (Sect. 8.4), and it does not do it either. Letting helium settle against binary diffusion and lifting it back with an eddy term redistributes the absorber by orders of magnitude and moves the inferred composition with it, but the required broadening stays at 22.19 km s^{-1} : it changes how much helium is there, not how fast it moves.
3. *Heating the code does not carry*. Taylor et al.'s trapped-ion turbulence, or any magnetically mediated dissipation, is outside a 1D hydrodynamic energy equation by construction. Sect. 8.8 sizes it: the field needed to make $v_A = 9.34 \text{ km s}^{-1}$ is a few tenths of a gauss, which is ordinary, but the wave power that field must carry is 12–870 times the XUV heating.
4. *Geometry*. A line-of-sight velocity field from day–night or zonal flow is a 3D effect; Rumenskikh et al. reach the observed width with $10\text{--}20 \text{ km s}^{-1}$ jets, which is the right order. For this planet, though, rotation gives $0.03\text{--}0.10 \text{ km s}^{-1}$ and the measured line shift caps any coherent line-of-sight flow at 4.7 km s^{-1} (Sect. 8.8).
5. *Adiabatic-cooling population bump*. Looked for (Sect. 8.6): the bump is present in these solutions, but it sits at $0.25\text{--}1.5 \text{ km s}^{-1}$, so it does not supply width.

The paper's composition constraint should be read in the same light: it inherits the isothermal assumption. Impose a 6100 K structure and only helium-rich gas absorbs enough; solve the energy balance on this spectrum and, in the well-mixed limit, the same absorption is reached near $\text{He}/\text{H} = 0.42$ — and near 0.048 on the other proxy spectrum, or near 1.6 once helium is allowed to settle against the adopted eddy coefficient. No answer on that list is secure while the width is unexplained, because the width is what tells us whether the line-forming gas is the gas either model thinks it is.

9. The best each model can do

Figure 11 puts only the two best models on the measurement: the paper’s p-winds fit at its own best fit (the MCMC medians), and the EXHALE solution whose equivalent width matches. They fail in different ways, and neither reaches the observed width. p-winds is the broader of the two, because its temperature is imposed where the line forms, but at $0.575 \text{ \AA}$ it is still 1.5 times narrower than the measurement, and it overshoots the blue component by a factor of about three. EXHALE carries the right total absorption but concentrates it into a line a third as wide as observed, so its core is too deep. Only the second failure can be undone by broadening alone (Sects. 8.2 and 8.5).

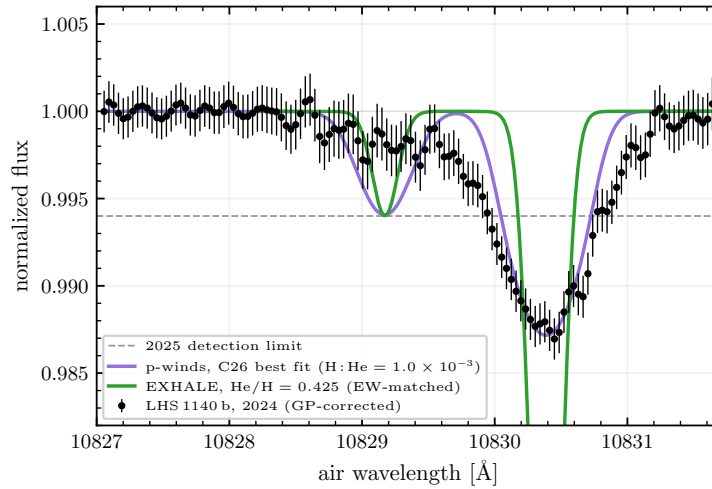


Figure 11: The two best models over the 2024 spectrum, in the presentation of the paper’s Fig. 4: p-winds at the C26 best fit (the MCMC medians) with the authors’ settings, and EXHALE at the equivalent-width-matched composition. Both are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

Figure 12 shows what that EXHALE solution actually is. It is not a fast, hot outflow: the temperature peaks at 4623 K near $1.45 R_p$ — comparable to the isothermal value p-winds imposes — but then falls by adiabatic expansion to 800 K by $10 R_p$ over the region that forms the line, and the velocity reaches only 1.2 km s^{-1} at $20 R_p$. Both elements stay mostly neutral through the line-forming region and helium is the *more* ionized of the two, not the less — $x(\text{H II}) = 0.05, 0.10$ and 0.22 at $2, 3$ and $5 R_p$ against $x(\text{He II} + \text{He III}) = 0.08, 0.16$ and 0.33 , since helium carries the electrons that make the gas and its ionization threshold is where this SED puts most of its photons. The metastable population peaks well out, at $5.3 R_p$ with $n(2^3S) = 48 \text{ cm}^{-3}$ in local equilibrium. That combination — a large, cold, slow, metastable-rich envelope — is precisely what produces the right equivalent width with the wrong profile: plenty of absorbers spread over most of the stellar disk, but almost no velocity spread to distribute the absorption in wavelength.

10. The other proxy: five times the XUV

The first item on the list above — more energy — can be tested without adding any physics, because the paper itself runs two proxy spectra. Both stand in for LHS 1140’s unobserved XUV: GJ 1132 (used everywhere above) and GJ 699, scaled to the orbit by the catalog X-ray flux ratios $A_{1132} = 0.59$ and $A_{699} = 0.056$ following the authors’ script, the

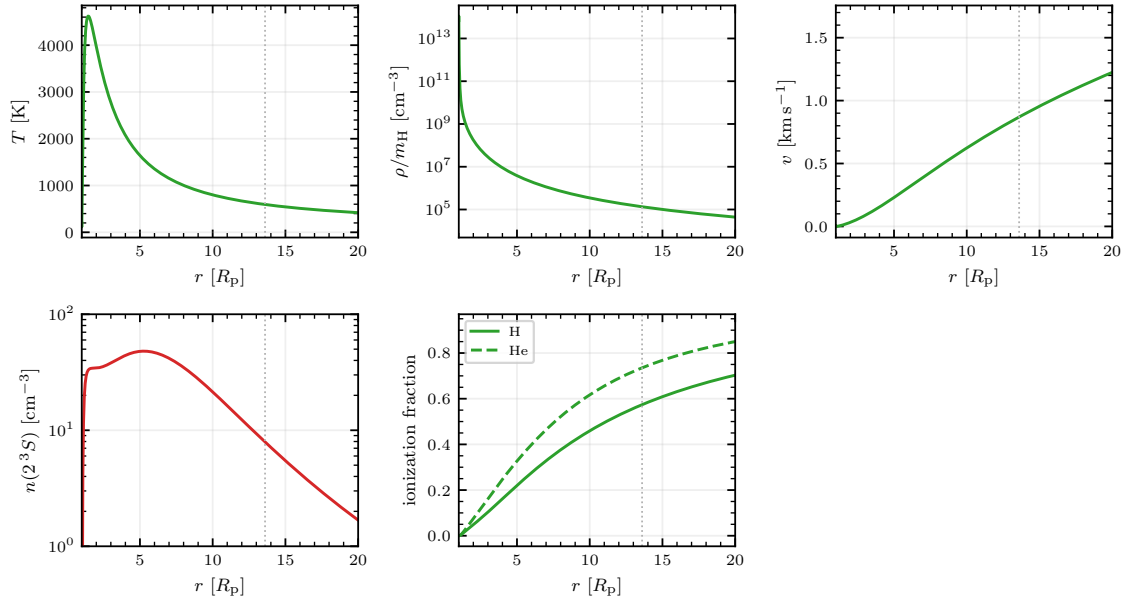


Figure 12: The EW-matched EXHALE solution ($\text{He}/\text{H} = 0.425$, the arm on the well-mixed crossing 0.4245): temperature, density, velocity, metastable helium density, and ionization fractions. Dotted vertical line: the stellar radius.

same convention adopted in Sect. 2.3. Integrated over 10–1300 Å at the orbit of the planet, the GJ 699 file carries $169.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ against 33.5 for the GJ 1132 one: a factor 5.1 in XUV input.

On the EXHALE side nothing else changes. Each GJ 699 case is the matching GJ 1132 input with one key replaced, `Spectrum file::`; the luminosity keys in the file are irrelevant because `sed_read` overwrites them with the integral of the spectrum it reads. Each case is started from the converged solution of the same composition on the GJ 1132 spectrum and finished with JFNK, and all seven reach `info = 0` with $\|R\| < 10^{-3}$. One case needed a different seed: $\text{He}/\text{H} = 0.55$ stalled when started from its GJ 1132 counterpart and converged when restarted from the already-solved $\text{He}/\text{H} = 0.25$ GJ 699 solution.

10.1 Escape rate: the retrieval value, on this spectrum

The brighter spectrum moves $\dot{M}$ where the composition never did:

	GJ 1132 proxy	GJ 699 proxy
$F_{\text{XUV}} (10\text{--}1300 \text{ Å}) [\text{erg s}^{-1} \text{ cm}^{-2}]$	33.5	169.9
$\log_{10} \dot{M} [\text{g s}^{-1}], \text{He}/\text{H} \leq 1$	7.75–7.80	8.57–8.61
$\dot{M} [\text{g s}^{-1}]$	6×10^7	4.0×10^8
C26 retrieval on the same proxy	2.03×10^8	$4.22^{+1.14}_{-1.00} \times 10^8$

A factor 5.1 in XUV buys a factor $\simeq 7$ in $\dot{M}$ — steeper than linear, as expected when a larger fraction of the input goes into the wind rather than into radiation. The result is that on this spectrum EXHALE’s escape rate agrees with the paper’s retrieved value, $\log_{10} \dot{M} = 8.58$ at the crossing against 8.63, inside 0.05 dex and far inside the retrieval’s

own ± 0.11 dex. It is also 1.4 times the energy-limited estimate the paper quotes for this proxy, $2.90 \times 10^8 \text{ g s}^{-1}$. On the GJ 1132 spectrum the same code fell a factor 3 short of the retrieval. Whether a solved wind and an imposed one agree on $\dot{M}$ therefore depends on which proxy is assumed, and the disagreement reported in Sect. 8. is not a fixed property of the code.

10.2 Structure and line

Seven compositions were solved on this spectrum, two of them ($\text{He}/\text{H} = 0.04$ and 0.06) placed specifically to bracket the equivalent-width crossing:

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	T_{max} [K] (at r/R_p)	$v(20R_p)$ [km s ⁻¹]	$n(2^3S)_{\text{max}}$ [cm ⁻³] (at r/R_p)	red [%]	FWHM [Å]	EW [% Å]
0.04	8.58	2599 (1.50)	3.60	44.2 (6.0)	3.14	0.281	0.938
0.06	8.59	2747 (1.48)	3.56	66.7 (5.9)	4.51	0.282	1.353
0.083	8.60	2902 (1.48)	3.51	93.1 (5.9)	6.00	0.284	1.806
0.25	8.61	3865 (1.43)	3.19	277 (6.1)	13.16	0.291	4.064
0.55	8.57	5099 (1.38)	2.79	511 (6.6)	16.74	0.300	5.337
1	8.58	6099 (1.37)	2.57	648 (7.1)	17.15	0.310	5.662
10^3	8.43	9352 (1.38)	2.07	1.4×10^4 (1.0)	9.26	0.352	3.479
observed	—	—	—	—	1.254	0.841	1.108 ± 0.030

The wind is hotter and faster than on the fainter spectrum — at $\text{He}/\text{H} = 0.06$ it reaches 3.6 km s^{-1} at $20R_p$ against 1.2 for the equivalent-width-matched GJ 1132 solution — and the metastable population still peaks far out, near $6R_p$. As in Sect. 5., the $\text{He}/\text{H} = 10^3$ row is a corrected value ($EW \ 35.87 \rightarrow 3.479 \% \text{ Å}$ after the advection-gate fix, `Update_EXHALE.md` Sect. 72); the six helium-poorer rows, which are the ones the crossing is interpolated in, are unaffected by *that* correction. All seven rows are re-solved on the current Penning coefficient (Sect. 2.5, `LHS1140b/exhale/misc_gm25/`). The crossing quoted below is still read off a separate ladder of five arms over $\text{He}/\text{H} = 0.042\text{--}0.050$ (`LHS1140b/exhale/ladder_gm25`, lineage G2), and that provenance now comes with a check rather than a warning: interpolated in log–log, the six rows with $\text{He}/\text{H} \leq 1$ cross the measured equivalent width at $\text{He}/\text{H} = 0.0481$ ($0.0466\text{--}0.0495$ on the $EW \ 1\sigma$) against the ladder’s 0.0479 ($0.0465\text{--}0.0493$) — agreement to 0.4 per cent, and to 0.3 per cent on both ends of the band. On the retired coefficient the same seven rows gave 0.0603 , which is why they were kept apart from the crossing before.

10.3 The composition estimate moves by an order of magnitude

Interpolating the equivalent width in log–log across the re-solved ladder and solving for the measured value gives

$$\text{He}/\text{H} = 0.0479 \quad (0.0465\text{--}0.0493 \text{ on the } EW \ 1\sigma), \quad \text{H}:\text{He} = 20.9, \quad (7)$$

chord and quadratic agreeing to 0.00 per cent. A run at 0.0479 itself gives $EW = 1.1073 \% \text{ Å}$, 0.02σ from the measurement, solved on the lineage the crossing was read on (`LHS1140b/exhale/confirm_gm25`). The bracketing runs

are what fix this: the arms at 0.046 and 0.048 straddle the measured band directly ($EW = 1.0672$ and 1.1094 against $1.108 \pm 0.030 \%$ Å), so the crossing is measured rather than extrapolated from the solar case. Two lineages were run here as well; the first, seeded from a stored solution, carries one off-trend arm at 0.044 and is the one discarded — re-seeded from a solution of the current binary the ladder is monotone and evenly spaced to four decimals, and that is the lineage of record. The turbulence-on crossing was not re-measured; on the earlier scan it moved the value by less than a per cent.

This is a shift of an order of magnitude from the $\text{He}/\text{H} = 0.4245$ the same code gives on the GJ 1132 proxy, and it crosses the solar value on the way: the brighter spectrum wants a helium-*poor* atmosphere, $0.58\times$ solar, where the fainter one wanted a helium-rich one, $5.1\times$ solar. The measurement cannot arbitrate, because the two proxies are two guesses at the same unobserved spectrum and both reproduce the observed equivalent width once composition is free. The conclusion to draw is therefore about the inference, not about the planet: what the helium line constrains is the product of composition and XUV input, and any composition quoted from it — the paper’s $\text{H}:\text{He} \lesssim 10^{-3}$ as much as the values here — is conditional on the assumed SED. On the present evidence the composition inference is SED-limited.

10.4 The width does not move

The width is the one thing the extra energy does not fix. Across the whole GJ 699 scan the red-pair FWHM is $0.283\text{--}0.352$ Å ($0.315\text{--}0.435$ with the turbulence term), against the measured 0.841 Å. Measuring the deficit the way Sect. 8.2 does — convolving with a Gaussian and asking which width reproduces the observation — the arm at the crossing needs 21.97 km s^{-1} FWHM of added velocity spread, against 22.22 km s^{-1} for the equivalent-width-matched GJ 1132 solution and 22.23 for the diffusive one of Sect. 8.4. A factor 5.1 in XUV input, a factor 7 in $\dot{M}$ and a factor 3 in outflow velocity at $20R_p$ change the required broadening by 1% .

The reason is visible in the table: the outflow velocity that grows is radial and small compared with the missing spread, while the line-forming region stays cold. The candidate “more energy” is therefore tested and rejected as an explanation of the width. It moves $\dot{M}$ and it moves the composition estimate; it does not move the profile. The remaining candidates — heating outside a 1D hydrodynamic energy equation, 3D velocity fields, and an adiabatic-cooling metastable bump — are unaffected by this test; the last of them has since been looked for on both spectra (Sect. 8.6) and does not help either.

11. Closing the loop: the composition as an output

Every number above took the helium abundance as an input. The scans of Sects. 5. and 8.4 chose it, solved the wind, and asked what the line did; the reservoir at the base was a parameter set by hand, and the lower atmosphere below the $1 \mu\text{bar}$ matching level was a handful of scalars. This section removes that freedom on one axis: the lower atmosphere is solved as a photochemical column, handed over as a profile, and the elemental fluxes are iterated until they are continuous across the matching level. The composition at the match is then a solution of the coupled problem rather than a number chosen for it. The implementation and its tests are recorded in docs/phase_e_flux_closure_design.md and docs/Update_EXHALE.md, Sects. 76–79; the runs are LHS1140b/exhale/flux_closure/. The three arms of this section are on Photochem 0.8.4; their winds have since been re-solved on the current Penning coefficient with the

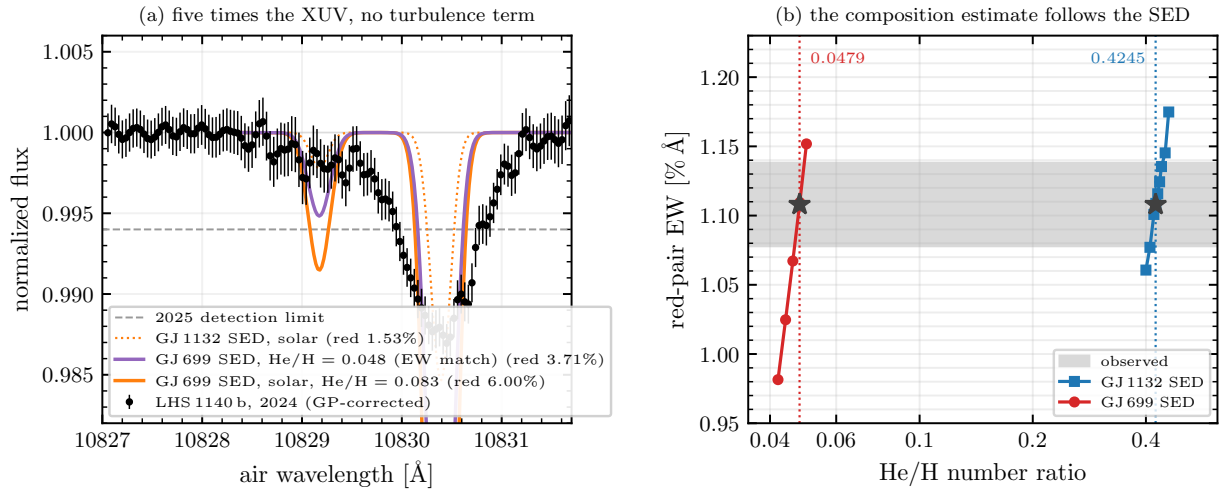


Figure 13: The GJ 699 proxy, five times the XUV of the GJ 1132 proxy used elsewhere in this memo. Left: the measurement with the EXHALE solution nearest the equivalent-width crossing 0.0479, the $\text{He}/\text{H} = 0.048$ arm of the ladder of record, and the solar case on this spectrum, with the GJ 1132 solar case dotted for contrast; all curves without the turbulence term and shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$. Every curve is on the current Penning coefficient, the two solar ones re-solved in LHS1140b/exhale/misc_gm25/. The lines are deeper and no wider: the GJ 699 solar curve is $3.9\times$ the depth of the GJ 1132 one (6.00% against 1.53%) while the added broadening the measurement demands of the two differs by 0.45 km s^{-1} , 2 per cent (21.95 against 22.40 km s^{-1}). Right: red-pair equivalent width against composition for the two spectra with the measured band and the two crossings; the composition the line implies moves from 0.4245 to 0.0479 when the assumed SED is changed.

photochemical columns held fixed (LHS1140b/exhale/misc_gm25/), which is what Table 11 now reports. What they establish — that the closure is single-valued and that at $K_{zz} = 10^9$ it hands back the reservoir it was given — is a statement about the iteration and not about the line, and the ladder of Sect. 11.7, which is what the composition is read from, was re-solved on both.

11.1 The setup

The column below the match is solved with Photochem (Wogan 2025) on the Zahnle reaction set restricted to H/He/N/O/C. The temperature is not prescribed: clima’s AdiabaticClimate is run with solve_for_T_trop on, so the stratospheric temperature is the skin temperature of the solution, and the background gas is chosen by abundance rather than by name — on this atmosphere that is helium, not H_2 . The climate solve is taken from a deep boundary at 20 bar, which is the shallowest one at which the photochemical model bottom fits inside the climate column. Chemistry and climate are run on the same elemental vector, He/H starting at 2.09 — the $K_{zz} = 10^9$ crossing of Table 2 — with Lodders C, N and O, on the GJ 1132 proxy spectrum at the planet’s orbit and at the adopted constant $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$. The result is handed to EXHALE as a profile of 102 levels from 16.73 bar to 1.10×10^{-8} bar, carrying p , r , T , n , ρ , K_{zz} , the molecular fractions and the elemental reservoirs He/H, C/H, N/H and O/H, matched to the wind at $1 \mu\text{bar}$; Sect. 11.2 reads that solution column by column, and says which of the columns the wind is actually built from. The loop is then closed in the other direction: the elemental escape fluxes EXHALE computes are imposed back on the photochemical model at its top — $\phi_{\text{El}} = \Phi_{\text{El}}/(m_{\text{El}} 4\pi r_{\text{top}}^2)$, helium on He and hydrogen on H_2 at $\phi_{\text{H}}/2$, the dominant carriers, the atomic-H share at that level being 10^{-4} — and the pair is iterated by damped Picard ($\omega = 0.5$) on the residual $|F_{\text{measured}} - \Phi_{\text{trial}}|/|F_{\text{measured}}|$ to a tolerance of 5%.

Two things about that measurement are worth stating, because they decide whether it means anything. The elemental flux is read in the *steady* window $r \geq r_{\text{esc}}$, not in the interval the two models share: at a steady state $4\pi r^2(\rho X v + J)$ does not depend on radius, so a flux measured there is the flux through the matching level. The overlap window cannot carry it — from the match to the profile top the column spans 1.96 decades of pressure in $0.0098 R_p$, entirely inside the region where the wind’s mass flux is not yet its own, and the flux across it has a radial spread of 13 in F_{H} and 35 in F_{He} . In the steady window the same solution holds 191 faces at spreads of 0.44% and 0.46%. And that flatness is a statement about how far the wind was converged, which was measured rather than assumed: at $\|R\|_{\text{tol}} = 10^{-3}$ the same solution spreads by 7.4%, which would fail the closure’s own validity test, and continuing it to 10^{-4} brings it to 0.50% and moves $\log_{10} \dot{M}$ from 7.47 to 7.50. The closure runs the wind at 10^{-4} .

Not verified: the cross-check with a second chemistry code. The same handoff was set up with VULCAN (Tsai et al. 2017) on the production profile’s own $T(p)$ and K_{zz} , and VULCAN did not converge on this cold helium-rich column: its convergence measure rose from 2.65 to 2.75 against a criterion of 0.01 with the timestep pinned, the limiting cell being H_2O at the cold trap, and the run exited on its step limit. No VULCAN composition at the match was produced and no wind was run on one, so the cross-code spread is not measured here. The record is LHS1140b/lower_profile_vulcan/vulcan_work/STOPPED.txt.

11.2 The column the chemistry returns

The settings above produce one file, and what EXHALE does on the profile branch rests entirely on it, so this section reads the solution itself. Figs. 14 and 15 draw every column of the schema except the two elemental-flux columns, which are constants carrying the trial fluxes the header states. The file measured here is LHS1140b/lower_profile/lower_atmosphere_profile.dat, the one-way handoff at iteration 0; the closed arms of Sect. 11.7 carry their own copies and are compared at the end of this section.

The grid and the thermal structure. 102 levels from 16.73 bar to 1.098×10^{-8} bar — the photochemical model's own domain, sitting inside the 20 bar column the climate step was taken from. Temperature runs from 520.3 K at the deep boundary to a floor of 185.05 K, which it reaches at 0.94 bar and holds at every level above: the stratosphere is isothermal at the skin temperature, and the tropopause the climate solve reports, 1.031 bar, is one level below the first isothermal level. K_{zz} is $10^9 \text{ cm}^2 \text{ s}^{-1}$ at all 102 levels, so the eddy coefficient the wind runs at and the one the column was mixed at are the same constant. The column spans 553.6 km in radius, 0.15608 to $0.16400 R_J$, with the 1 μ bar match at $0.16240 R_J = 1.782 R_{\oplus}$ — which is what fixes the base radius of the wind. Density falls from $n = 2.33 \times 10^{20} \text{ cm}^{-3}$ ($\rho = 1.40 \times 10^{-3} \text{ g cm}^{-3}$) to $4.30 \times 10^{11} \text{ cm}^{-3}$ (2.58×10^{-12}).

The column is chemically almost frozen. Between the deep boundary and the match, 17 decades of pressure, q_{H_2} moves from 0.19258 to 0.19264, q_{CH_4} from 1.072×10^{-4} to 1.025×10^{-4} and q_{NH_3} from 3.158×10^{-5} to 2.919×10^{-5} . What changes at all is water, which condenses, and a set of trace photochemical products at the very top. In an eddy-mixed column at $K_{zz} = 10^9$ that is what one expects, and it is the same statement as Sect. 11. makes from the other side — the eddy term homogenizes the column all the way to the match.

Hydrogen stays molecular to the top. q_{H} rises from 5.9×10^{-21} at the deep boundary to 6.5×10^{-6} at the match and 4.0×10^{-5} at the model top, four to five decades below q_{H_2} throughout: there is no level in this column at which atomic hydrogen overtakes H_2 . The base EXHALE is handed is an H_2 base in that sense. It does not enter the wind that way in these runs — EXHALE's molecular chemistry is off, and q_{H_2} is read but inert (below) — which is worth stating rather than leaving implicit.

The composition is cold and reducing. At 520 K and 17 bar CO is already negligible, $q_{\text{CO}} = 8.1 \times 10^{-18}$, and CO_2 smaller still: carbon is CH_4 and nitrogen is NH_3 , the opposite of the CO/N_2 -dominated deep chemistry of a hot Jupiter. Counting carriers at the match against the hydrogen-nucleus fraction of the gas, $2q_{\text{H}_2} + q_{\text{H}} = 0.38532$:

- carbon: CH_4 gives 2.661×10^{-4} per H, 99.7% of the carbon the profile's carrier columns hold, with C_2H_2 and HCN supplying the rest;
- nitrogen: NH_3 gives 7.576×10^{-5} per H, 96.3%, with N_2 at 3.4%;
- oxygen: the water that survived the cold trap gives 4.831×10^{-7} per H, 99.9%.

Those carrier sums come to 96.0, 96.0 and 97.6 per cent of the elemental columns $\text{C}/\text{H} = 2.778 \times 10^{-4}$, $\text{N}/\text{H} = 8.187 \times 10^{-5}$ and $\text{O}/\text{H} = 4.957 \times 10^{-7}$; the remainder sits in species the profile does not carry as columns, so the carrier accounting here is a decomposition of the elemental ratios and not an independent measurement of them.

This is the direct link to Sect. 11.6. The C and N that cool the wind and take its line down by $2.6\times$ are held in the column as CH_4 and NH_3 , both of which the solution keeps in the gas phase across the trap, so they cross the tropopause intact and arrive at the match at essentially their deep values. Oxygen, whose carrier is water, does not.

Photochemistry acts only in the top decades. From the tropopause to the match, HCN goes $1.3 \times 10^{-13} \rightarrow 6.5 \times 10^{-8}$, N_2 $9.2 \times 10^{-9} \rightarrow 5.2 \times 10^{-7}$, C_2H_2 $2.5 \times 10^{-13} \rightarrow 1.1 \times 10^{-7}$, CO $5.2 \times 10^{-16} \rightarrow 1.8 \times 10^{-10}$ and OH $4.5 \times 10^{-27} \rightarrow 7.1 \times 10^{-11}$ — several decades each, and all of them still trace. They do not move the elemental budget: the carrier accounting above puts every one of them below 4% of its element.

What EXHALE actually consumes from this file. Read in the code rather than inferred: the reader in files_I0/lower_atmosphere_profile.f90 requires the columns p, r, T, n_tot, rho, Kzz, q_H2, q_H and X_He to be present, and apply_lower_atmosphere_profile (input_read.f90) then takes, *at the matching level only*, T into T_0 , r into R_0 , q_H2 into q_h2_base, X_He into He/H, and X_C, X_N, X_O (and any other $X_{<\text{El}>}$ the file carries) into the metal abundances, through the same door metals.inp uses. p_match_bar comes from the header. The Kzz column is the one thing read over the whole profile, interpolated onto the radial grid through the file's own r column (eddy_diffusion_on_grid); a scalar He_Kzz: in input.inp is then announced as inert. Everything else is *not* read: n_tot and rho are required by the schema but never consumed — the base density is n_0 of input.inp and the base particle count follows from T_0 , He/H and the elemental ratios — q_H is required and never consumed, the two flux columns F_H and F_He are not read by the Fortran at all (the closure driver reads them on the Python side), and none of the molecular columns beyond q_H2 is read. And q_H2 itself only acts when a molecular base is switched on: it reaches the base EOS through h2_mixing_ratio_base, which comp_ntot_bc calls only under molecular_base. In the runs of this memo that flag is off, and the resolved report confirms it — mass_per_H = 9.37393 and ntot_bc = 1.000116 are exactly $1 + 4\text{He}/\text{H} + \sum X_{\text{El}}A_{\text{El}}$ and $(1 + \text{He}/\text{H} + \sum X_{\text{El}})/(1 + \text{He}/\text{H})$ with no H_2 term. So of the 23 columns, six numbers at one level plus one full column are what the wind is actually built from.

What the reservoir does to the column. The closure arms each carry their own solved column, and raising the reservoir changes it in three measurable ways (Fig. 15d–f). The deep boundary cools, $520.3 \rightarrow 439.4 \rightarrow 387.9$ K at 16.73 bar for He/H = 2.09, 5.01 and 10.31, while the stratosphere stays at 185 K. H_2 is diluted by number, $q_{\text{H}_2} = 0.1926 \rightarrow 0.0907 \rightarrow 0.0462$ at the match, with C/H and N/H unchanged to four digits because they are stated per hydrogen nucleus. And the cold trap loosens: O/H at the match goes $4.96 \times 10^{-7} \rightarrow 6.43 \times 10^{-7} \rightarrow 9.09 \times 10^{-7}$, the trend the O/H column of Table 13 continues over 8.45–11.0. Oxygen is the only element the reservoir moves here, and it is the one whose carrier condenses.

11.3 What the constant K_{zz} carries: the pressure form, tested

The eddy coefficient of the last section is a stated number rather than a solved one, and it is stated in a particular way: constant, $1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1}$ at every one of the 102 levels, which is what the adapter's --kzz-const 1.0e9 writes into the Kzz column. The form is an assumption laid on top of the value. What the *value* carries is already measured — the equivalent width moves by a factor $\simeq 9$ between $K_{zz} = 10^8$ and 10^{10} (LHS1140b/kzz_decision.md, Sect. 3) — so the open question is the different one of whether the *shape* carries anything. This section measures that, and the answer is that it does not carry the conclusions of this memo, for a reason that can be stated in one line: of the profile, EXHALE

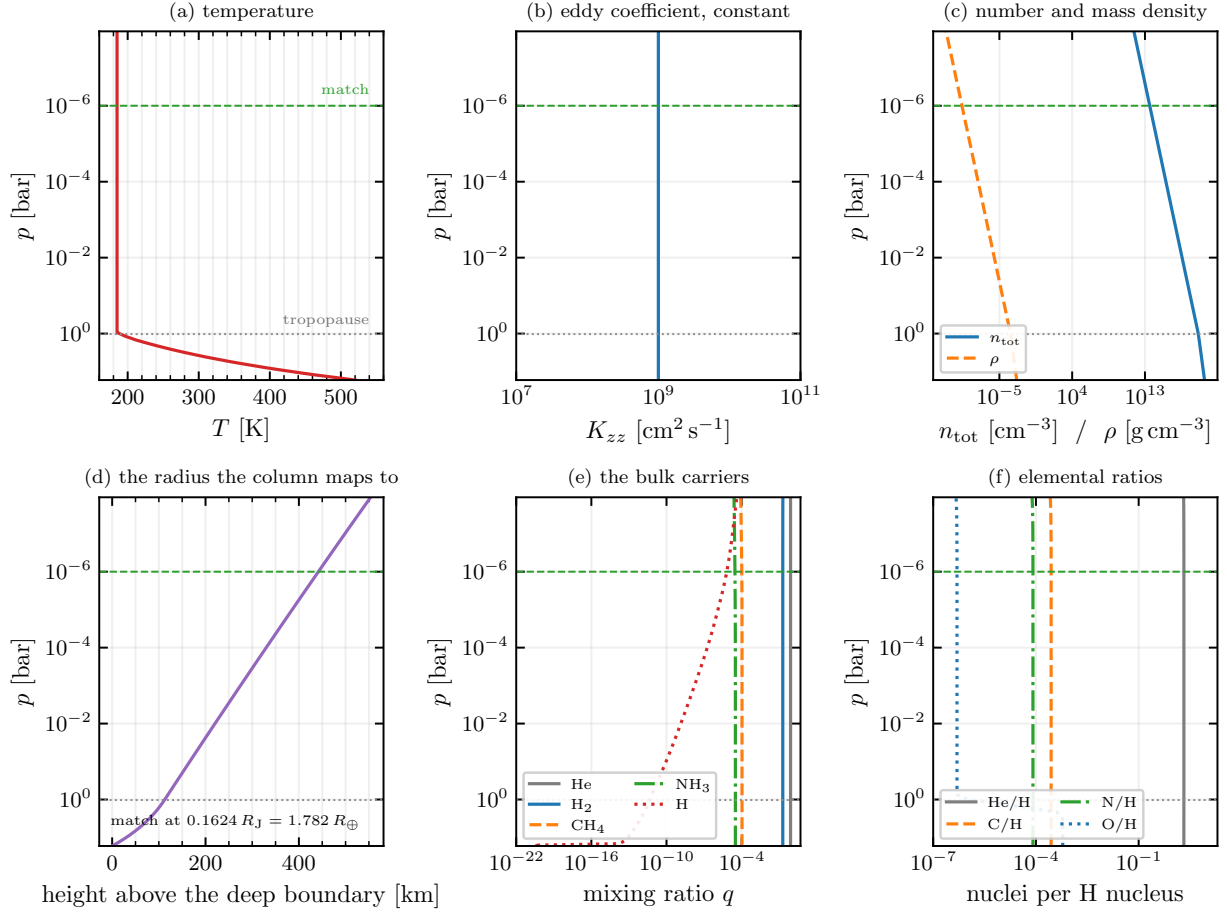


Figure 14: **The photochemical column, structure and bulk carriers.** (a) the solved temperature; (b) the eddy coefficient, constant at $10^9 \text{ cm}^2 \text{ s}^{-1}$ over all 102 levels; (c) number and mass density; (d) the radius the pressure grid maps to, which is what fixes the base radius of the wind; (e) the carriers that hold the bulk — helium reconstructed as $\text{He}/\text{H}(2q_{\text{H}_2} + q_{\text{H}})$, and atomic H four to five decades below H_2 everywhere; (f) the elemental ratios the profile hands over. The tropopause (1.031 bar, dotted) and the $1 \mu\text{bar}$ matching level (dashed) are marked on every panel. Fig. 19b,c shows the same climate and elemental ratios in the context of the closure; this figure is the file itself, column by column.

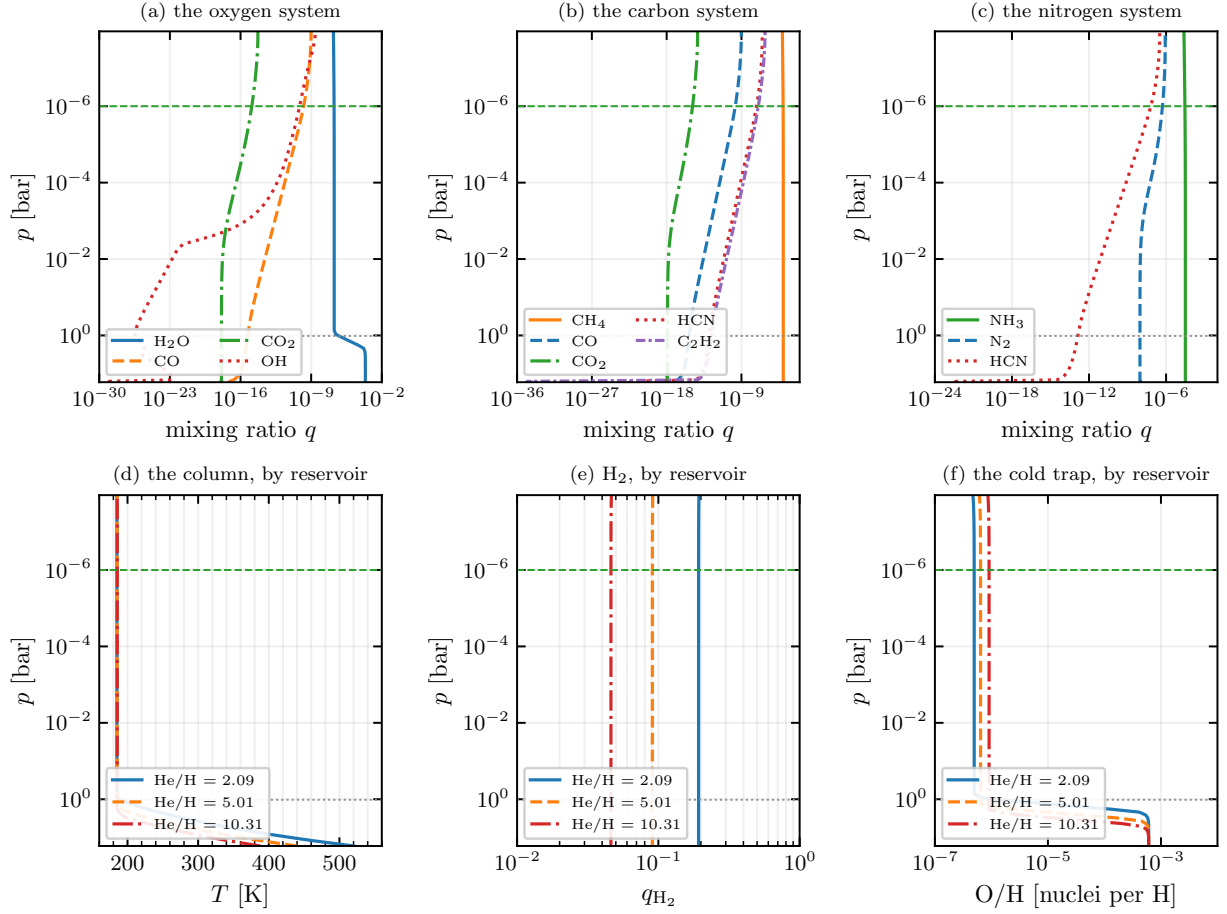


Figure 15: **The molecular systems, and what the reservoir does to them.** (a)–(c) the oxygen, carbon and nitrogen carriers of the same file: water condenses out at the tropopause while CH_4 and NH_3 cross it unchanged, and the photochemical products (HCN , C_2H_2 , N_2 , CO , OH) climb several decades in the top of the column while staying trace. (d)–(f) the same column solved at three reservoirs of the earlier ladder: the deep boundary cools as helium is added, H_2 is diluted by number, and the cold trap loosens, so oxygen is the one element the reservoir moves. Levels marked as in Fig. 14.

consumes elemental ratios and thermodynamics, and both are the same at the match to better than 1.2 per cent under either form.

What was tested. The saturated gravity-wave form,

$$K_{zz}(p) = 1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1} \left(\frac{p}{1.1 \times 10^{-8} \text{ bar}} \right)^{-1/2}, \quad (8)$$

anchored at the top of the profile and not at 1 bar. The anchor is the point of the test rather than a detail of it: EXHALE gives every cell above the shallowest tabulated level that level's value (`eddy_diffusion_on_grid`, Sect. 11.2), so anchoring at the top leaves the number the wind inherits above the file exactly what it was and confines the change to the interval the file describes. Equation (8) then implies 1.05×10^8 at the $1 \mu\text{bar}$ match, 1.05×10^5 at 1 bar and 2.56×10^4 at the 16.73 bar deep boundary: a factor 9.5 below the constant at the match, and four decades below it at the bottom.

Two arms separate the chemistry from the wind, because the profile hands over both:

- *A, chemistry only.* The column is solved on Eq. (8) and then its K_{zz} column alone is written back to the constant 10^9 before the handoff, so the wind sees the eddy coefficient it saw before and only the composition it is handed changes.
- *B, chemistry and wind.* The profile as produced; the wind reads $K_{zz}(p)$ as well — but only where the file speaks. The column reaches $1.0091 R_p$ above the match, and every cell above that takes the top level's value, so the wind-side change is a shell nine thousandths of a radius deep in which the coefficient is 9.5 times lower at the base and equal at the top. The homopause under the constant sits at $1.056 R_p$ (`kzz_decision.md`, Sect. 3), outside that shell. *Arm B is therefore a bounded test of the wind side and not a test of lowering K_{zz} where helium settles;* the unbounded version of that is the magnitude scan, which is already measured.

The reference is the constant- K_{zz} profile re-solved with the same seed and the same binary, so the comparison is not against a number recorded on an earlier occasion; the recorded number is carried in the tables anyway, because the difference between it and the reference is the size of the workflow's own path spread. Two reservoirs of the earlier closed ladder were run, one at each end of it: the 11.11 rung (`flux_closure/heh11p1`, $k=1$) and the 2.09 one on its high-trial-flux closure branch (`flux_closure/hi`, $k=6$). Runs and the measurement script: `LHS1140b/exhale/kzz_power/`. The test has since been repeated run by run on the current Penning coefficient (`LHS1140b/exhale/xuvkzz_gm25/`), and the reasoning it was first reported under — that the coefficient is common to both sides of every entry, so a ratio between two chemistries at fixed everything else does not turn on it — is confirmed rather than overturned: every ratio the test reports moves by less than the workflow's own path spread. The table below is the repeated version.

What did not move. The climate is identical: deep boundary 411.4 K, tropopause 2.636 bar at 185.5 K, cold-trap $f_{\text{H}_2\text{O}} = 5.590 \times 10^{-8}$ and match temperature 185.478 K, the same digits under either form. That is structural rather than coincidental — the adapter's `--climate` path solves the radiative-convective column with no eddy coefficient entering it, and hands the photochemistry a (P, T) pair with the K_{zz} slot empty — so the thermal structure, the tropopause and the trap the profile carries cannot depend on the eddy form at all.

The elemental ratios at the match, which with the temperature are what EXHALE consumes, move by at most 1.2 per cent:

Table 8: The pressure form of K_{zz} against the constant, at two reservoirs. *A* solves the chemistry on Eq. (8) and hands the wind the constant; *B* hands it Eq. (8) as well. The reference row is the constant- K_{zz} profile re-solved with the same seed, and the row above it is the same profile solved on an earlier occasion from a different seed — the path spread of the workflow, against which the effect of the eddy form is to be read. All eight arms are on the current Penning coefficient. *EW* is on this memo’s red-pair window, comparable with the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$.

run	red [%]	FWHM [$\text{\AA}$]	<i>EW</i> [% $\text{\AA}$]	<i>EW</i> /obs	He/H at $2R_p$	$\log_{10} \dot{M}$
<i>reservoir</i> He/H = 11.11						
recorded, other seed	3.8442	0.2840	1.1661	1.052	1.5498	7.400
reference, $K_{zz} = 10^9$	3.8920	0.2845	1.1813	1.066	1.5699	7.410
A, chemistry only	3.9063	0.2845	1.1855	1.070	1.5743	7.410
B, chemistry and wind	3.9071	0.2840	1.1849	1.069	1.5582	7.410
<i>reservoir</i> He/H = 2.09						
recorded, other seed	1.9855	0.2467	0.5218	0.471	0.2273	7.500
reference, $K_{zz} = 10^9$	1.9925	0.2467	0.5237	0.473	0.2280	7.500
A, chemistry only	2.0079	0.2467	0.5278	0.476	0.2296	7.500
B, chemistry and wind	1.9910	0.2467	0.5230	0.472	0.2267	7.500

Table 9: At the $1 \mu\text{bar}$ match, the ratio of the $K_{zz}(p)$ column to the constant- K_{zz} one. The top block is what EXHALE reads; the bottom block is what it does not.

quantity at the match	11.11 arm	2.09 arm
<i>consumed by the wind</i>		
X_{He}	0.9985	0.9985
X_{C}	0.9899	0.9903
X_{N}	0.9878	0.9875
X_{O}	0.9894	0.9821
T	1.0000	1.0000
<i>carried, not consumed</i>		
q_{CH_4}	0.443	0.665
q_{CO}	54.7	47.6
q_{CO_2}	9.3×10^3	4.3×10^3
q_{NH_3}	0.0038	0.0496
q_{N_2}	28.7	28.1
q_{HCN}	0.566	2.95
$q_{\text{H}_2\text{O}}$	0.714	0.804
q_{H}	4.99	5.55

What did move: the carriers, by up to four decades. The lower part of Table 9 is the real effect of the eddy form, and it is large. Nitrogen changes carrier outright: on the 11.11 arm q_{NH_3} at the match falls from 6.32×10^{-6} to 2.43×10^{-8} while q_{N_2} rises from 1.21×10^{-7} to 3.46×10^{-6} , so a column that was NH_3 -dominated becomes N_2 -dominated. Carbon partly follows, CH_4 down by 2.3 and CO up by 55; CO_2 rises by nearly four decades from a negligible base. Deeper the redistribution is more extreme still — at 1 bar the same ratios are 1.3×10^5 for CO, 3.0×10^7 for CO_2 and 8.8×10^3 for N_2 — because that is where the two forms differ most. *Elemental* nitrogen at the match nevertheless moves by -1.2 per cent, because a carrier swap conserves nuclei, and elemental nuclei are what the handoff transmits.

The crossing, against the path spread. Converted with the local slope $d \ln EW / d \ln(\text{He}/\text{H}) = 0.382$ measured over the 11.11–12.01 interval of the ladder these arms belong to, the four equivalent widths move that ladder’s crossing from 11.73 to 11.62 and 11.64 on the 11.11 arm and to 11.49 and 11.77 on the 2.09 one: -0.24 to $+0.04$ in He/H , under a tenth of the 10.92–12.58 interval the measurement’s own 1σ band allowed there. In relative terms that is -2.0 to $+0.4$ per cent, and it is the relative figure that carries over to the present crossing. It is also no larger than the workflow’s path spread: re-solving the *same* profile from a different seed moves the equivalent width by 1.29 per cent on the 11.11 arm and 0.36 per cent on the 2.09 one, against the 0.13–0.79 per cent the eddy form produces. The signal is buried in the spread of the procedure that would have to measure it.

An objection to the test itself, which has to be stated. The deep half of Eq. (8) is not the right law. The $P^{-1/2}$ growth is the saturated gravity-wave scaling of a *stratified* region (Lindzen 1981), and the climate solve puts this planet’s tropopause at 2.636 bar: below it the column is convective, where the mixing-length form of Gierasch & Conrath applies and K_{zz} should flatten or grow with depth, not collapse. Nor is the amplitude defensible there. At 1 bar Eq. (8) gives $1.05 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$, which is at or below the floor of the modelling literature — one and a half to two and a half decades under the 3×10^6 – 3×10^7 Charnay et al. (2015) use for GJ 1214 b, 3.7 decades under the 5×10^8 of Parmentier et al. (2013) for HD 209458 b, and equal to the 10^5 floor Ackerman & Marley (2001) impose by hand. The physically correct construction is two-branch, a wave law above the radiative–convective boundary joined to a mixing-length law below it; this test ran one branch all the way down to 16.73 bar. *Read as interpretation:* that is very likely why the carriers moved as far as they did, since the 0.1–10 bar interval the single branch over-quenches is the one that fixes where CH_4 and NH_3 quench. But the objection does not reach the verdict — a form this wrong in the deep column still failed to move an elemental ratio at the match by more than 1.2 per cent.

At the other end the same equation is on firmer ground than the adopted constant. Its value at the match, $1.05 \times 10^8 \text{ cm}^2 \text{ s}^{-1}$, falls inside the 2.8×10^7 – 1.4×10^8 that the Arfaux & Lavvas homopause scaling gives when it is evaluated with *this* planet’s gravity and temperature — the one estimate in the survey derived from LHS 1140 b itself, and the direction in which docs/eddy_diffusion_kzz.tex already records the adopted 10^9 as having no planet-specific basis.

Verdict. The constant assumption was carrying a great deal of the lower atmosphere’s carrier chemistry — up to four decades — and none of this memo’s conclusions. What EXHALE consumes from the profile is the elemental ratios and the thermodynamics; those agree at the match to 1.2 per cent under a form that differs from the constant by a factor 9.5 there and four decades at the bottom, because the match is still well mixed under either (1.05×10^8 is 200 times the molecular coefficient of $\simeq 4.6 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$ at the base, kzz_decision.md Sect. 1). *What the results carry is the magnitude of K_{zz} — the 10^8 – 10^{10} bracket, across which the equivalent width moves by a factor 9 — and not its pressure*

form. One limit on the statement: it is a statement about this anchor. Anchoring Eq. (8) at 1 μ bar instead would raise by 9.5 the value the wind inherits over its whole domain, which is a change of magnitude and therefore a different test; it was not run.

11.4 The cold trap the climate solve puts in

Solving the temperature rather than prescribing it puts a cold trap into the handoff, and the trap is where the climate solution says it is rather than where a closed form assumes it.

Table 10: The climate solve over its one free parameter, the deep boundary. $f_{\text{H}_2\text{O}}$ is the water mixing ratio above the tropopause, i.e. what crosses into the photochemical column. The stratospheric temperature is the robust part; the tropopause pressure is not, and the trapped water follows it, since at fixed T_{trop} the trapped ratio is $p_{\text{sat}}(T_{\text{trop}})/P_{\text{trop}}$.

deep boundary [bar]	T_{deep} [K]	P_{trop} [bar]	T_{trop} [K]	$f_{\text{H}_2\text{O}}$
1	242.4	0.488	186.0	3.33×10^{-7}
3	308.5	0.765	185.8	2.06×10^{-7}
10	439.9	0.974	185.4	1.49×10^{-7}
20 (adopted)	555.9	1.031	185.0	1.32×10^{-7}
30, 50	refused: the pseudoadiabatic leaves the thermodynamic data			

T_{trop} moves by 1 K over a factor of 20 in the deep boundary, against the skin temperature $T_{\text{eq}}/2^{1/4} = 190$ K; P_{trop} moves by a factor of two, and the water moves with it. The sensitivity is measurable from the scan itself, since $f_{\text{H}_2\text{O}}P_{\text{trop}}$ is $p_{\text{sat}}(T_{\text{trop}})$: across the rows it gives 19% per K at 185 K, so the trapped water is conditional on the tropopause temperature at about a fifth per kelvin and on the tropopause pressure linearly.

Against the published estimate. Cherubim et al. (2026) quote $f_{\text{H}_2\text{O}} \sim 7$ ppm from a closed-form cold trap at a 0.1 bar tropopause with $T_{\text{skin}} = 194$ K. The climate solution gives 0.13–0.33 ppm at a tropopause of 0.5–1.0 bar: a factor 20–50 less water at a tropopause 5–10 $\times$ deeper. The saturation physics agrees — $p_{\text{sat}}(185 \text{ K})/1 \text{ bar}$ is 1.3×10^{-7} and $p_{\text{sat}}(194 \text{ K})/0.1 \text{ bar}$ about 5×10^{-6} , so each number follows from its own tropopause — and what differs is where the tropopause is put. Ours is a solved level in a helium-dominated column ($\mu = 3.6$ at $\text{He}/\text{H} = 2.09$), theirs an assumed one. This is reported as a result about the climate model and was not tuned toward the published number.

What the trap does to the handoff is a depletion with height, not a condensate sitting at the base: gas-phase oxygen falls from $\text{O}/\text{H} = 6.062 \times 10^{-4}$ at the deep boundary to 4.957×10^{-7} at the match, a factor 1223, while C/H (2.778×10^{-4}), N/H (8.187×10^{-5}) and He/H cross the tropopause unchanged — the carriers of carbon and nitrogen in this column are CH_4 and NH_3 (Sect. 11.2), and this solution keeps both of them in the gas phase across the trap (Figs. 19c and 15a–c).

11.5 The closure, and what it returns

Three arms were run, from the reference trial pair $(\Phi_{\text{H}}, \Phi_{\text{He}}) = (1.8, 1.3) \times 10^7 \text{ g s}^{-1}$ and from 0.3 $\times$ and 3 $\times$ it, all seeded from the same converged wind and all starting from the same $\text{He}/\text{H} = 2.09$ reservoir.

Table 11: The elemental-flux closure on LHS 1140 b. k is the iteration at which the 5% residual criterion was met. The equivalent width is re-measured here on this memo’s own red-pair window (10832.60–10834.20 Å vacuum), so it is directly comparable with the observed 1.108 ± 0.030 % Å; the same curves integrated over the whole synthesized band, which is how Update_EXHALE.md Sect. 79 records them, give 4.74–4.82 mÅ on the retired coefficient. The k , F_{H} , F_{He} and He/H columns are outputs of the Picard iteration over the photochemical column and are as the stored closure left them; the last three columns are re-solved on the current Penning coefficient. *Both line columns are now at $R = 68,000$, the resolving power of the measurement and of the rest of this memo. An earlier version of this table printed its red column from the $R = 80,000$ synthesis while its EW column and the observed row were at $R = 68,000$; that mixture is corrected here.*

start	k	F_{H} [g s ⁻¹]	F_{He} [g s ⁻¹]	He/H at match	$\log_{10} \dot{M}$	red [%]	EW [% Å]
1.0×	0	1.8143×10^7	1.3204×10^7	2.09235	7.490	1.938	0.509
0.3×	5	1.8203×10^7	1.3427×10^7	2.09235	7.500	1.981	0.520
3.0×	6	1.8212×10^7	1.3471×10^7	2.09236	7.500	1.986	0.522
observed	—	—	—	—	—	$1.254^{+0.22}_{-0.23}$	1.108 ± 0.030

The three arms agree on He/H to 5.5×10^{-6} , on $\log_{10} \dot{M}$ to better than the 0.005 the log prints, and on the fluxes to 0.4% and 2%, over a factor of ten in where they started: the closure is single-valued on this planet. The damping was never exercised — the residual fell on every iteration — but the 3× arm used six of its eight allowed steps, so a start further than an order of magnitude out would need a larger iteration budget.

The composition is an output for the first time, and at this eddy coefficient it is the well-mixed value. He/H at the match comes back at 2.09235 against the 2.09 the reservoir was started from, a change of 1.1×10^{-3} , of which only 2×10^{-4} comes from the escape flux itself. The reason is measured: at $K_{\text{zz}} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ eddy mixing homogenizes the column all the way to 1 μbar, so a $1.8 \times 10^7 \text{ g s}^{-1}$ hydrogen escape imposed at the model top moves He/H by 1.8×10^{-2} at that top and by 2×10^{-4} at the match. *It is the eddy coefficient, not the escape flux, that sets the composition at the matching level.* Closing the loop therefore does not retire the conditionality of Sect. 8.4; it relocates it, onto the same adopted K_{zz} .

Above the match the wind separates the two elements as strongly as before. The elemental fluxes carry $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir’s 8.31, and He/H falls from 2.09 at the base to 0.183 at $30R_p$. The closure does not weaken that: it says the fractionation happens in the wind and not below the match. One caveat on the budget, stated rather than corrected: the diffusion operator’s mass fraction is that of the H + He mixture and the trace metals ride inside ρ , so the measured F_{H} is hydrogen plus metals and overstates elemental hydrogen by 4.8×10^{-4} , two decades below the tolerance.

11.6 What the base metals do, and what the closed solution does not reproduce

The profile brings something the scalar base never had: the carbon, nitrogen and oxygen of the photochemical column. Isolating it takes a ladder of controls in which the scalar-base run is moved one key at a time onto the profile’s own

base state, every rung a converged solution (Update_EXHALE.md Sect. 78). Setting the base helium ratio to the profile's 2.0921 moves $\log_{10} \dot{M}$ by less than 0.01; setting the base temperature to the solved 185.05 K, from the 226 K the scalar runs used, moves it by less than 0.01 more — the base temperature is not the lever. Adding the base radius moves it to 7.85 (the molecular fraction does not enter: `q_H2_base` is inert unless `Molecular_base` is set, which neither route sets). Adding the profile's C/H, N/H and O/H takes it to **7.48**, and the red-pair depth down by a factor **2.9** on that rung ($5.53 \rightarrow 1.91$ per cent): -0.37 dex in $\dot{M}$ and that much of a shallower helium line, from metal-line cooling of a 185 K base at $C/H = 2.8 \times 10^{-4}$. Over the ladder end to end the same pair is -0.32 dex and a factor 2.6. The profile route reproduces that rung to 0.01 dex in $\dot{M}$ and 1 per cent in the line, inside the workflow's own path spread, which is the consistency check the two routes exist to give. The six rungs are re-solved in LHS1140b/exhale/basemeta1s_gm25/, at $\|R\|_{\text{tol}} = 1.0 \times 10^{-4}$ on the four scalar rungs and 4.0×10^{-4} and 5.0×10^{-4} on the two metal ones; at that tolerance the scalar arm's equivalent width is 1.353 % Å against the 1.340 of Table 12, whose arm carries the looser 5.0×10^{-3} of the stored run — the two tables are each internally consistent and differ by 1 per cent for that reason.

The consequence for the line is that the flux-closed solution does not reproduce the measured equivalent width.

At $K_{zz} = 10^9$ and a metal-free base, $\text{He}/\text{H} = 2.13$ gives $EW = 1.340$ % Å. The flux-closed solution sits at the same eddy coefficient and essentially the same reservoir, 2.09235, and gives 0.522 % Å — a factor 2.1 below the observed 1.108 ± 0.030 , and 19.5σ from it. The difference is entirely the base metals beneath it. Both equivalent widths are re-solved on the current Penning coefficient (Sect. 2.5, LHS1140b/exhale/thermostat_gm25/), and the metal-free-to-closed ratio between them, 2.6, is unchanged by the replacement: the two rose together, by 19 and 22 per cent, while the 11.11 rung of the earlier closure ladder (`.../flux_closure/heh11p1/k01`) rose by 7.4 per cent. What the comparison establishes — that at one reservoir the photochemical base takes the line down by a large factor — is a comparison at fixed reservoir and does not turn on the coefficient.

So closing the elemental fluxes and matching the equivalent width are two different constraints, and the solution presented at 2.09 satisfies the first only. Fixing the reservoir by the line, as Sects. 7. and 8.4 do, was done above a base carrying no metals; with the photochemical column beneath it the same reservoir absorbs several times less, so the composition that satisfies both conditions at once is larger than the metal-free crossing. Sect. 11.7 solves for it, at the adopted K_{zz} , and returns $\text{He}/\text{H} = 9.05$ against the 1.63 the same line returns above the metal-free base — a factor 5.6, the same factor the earlier draft measured between 11.7 and 2.09. So the lower boundary is an axis of its own, of the same standing as the assumed spectrum, the assumed mixing and composition itself: what is placed beneath the wind is not a free parameter once the lower atmosphere is solved, but is set by the elemental vector and the cold trap of the column below.

The mechanism: helium as a thermostat

Why the same reservoir absorbs differently over the two bases can be read off the solved structures directly. Table 12 measures the three solutions that bracket the question: the metal-free scalar base at $\text{He}/\text{H} = 2.13$, which reproduces the line (LHS1140b/exhale/heh2p13_diff_kzz1e9/); the flux-closed solution on the photochemical column at 2.09, which does not (`.../flux_closure/hi/k06/`); and the closed solution at 10.31, the rung just below the crossing (`.../flux_closure/heh10p3/k01/`). All three run with H/He element diffusion active at $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ — constant

through the profile in the two profile runs, the scalar value in the third — and with EXHALE’s molecular chemistry off, so no molecular carrier appears in the wind. Mean mass per particle is $(n_{\text{H}} + 4n_{\text{He}})/(n_{\text{H}} + n_{\text{He}} + n_e)$; nuclei are counted as H I + H II and He I + He II + He III. The values are from the advection-corrected profiles; the reference arm of the closure (`.../flux_closure/ref/k00/`) reproduces the 2.09 column to 2.5% or better in every entry — the two largest departures being the equivalent width and the $2\text{--}10R_p$ metastable share, both at 2.5%.

Table 12: The two directions the thermostat runs, measured on the solved structures. Columns are the metal-free scalar base, the flux-closed solution on the photochemical column at the same reservoir, and the closed solution at 10.31. *EW* is on this memo’s red-pair window against the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$. All three solutions were re-solved on the current Penning coefficient of Sect. 2.5, each to four JFNK passes from its own output with the lower atmosphere held fixed (`LHS1140b/exhale/thermostat_gm25/`), so the table is read off one binary; the scalar arm is at its own $\|R\|_{\text{tot}} = 5.0 \times 10^{-3}$ and the two profile arms at 1.0×10^{-4} and 4.0×10^{-4} . The replacement raised the last four rows by 8 to 39 per cent, not by a uniform factor. The table remains a comparison at fixed reservoir, which is what the thermostat argument needs.

	scalar base	photochem base	photochem base
	no metals	C, N, O	C, N, O
reservoir He/H	2.13	2.09	10.31
T_{max} [K]	5498	3943	5914
$T(2R_p)$ [K]	4276	2979	5169
mean mass per particle at $2R_p$ [m_{H}]	1.98	1.50	2.53
$n(\text{H I})$ at $2R_p$ [cm^{-3}]	5.81×10^7	5.41×10^7	1.32×10^7
He/H by nuclei at $2R_p$	0.550	0.228	1.415
$\max n(2^3S)$ [cm^{-3}]	50.8	57.7	163.4
$\int n(2^3S) dr, 1\text{--}10R_p$ [$\text{cm}^{-3}R_p$]	81.3	47.9	114.6
of which $1\text{--}2R_p$	34.6	29.1	73.3
of which $2\text{--}10R_p$	46.7	18.9	41.3
<i>EW</i> [% $\text{\AA}$]	1.340	0.522	1.144

Adding the base metals cools the wind, and the line loses its outer column. With no metals, radiative cooling is 6.6–9.6% of the heating over $1.2\text{--}2R_p$ and the wind is limited by expansion rather than by radiation. With the column’s C, N and O beneath it, cooling reaches 71% of the heating at $1.2R_p$, of which 95% is metal-line cooling and the largest single contributor is neutral C I; T_{max} falls from 5498 to 3943 K. The metastable peak does not fall with it — it rises, $50.8 \rightarrow 57.7 \text{ cm}^{-3}$. The column inside $2R_p$ does fall, but only by 16% (34.6 against 29.1), against the 2.5 the column outside $2R_p$ loses: $46.7 \rightarrow 18.9 \text{ cm}^{-3}R_p$, close to the factor 2.6 the equivalent width falls by ($1.340 \rightarrow 0.522 \text{ \% } \text{\AA}$). What the metals take is therefore overwhelmingly the outer column: the cooler wind has the smaller scale height, so it does not carry the metastable population out to where it adds absorbing area.

Raising the reservoir reheats it, by two terms that are both measured. From 2.09 to 10.31 on the same base, T_{max} goes $3943 \rightarrow 5914$ K. Two things change together. First, there are fewer particles to share the deposited energy: the mean mass per particle at $2R_p$ rises from 1.50 to $2.53m_{\text{H}}$, a factor 1.68, and the heating rate at $2R_p$ divided by the

particle density rises by 2.01 — with n_{tot} on the code's `calc_ntot` convention, one particle per species and electrons excluded, as elsewhere in this section — while $T(2R_p)$ rises by 1.74. The three factors agreeing to within 20% is a consistency check on that reading, not a proof of it — the cooling term is not held fixed between the two runs. Second, the coolant is diluted along with the hydrogen it is referred to: the profile states C, N and O per hydrogen nucleus (schema column `X_*[El/H nuclei]`), and in the code $n_X = \text{melem_ab}(e)n_H$, and the two profiles carry the same $C/H = 2.778 \times 10^{-4}$ to four digits, so when $n(\text{H I})$ at $2R_p$ falls by 4.1 the carbon density falls with it by the same factor. The volumetric metal cooling nonetheless *rises*, $3.72 \rightarrow 5.96 \times 10^{-10} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $1.5R_p$, because the cooling function is steeply temperature dependent; what the dilution does is let the gas settle at the higher temperature at which that happens.

What the metastable balance does with that. The 2^3S population in these solutions is fed almost entirely by recombination and drained by Penning ionization on neutral hydrogen and by electron collisional de-excitation. Reconstructing the residual EXHALE solves — source $n_e n_{\text{He II}} \alpha_{2^3S}$ plus collisional excitation from the singlet ground state, against radiative decay A_{31} , photoionization, destruction on H I, electron collisional de-excitation to 2^1S and 2^1P , and electron collisional ionization (`ion_residual_core.f90`, `tr_triplet_row`) reproduces the tabulated $n(2^3S)$ to five digits, so the budget can be apportioned. The apportionment below is measured on the three re-solved solutions, i.e. on the García Muñoz rate. The direction the retired rate implied is confirmed: relative to the Taylor et al. (2025) fit both lifetimes lengthen and both Penning shares fall. At the metastable peak, collisional excitation supplies less than 10^{-13} of the source, and radiative decay, photoionization and collisional ionization together take 0.01% of the sink. What is left is: at $\text{He}/\text{H} = 2.09$, Penning 88% and electron de-excitation 12% of the loss; at 10.31, Penning 54% and electron de-excitation 46%. Between the two, the source rate rises by 1.77 (more He II and more electrons in the hotter gas, against a recombination coefficient that falls with temperature) and the metastable lifetime lengthens by 1.60, from 0.54 to 0.87 s: hydrogen has been thinned by 2.6 at the peak, so Penning removes the level more slowly, and the faster electron de-excitation of the hotter gas takes back only part of it. The product, 2.83, is the peak ratio $57.7 \rightarrow 163.4$. *There is no charge-exchange term on the metastable in this code* — the He/H charge exchange acts on ground-state He and He^+ — so it is not part of this balance.

Read as an interpretation, the thermal structure of the 10.31 solution ($T_{\text{max}} = 5914 \text{ K}$, column 114.6) has recovered and slightly passed that of the metal-free 2.13 solution (5498 K, 81.3), which is why the equivalent width comes back to the measured value there. The ladder therefore looks less like a determination of composition than like a thermal trade between the metal content of the base and the helium content above it, and one line cannot separate the two. Helium in these solutions is not only the absorber; it is also the thermostat that sets how far the absorber is carried.

What the wind is made of, and how ionized it is. Fig. 16 carries the elemental fractions and the ionization state of the same three solutions, on the same advection-corrected profiles as Table 12. Two things follow from it that the thermostat argument rests on.

Diffusion keeps the wind far less helium-rich than the reservoir. On both branches the helium fraction falls steeply through the line-forming region and then flattens: $\text{He}/\text{H} = 2.09$ at the base gives 0.228 by nuclei at $2R_p$ and 0.184 beyond $5R_p$ on this arm (0.180 on the reference arm quoted in Sect. 11.); 10.31 gives 1.415 and 1.153. That is a factor 9.2 and 7.3 between the reservoir and the wind at $2R_p$, and 11.4 and 8.9 beyond $5R_p$, in a solution where the reservoir

is an output of the closure rather than a choice. The consequence is worth stating in the composition's own terms: the flux-closed solution at 2.09 — a reservoir that is 68 per cent helium by number — has a *hydrogen-dominated* line-forming region, $n_{\text{H}}/n_{\text{tot}} = 0.81$ against $n_{\text{He}}/n_{\text{tot}} = 0.19$ at $2R_p$. At 10.31 helium still leads there, 0.59 against 0.41, but by a factor 7 less than the reservoir states. A composition quoted for this planet therefore has to say at which level it is quoted; the reservoir and the absorbing layer are not the same number, and the memo's crossings are reservoir values.

The hotter solution is also the more ionized one, and it is not shielding. At $2R_p$, $x(\text{H II}) = 0.030$ on the photochemical base at 2.09 and 0.104 at 10.31 — a factor 3.4, in the run whose neutral hydrogen density is 4.1 times lower. The obvious reading, that thinner hydrogen shields itself less, is contradicted by the outputs: the H I column above $2R_p$ gives $\tau(13.6\text{eV}) = 0.23$ and 0.054, so both are optically thin there, and the H I photoelectric heating per neutral hydrogen atom is the same in the two, 1.21 against $1.20 \times 10^{-18} \text{ erg s}^{-1}$ per atom. The recombination sink moves little: with α_B as the code defines it (rec_HII_B, the Badnell radiative rate less the Mao & Kaastra $n = 1$ term), $\alpha_B(T)n_e$ is 1.70 against $1.42 \times 10^{-6} \text{ s}^{-1}$, a factor 1.20. What grows is the ionization of hydrogen *by helium*. Measured on the equilibrium state at $2R_p$, the two channels that can be computed in closed form from the outputs are Penning ionization $\text{He}(2^3\text{S})+\text{H}$, which supplies 30% of what case-B recombination removes at 2.09 and 20% at 10.31 — a falling share but a rate per neutral hydrogen atom 2.0 times higher — and the charge exchange He^++H , at 3.5% and 2.6%. The rest, 66% and 77%, is photoionization, and inside it the helium-fed parts are what scale: the He-recombination-radiation channel (He II recombining to ground, whose $\geq 24.6 \text{ eV}$ continuum ionizes H) carries 6.4 times more heating per neutral hydrogen atom in the 10.31 solution, and the energy deposited by He I photoionization per neutral hydrogen atom, part of which secondary ionization turns into hydrogen ionizations, is 7.0 times higher. *What could not be settled from the outputs* is the split inside that 66–77%: the individual rates — direct $\Gamma_{\text{H I}}$, the He-recombination-driven addition to it, and the secondary-ionization yield — are not written to any output file, and the factors quoted here are read off the heating columns on the assumption that the mean photoelectron energy is similar between the two runs. The one channel that can be excluded outright is $n = 2$ photoionization: heat_Hpe[excitedH] is exactly zero at every cell of all three runs. Helium ionization itself tracks the temperature more simply — $x(\text{He II})$ is 0.055, 0.065 and 0.081 at $2R_p$ across the three, and $x(\text{He III})$ stays below 2×10^{-4} at $2R_p$ (9.7×10^{-5} , 1.4×10^{-4} and 2.0×10^{-4}), rising to at most 3.0, 4.4 and 6.2×10^{-4} by $2.98R_p$.

Fig. 12 shows the same kind of structure for the single well-mixed solution that reproduces the line with no diffusion and no base metals; Fig. 16 is the set the lower-boundary axis is built on, so the two are complementary rather than alternative views.

11.7 The composition the line needs above a photochemical base

Sect. 11.6 left the composition that closes the fluxes *and* reproduces the line unsolved. This section solves for it on the composition axis, at the adopted eddy coefficient. What comes out is the answer on the solved-column branch alone: it is the counterpart of the 1.63 that the same line returns above a prescribed metal-free base (Sect. 8.4), not a replacement for it, and the two differ by the base and not by the mixing, which is $K_{zz} = 10^9$ with diffusion active in both. The setup: the closure driver is run at eleven reservoirs between 8.45 and 11.0, *all from one common parent* (the converged 9.71 arm of the earlier ladder, so no reservoir step exceeds $1.21\times$), and each converged iterate's wind

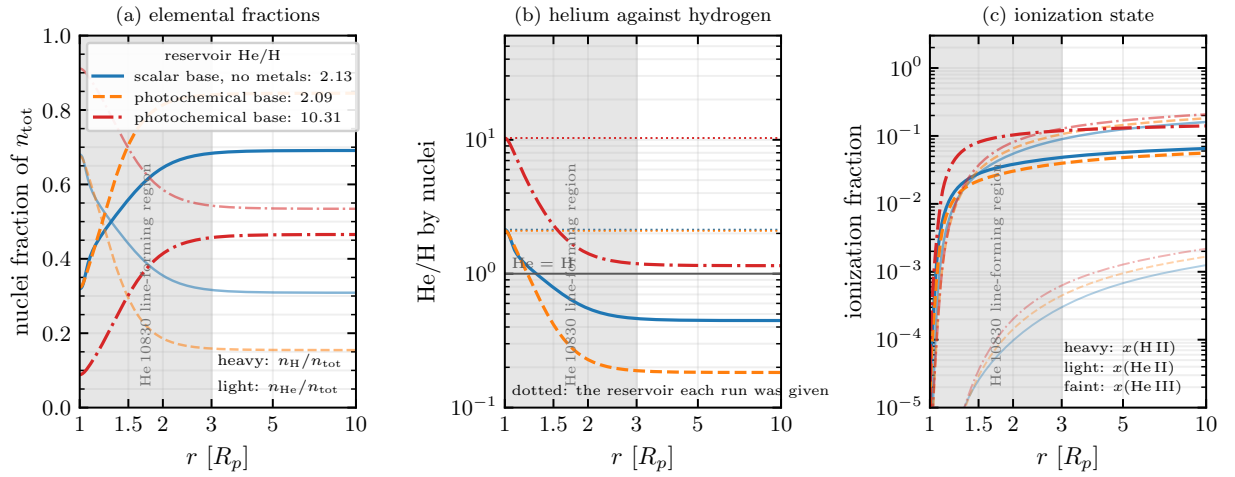


Figure 16: **Composition and ionization against radius**, for the three solutions of Table 12, from the same advection-corrected `Ion_species_adv.txt` profiles. **(a)** the hydrogen and helium nuclei fractions of n_{tot} ; **(b)** He/H by nuclei, with the reservoir each run was given as a dotted line, so the vertical drop from dotted line to curve is what element diffusion does; **(c)** the ionization fractions. n_{tot} follows the code’s `calc_ntot` convention — one particle per species, H I, H II, He I, He II, He III and every metal ion stage, electrons excluded, and He 2^3S not added because it is an excited level inside He I — so (a) does not sum to one on the photochemical base. The metals it misses are negligible: their largest share of n_{tot} anywhere in $1\text{--}30R_p$ is 3.0×10^{-4} at He/H = 2.09 and 1.7×10^{-4} at 10.31, and zero on the metal-free base. Nuclei are counted as H I + H II and He I + He II + He III. All three solutions are on the current Penning coefficient (Sect. 2.5), and the curves this figure draws barely felt the replacement: the elemental fractions moved by under 2 per cent and the ionization fractions by under 3. The re-solved ladder crosses the measurement at He/H = 9.048 (Sect. 11.7), between the two profile runs drawn here. The He 10830 line-forming region $1\text{--}3R_p$ is shaded.

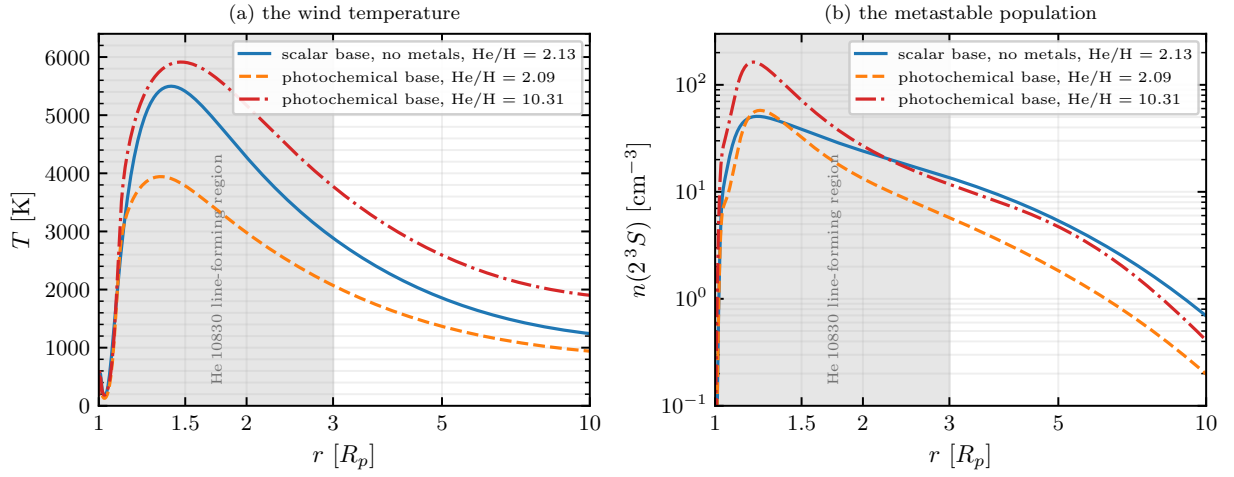


Figure 17: The thermostat, on the three solutions of Table 12. (a) temperature and (b) the metastable density, against radius, with the He 10830 line-forming region $1\text{--}3R_p$ shaded. Adding the photochemical column’s C, N and O at fixed reservoir (solid to dashed) cools the wind and strips the metastable population outside $2R_p$ — by a factor 2.5 — while the peak itself rises and the column inside $2R_p$ falls by only 16 per cent; raising the reservoir on that same base (dashed to dot-dashed) reheats it past where it started and carries the population back out. All three solutions are on the current Penning coefficient (LHS1140b/exhale/thermostat_gm25/).

is then re-solved on its own output with the profile held fixed, so that arms which stopped at different k are read at the same place. Everything else is the configuration of Sect. 11. — Photochem with the `cLima` step solved, $K_{zz} = 10^9$, a $1\mu\text{bar}$ match, the GJ 1132 proxy spectrum, a 5% residual tolerance and $k_{\text{max}} = 8$. The runs and the measurement scripts are LHS1140b/exhale/crossings_pc090/, and the ladder was solved twice — once with Photochem 0.8.4 (`crossings_gm25`) and once with 0.9.0 and the local solver corrections, which is what the chemistry-version caveat of Sect. 1 is measured on.

The earlier ladder of nine arms between 2.09 and 12.0, seeded one step at a time, is superseded by this one and is not re-tabulated. What it established and this ladder confirms is recorded where it belongs: Sect. 11.8 keeps the discarded arm that showed a converged wind inheriting its outer region from its seed.

Every arm returns a matching-level He/H equal to its reservoir to four digits, for the reason Sect. 11. measured at 2.09: at $K_{zz} = 10^9$ the eddy-mixed column reaches the microbar match, so the escape flux does not set the composition there. What the ladder adds is the line.

The crossing. The arms at 9.01 and 9.11 straddle the measured equivalent width directly, 1.1069 below it and 1.1092 above against $1.108 \pm 0.030\%$ Å, and every rung from 8.51 to 11.01 is itself inside the measurement’s 1σ band. The crossing is therefore read inside a bracket the runs measured, not off the end of a scan:

$$\text{He}/\text{H} = 9.048 \quad (8.465\text{--}10.736 \text{ on the } EW \text{ } 1\sigma \text{ interval}), \quad (9)$$

where the range carries the measurement error alone and none of the model uncertainty, and both of its edges are bracketed by solved arms as well. Two readings of the same ladder agree on the center: chord interpolation in log–log between the two straddling rungs gives 9.0484 and a quadratic in log–log gives 9.0461, 0.03 per cent apart. The

Table 13: The reservoir ladder under flux closure, on Photochem 0.9.0 with the local solver corrections. One row per arm, at the iteration k where the 5% residual criterion was met; fluxes in 10^6 g s^{-1} , the cold-trap oxygen in 10^{-7} nuclei per H interpolated to the matching level. The equivalent width is on this memo’s red-pair window, directly comparable with the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$. The 9.01 and 9.11 rungs straddle that measurement, and every rung from 8.51 to 11.01 lies inside its 1σ band. ε is the larger of the two elemental closure residuals.

reservoir	k	ε	F_{H}	F_{He}	He/H at match	$\log_{10} \dot{M}$	red [%]	FWHM [$\text{\AA}$]	EW [% $\text{\AA}$]	O/H trap
8.45	2	0.028	5.38	19.06	8.4595	7.390	3.628	0.2775	1.0743	8.33
8.5	2	0.026	5.36	19.36	8.5095	7.400	3.665	0.2775	1.0866	8.38
8.6	2	0.023	5.30	19.40	8.6096	7.400	3.672	0.2780	1.0898	8.50
8.7	1	0.049	5.26	19.59	8.7097	7.400	3.697	0.2785	1.0986	8.59
8.75	1	0.045	5.22	19.61	8.7598	7.400	3.700	0.2785	1.1001	8.67
8.9	1	0.037	5.13	19.67	8.9100	7.400	3.708	0.2785	1.1040	8.82
9.0	1	0.031	5.08	19.71	9.0101	7.400	3.714	0.2795	1.1069	8.85
9.1	1	0.026	5.02	19.74	9.1102	7.400	3.719	0.2795	1.1092	8.82
9.3	0	0.030	4.91	19.83	9.3104	7.400	3.743	0.2800	1.1187	8.72
10.6	1	0.049	4.35	20.14	10.6120	7.390	3.763	0.2825	1.1354	9.30
11.0	2	0.034	4.20	20.26	11.0124	7.390	3.777	0.2835	1.1430	9.62
observed	—	—	—	—	—	—	$1.254^{+0.22}_{-0.23}$	0.841	1.108 ± 0.030	—

crossing is also the one number that did not move when the chemistry was corrected: the 0.8.4 ladder and the 0.9.0 one give 9.0484 and 9.0461 to every digit either prints.

The center was read as an interpolation between 9.01 and 9.11, a bracket 0.10 wide in reservoir, and has since been solved directly: the closure run at reservoir He/H = 9.048 — chemistry, wind, elemental-flux measurement and under-relaxed update, then its converged iterate’s wind re-solved to its own fixed point as every arm of record was — gives $EW = 1.10800 \text{ \% } \text{\AA}$ against the 1.1079872 the chord asserts, a difference of $1.3 \times 10^{-5} \text{ \% } \text{\AA}$ or 0.0004σ (LHS1140b/exhale/confirm_gm25). *And the ladder is not smooth at this resolution.* The local power-law slope $d \ln EW / d \ln (\text{He}/\text{H})$ over the successive intervals is 1.92, 0.25, 0.70, 0.24, 0.21, 0.23, 0.19, 0.39, 0.11 and 0.18: the equivalent width rises by $0.0122 \text{ \% } \text{\AA}$ between 8.46 and 8.51 and by only 0.0048 over the three times wider interval 8.51–8.61. That step is not a lineage artifact — two arms of record re-run in a second directory reproduce to every printed digit — and it tracks the iteration count: the arms that stop at $k = 1$ carry a looser residual (0.045–0.049) than those that reach $k = 2$ (0.022–0.028). *It is the closure’s own tol = 0.05, made visible.* The trend across the ladder as a whole is a mean slope of 0.24 over 8.46–11.01, and what the sublinearity settles is that the equivalent width does *not* saturate before the measurement is reached, which the helium-rich end of the metal-free ladder of Sect. 8.4 had made a live possibility.

At the crossing the escape rate is $\log_{10} \dot{M} = 7.40$, i.e. $2.5 \times 10^7 \text{ g s}^{-1}$, a factor 8 below the $2.03^{+0.58}_{-0.67} \times 10^8 \text{ g s}^{-1}$ the p-winds retrieval imposes, and the elemental fluxes are $F_{\text{H}} = 5.1 \times 10^6$ and $F_{\text{He}} = 2.0 \times 10^7 \text{ g s}^{-1}$.

Two statements follow, and they are worth separating.

- **The closure by itself does not select a reservoir on this planet.** Every rung of Table 13 satisfies it, at elemental residuals of 2.3–4.9% against the 5% tolerance. What selects 9.05 is the line, exactly as in Sects. 7. and 8.4. What the closure contributes is that the value is now consistent with a photochemical lower atmosphere carrying its own C, N and O, rather than resting on a bare H/He base — and the price of that consistency is a helium fraction of 90 per cent by number. Read against the 1.63 the same line returns above a metal-free base, the pair measures the thermal trade of Sect. 11.6 rather than two attempts at one number.
- **The escape rate is flat along the whole ladder,** 7.39 to 7.40 in $\log_{10} \dot{M}$ over a factor 1.3 in reservoir, and 7.39–7.50 once the earlier arms down to 2.09 are included. So here too the composition is fixed by the line and not by the wind, and the $\dot{M}$ this memo quotes does not depend on where in the ladder the crossing lands.
- **A seed systematic sits under all of it.** Three arms re-closed from the neighbour below instead of from the common parent move by 0.2–1.9 per cent in equivalent width, which at the local slope is of order half a unit in reservoir — larger than the 0.10-wide bracket the crossing is read in, and invisible to every test the driver applies. The crossing quoted here is the crossing of the single-parent ladder.

The wind residual the JFNK finish reaches degrades as the reservoir grows — the line search stalls on the base contact mode and returns its best iterate — so the target was raised arm by arm and the achieved values recorded rather than smoothed over: $\|R\| = 1.5\text{--}3.1 \times 10^{-4}$ at $\|R\|_{\text{tol}} = 4.0 \times 10^{-4}$, all at $\text{info} = 0$. The test the closure actually applies is not that residual but the elemental residual ε of Table 13; the radial spread of the flux over the steady window it is measured on is 0.9–2.7%.

What the crossing does not fix, and what it inherits. The line is still too narrow: FWHM 0.280 Å at the crossing against the measured 0.841, unchanged from Sect. 8.4 and from the diffusion-off solutions, so the modelled red depth of 3.7 per cent overshoots the measured 1.254 per cent by the same factor the width falls short — which is why the composition here is inferred from the equivalent width and not from the depth. And the winds these numbers come from carry the caveat of Sect. 12. in full: they are subsonic to the $30R_p$ domain edge, their mass flux is set at the outer boundary rather than at a critical point, and the critical region is transitional, so the $\log_{10} \dot{M} = 7.40$ quoted above is *unvalidated as a continuum solution*. The He 10830 layer that fixes 9.05 is not: it sits at $\text{Kn} \leq 0.019$.

11.8 The outer region a converged wind inherits

The effect was found on the earlier ladder, at its 12.01 rung, and it is why the ladder of Table 13 is climbed from one common parent. That rung was first reached in one step from the 8.0 arm, a $1.50\times$ jump in reservoir; a control run reached the same reservoir from 11.11, a $1.08\times$ step. The two solutions agree on everything the closure and the chemistry set, and disagree on the line:

- *Same reservoir:* He/H at the match 12.013522 against 12.013520, seven digits.
- *Same lower atmosphere:* cold-trap O/H 1.0656×10^{-6} against 1.0660×10^{-6} nuclei per H.
- *Both converged:* $\text{info} = 0$ at $\|R\| = 2.533 \times 10^{-4}$ (jump-seeded) and 1.249×10^{-4} (control), both inside the 4.0×10^{-4} target the arm was run to, and both inside the closure’s own flux-spread tolerance.

- *Different line:* $EW = 1.4141$ against 1.1953% Å and a red depth of 4.741 against 3.914 per cent, an 18 per cent difference in equivalent width at fixed composition.

The difference sits entirely in the outer wind. Two diagnostics that do not depend on where a spectral window is placed separate the two solutions, and both are monotonic along the ladder except at the discarded point:

Table 14: Two sampling-free measures of the outer wind along the earlier closed ladder, from the advection-corrected profiles. r_{drop} is the first radius outside $1.5R_p$ at which T falls below half of $T(12R_p)$; the last column is the share of the He 2^3S radial column that lies outside $10R_p$. Every entry was recomputed from the run directories for this table. These arms are on the current Penning coefficient (Sect. 2.5), re-solved arm by arm in LHS1140b/exhale/xuvkzz_gm25/; the same two diagnostics on the single-parent ladder of Table 13 are $r_{\text{drop}} = 20.46$ falling to $19.47R_p$ and an outer column share of $0.0091\text{--}0.0095$, flat across every arm, with no arm standing out.

case	$r_{\text{drop}} [R_p]$	$T(12R_p)$ [K]	He 2^3S column outside $10R_p$
2.09	30.00	892	0.0117
3.0	27.63	1028	0.0100
5.0	23.76	1300	0.0087
8.0	20.80	1634	0.0088
9.7	19.79	1797	0.0096
10.3	19.47	1843	0.0097
10.7	19.47	1872	0.0098
11.1	19.47	1900	0.0098
12.0	19.47	1960	0.0102
12.0 discarded	25.83	1838	0.0207

r_{drop} moves inward monotonically as the reservoir rises, from 30.0 to $19.5R_p$, and the outer share of the metastable column stays in a narrow band, $0.0087\text{--}0.0117$, across the whole ladder. The discarded solution departs from both: $25.8R_p$ where its own reservoir gives 19.5 , and about twice the outer column share of any other rung (2.03 times its control's, 1.76 times the largest any other rung reaches). Its $T(12R_p)$ does *not* match its control's — 1838 K against 1960 , 6.2 per cent apart — so that quantity no longer carries the point that the two agree on the base; what carries it is the matching-level composition, identical to seven digits, and the profile ratio below. It is a hotter, more extended outer atmosphere sitting on an identical base.

Read as the mechanism: the outer state is inherited from the seed and the JFNK finish does not re-solve it. The density there is low enough that it contributes almost nothing to the residual norm, so $\|R\|$ cannot tell the two states apart — both report $\text{info} = 0$ at residuals inside the target, and which of the two reports the smaller one is not a property of which is right: on the current coefficient the discarded solution reports the larger residual, on the retired one it reported the smaller. He 10830 is optically thick at these columns, so a hot, extended outer region fills the line wings and raises the equivalent width even though it carries about one per cent of the metastable column.

The physical judgement has to be stated plainly. Measured on the metastable profiles, the two solutions agree to

2.9 per cent out to $11.9R_p$ and then separate abruptly — the departure passes 10 per cent at $11.95R_p$ and 50 per cent at 12.14, reaching a factor 5.0 — so the region in which they differ is $12\text{--}30R_p$, not the $20\text{--}30R_p$ an earlier draft of this paragraph gave. The exobase of these two arms is not the $18.8\text{--}25.9R_p$ of Sect. 12. either: that is the three-case range of Table 16, whose cases are two non-closure runs and the 2.09 closure arm. For this pair it is $11.81R_p$ (jump-seeded) and 14.75 (control). Putting the corrected radii together: for the solution being discarded the whole differing region lies outside its own exobase, where $\text{Kn} > 1$; for its control the innermost $2.6R_p$ of that region is nominally still collisional. *The fluid solution is not valid in most of the region that makes the difference, and in the discarded solution it is not valid anywhere in it.* The ground for discarding that solution is therefore continuity of the trend, not a physical criterion; and by the same token the absolute equivalent widths of the whole ladder carry a systematic error that the closure tolerance — a 5% flux spread — does not catch. Its size is small, because the outer region is about one per cent of the metastable column, but it is not zero. The exobase itself does not move with the Penning coefficient: across the ladder it is $11.8\text{--}18.8R_p$ re-solved against $12.1\text{--}18.8$ stored.

What was and was not tested. Path independence was checked directly at $\text{He}/\text{H} = 3.0$, from a seed $3.7\times$ away in composition and in the downward direction the ladder never takes: it returns the same r_{drop} to the digit, with everything else inside the closure tolerance. The 5.0, 8.0 and 2.09 arms were not checked this way. Separately, the 10.31 arm was re-run from the same seed in its own directory and reproduced bitwise, which fixes determinism but says nothing about the seed dependence. The practical rule the episode leaves is that a ladder scan must be climbed one step at a time; the record is docs/Update_EXHALE.md Sect. 83.

11.9 The 2025 non-detection

Cherubim et al. detect the line in 2024 and not in 2025, at a limit of 0.6 per cent in depth. The grid below asks what fraction of the fiducial XUV reproduces that. The spectrum is the fiducial GJ 1132 proxy with its flux column multiplied by the stated factor (sed/lhs1140_sed_gj1132_at_b_xuv*.txt; read_sed recomputes L_X and L_{EUV} from the file, so the luminosity lines of input.inp are provenance only). The lower atmosphere is held fixed in every case — the closure loop is not re-run, and neither is the chemistry that made the profile — so the grid isolates the wind's response to the irradiation and nothing else.

It is run from both models that reproduce the 2024 equivalent width, because neither is the other: the scalar-base solution at $\text{He}/\text{H} = 2.13$ ($EW = 1.340$, at its own $\|R\|_{\text{tol}} = 5.0 \times 10^{-3}$) and the flux-closed solution at $\text{He}/\text{H} = 9.71$ ($EW = 1.130$). Both grids are on the current Penning coefficient (Sect. 2.5), every arm re-solved to its own fixed point in LHS1140b/exhale/xuvkzz_gm25/. The replacement does not move the grid uniformly, and the direction is not the one an earlier draft of this section predicted: only the two fiducial points rise, by 18 and 8 per cent, while every scaled point *falls*, by up to a factor 4.6, so both models cross the 0.6 per cent limit at a *higher* XUV fraction than the retired coefficient gave. The result the section rests on is the comparison between the two grids, and both are made the same way.

The 0.6 per cent limit is met at about a third of the fiducial XUV in the scalar model and at about 0.42 in the flux-closed one. The two models no longer put the non-detection in the same place. In the scalar model the depth crosses 0.6 between 0.30 (0.439 per cent) and 0.33 (0.712), at 0.32; in the flux-closed model no row of the grid reaches 0.6 —

Table 15: The He 10830 line against the XUV scaling, from the two models that reproduce the 2024 measurement, every arm re-solved on the current Penning coefficient (LHS1140b/exhale/xuvkzz_gm25/). Bold marks the row at which each model is nearest the 0.6 per cent depth limit of the 2025 epoch from below. In the scalar model that row is a genuine bracket — 0.33 is above the limit and 0.30 below it — while in the flux-closed model no row of the grid reaches 0.6 at all, so its bold marks the top of the grid and its crossing lies above it.

$F_{\text{XUV}} / \text{fiducial}$	scalar base, He/H = 2.13			flux-closed, He/H = 9.71		
	red [%]	EW [% Å]	$\log_{10} \dot{M}$	red [%]	EW [% Å]	$\log_{10} \dot{M}$
1.00	4.901	1.340	7.80	3.768	1.130	7.40
0.33	0.712	0.177	7.32	0.455	0.127	6.73
0.30	0.439	0.109	7.27	—	—	—
0.25	0.067	0.018	7.23	0.320	0.084	6.55
0.20	0.010	0.0026	7.04	0.216	0.055	6.44
0.15	0.0055	0.0013	6.93	0.157	0.039	6.28
0.10	0.0023	0.0006	6.73	0.105	0.026	6.05
0.01	0.0004	0.0001	5.86	0.173	0.053	5.42

it is 0.455 at 0.33 and 0.320 at 0.25 — so its crossing sits *above* the top of the grid, at 0.42 if the top chord is continued. The two are a factor 1.31 apart, against the factor 4.6 in reservoir and 2.5 in escape rate that separates the models themselves; so the two readings remain of the same order, but the agreement is no longer close enough to be quoted as one number. On the paper’s fiducial $F_{\text{XUV}} = 33 \text{ erg cm}^{-2} \text{ s}^{-1}$, the 2025 epoch would need about 10–14 $\text{erg cm}^{-2} \text{ s}^{-1}$ or less. Both crossings are read linearly in both variables on the scaled grid alone, the fiducial being the model rather than a grid point; the same reading on the retired coefficient gave 0.31 and 0.36, so the re-solve moves both *up*, the closed one by much the more.

Two caveats belong with those numbers. The decline is far steeper in the scalar model than in the closed one — over the same 0.33 to 0.10 interval the depth falls by a factor 304 against a factor 4.3 — so the *shape* of the XUV dependence is a property of the model rather than a result, and on the current coefficient that contrast is if anything sharper than it was (290 against 3.7 before). And the lowest closed-model point, at 0.01, is not monotonic with the 0.10 one (0.173 against 0.105 per cent) on a wind that has fallen to $\log_{10} \dot{M} = 5.4$; that solution is not to be read as a limit on how faint the line can get.

11.10 The width, unmoved again

Measured the way Sect. 8.2 measures it — convolve the solved profile with a Gaussian and ask which one reproduces the observed width — the arm at the closure crossing asks for 21.95 km s^{-1} FWHM ($\sigma = 9.32 \text{ km s}^{-1}$), against 22.22 in the well-mixed limit, 22.23 with diffusion at the adopted K_{zz} , and 21.97 on the $5.1\times$ brighter proxy. Those four solutions span a factor 189 in composition and two different lower boundaries, and their kernels differ by 1.3 per cent. As solved, the red-pair FWHM at the crossing is 0.280 Å against the measured 0.841.

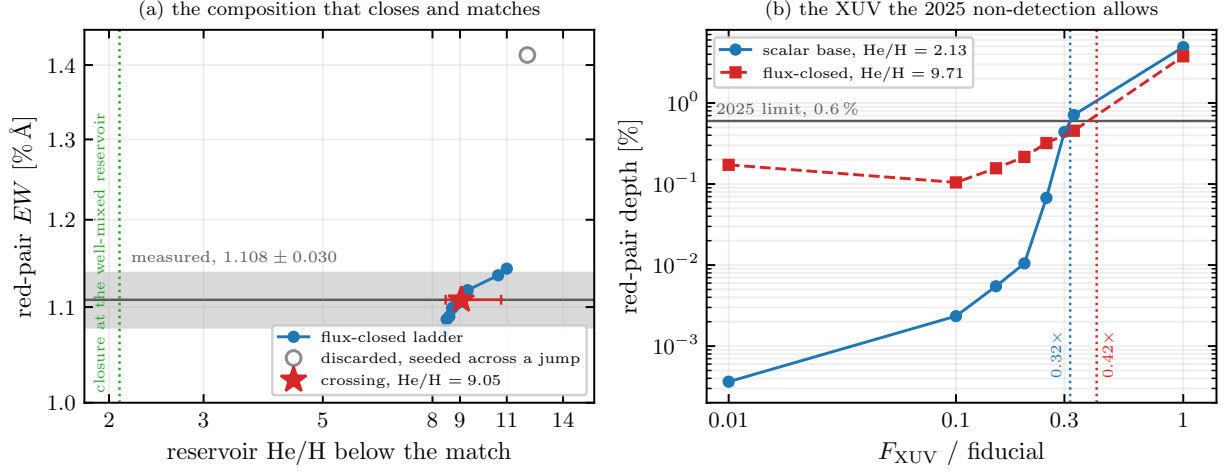


Figure 18: **The two constraints, and the epoch.** (a) The red-pair equivalent width of the flux-closed solution against the reservoir He/H it was closed at (Table 13), with the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$ as the band and the crossing at $\text{He}/\text{H} = 9.05$ as the star; the dotted line is the reservoir at which Sect. 11. closed the loop, 4.3 times lower. The reservoir runs 9.0 and 9.1 — the axis here is the reservoir, and they return $\text{He}/\text{H} = 9.01$ and 9.11 at the matching level — straddle the band, so the crossing is read inside the scanned range, and so are both edges of its 1σ interval. The star is placed at the target this memo and `results.txt` both use, $1.108 \pm 0.030 \text{ \% } \text{\AA}$, which is the digitized 1.10843 ± 0.02948 rounded; read at the unrounded value the crossing is 9.07 rather than 9.05. The open grey symbol is the discarded 12.01 solution of Sect. 11.8, from the earlier ladder: same reservoir, same base, same matching-level composition as its control, reached by a $1.50\times$ seed jump instead of a $1.08\times$ step, and differing only in an outer region that lies beyond the exobase. Every point on the curve satisfies the flux-closure condition, so it is the band and not the closure that picks a reservoir. (b) The red-pair depth against the XUV scaling (Table 15), from the scalar-base and flux-closed models that both reproduce the 2024 equivalent width. Both fall below the 0.6 per cent limit of the 2025 non-detection, the scalar-base model at about 0.32 of the fiducial irradiation and the flux-closed one at about 0.42 (dotted, one line per model), while the slope with which they get there differs by two orders of magnitude. Both crossings are read on the scaled grid alone: the scalar one falls between two drawn points, the closed one is the 0.25–0.33 chord continued past the top of the grid, so it does not coincide with where the segment drawn from 0.33 to the fiducial happens to cut the limit.

The lower-boundary axis therefore behaves like the other three: it moves the composition inference and the escape rate, and it does not move the width. At the kernel that reproduces the width the closed solution sits at a red depth of 1.235% against the observed $1.254^{+0.22}_{-0.23}$ and a blue component of 0.164% — inside the observed interval, as the crossing solutions of the other branches are, because the crossing is defined by the equivalent width. *That is a change from the earlier draft*, where the flux-closed solution presented was the one at reservoir 2.09, whose equivalent width falls a factor 2.6 short and which therefore could not be convolved into agreement at any kernel. The statement that survives is Sect. 8.5's: a solution short of absorbing power cannot be convolved into agreement, because broadening conserves equivalent width.

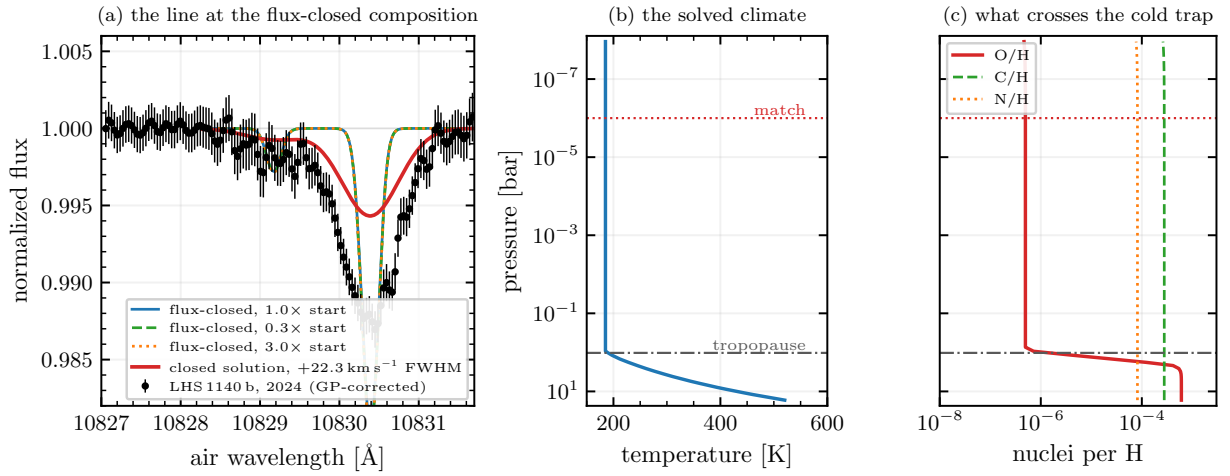


Figure 19: **The flux-closed solution and the column it stands on.** (a) The three closure arms — started a factor of ten apart in the trial escape flux — are indistinguishable, and all three fall well short of the measured line in absorbing power; the thick curve is the reference arm convolved with the 22.3 km s^{-1} FWHM Gaussian that reproduces the observed width, which then lands below the observed depth because the equivalent width was already short. Model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$, as elsewhere in this memo. (b) The solved climate of the column handed over, with the tropopause at 1.031 bar and the $1 \mu\text{bar}$ matching level marked; the stratosphere sits at the skin temperature, 185 K. (c) The elemental abundances the profile carries. Oxygen is cut by a factor 1223 at the tropopause by the water cold trap, while carbon and nitrogen cross it unchanged — and it is those two, not the helium, that make this solution's line $2.6\times$ weaker than the metal-free base at the same reservoir (Sect. 11.6; both sides of that factor are re-measured on the current Penning coefficient, and it is unchanged by the replacement). Panels (b) and (c) place the column in the closure; the file itself, every column of it, is Figs. 14 and 15.

12. Is the wind collisional where it is set?

Everything above reads the solutions as continuum solutions. A hydrodynamic wind is one only where the gas is collisional on the scale over which the flow varies, and above all through its critical point, which is where the topology — and with it the mass flux — is fixed. This section measures that. The tool is

src/utils/collisional_validity.py; it reads an existing run directory and changes nothing, and the definitions and full tables are in docs/collisional_validity.md (record: docs/Update_EXHALE.md Sects. 81 and 85). The numbers below were measured on the outputs the rest of this memo uses; nothing was re-run.

12.1 What is measured

The collisions. The four momentum-transfer limits are the ones the binary element-diffusion operator of Sect. 8.4 already uses, in the same Chapman–Enskog first approximation, so a mean free path quoted here and a diffusion coefficient used by the wind cannot drift apart: rigid-sphere neutral–neutral, non-resonant ion–neutral with the induced-dipole and rigid-core frictions added, and screened Coulomb with the Spitzer $\ln \Lambda$ for ion–ion and ion–electron. The step from a binary diffusion coefficient to a collision frequency is the definition of that coefficient itself, $\nu_{st} = n_t kT / (\mu_{st} \hat{D}_{st})$ with $\hat{D} = nD$, in which the total carrier density cancels. Species are H I, H II, He I, He II, He III and e , plus the molecular carriers where a run tracks them; He I 2^3S is a level inside the He I column and is not counted twice, and metals are dropped as trace ($< 10^{-3}$ of the particles) except in the electron budget. Two channels are deliberately absent — resonant charge exchange and electron–neutral momentum transfer — and both lengthen the mean free path, so the verdict they produce is the conservative one.

The numbers. $\lambda_s = \bar{v}_s / \nu_s$ with the harmonic sum over partners; the local structure scale is the length over which the state vector of the local Maxwellian changes,

$$\frac{1}{L^2} = \left(\frac{d \ln \rho}{dr} \right)^2 + \left(\frac{d \ln T}{dr} \right)^2 + \left(\frac{|dv/dr|}{c_s} \right)^2 + \frac{1}{r^2}, \quad L = \max \left(\frac{1}{\sqrt{\cdot}}, \Delta r_{\text{cell}} \right), \quad (10)$$

and $\text{Kn}_s = \lambda_s / L$. *The pressure is deliberately not in it.* The continuum closure expands about a local Maxwellian, which is fixed by (n, T, u) ; $p = nkT$ is a derived field, so including it would count the density and temperature gradients twice wherever they do not cancel and contribute nothing exactly where they do. It is the second half of that which bites: across a heating front ρ falls and T rises with nearly the same logarithmic slope, so $|d \ln p / dr|$ passes through a broad near-zero while the gas is changing fast, and the earlier definition $L = \min(H_p, L_v, r)$ read a scale height fifteen times longer than the density and temperature scales there. The sum is taken in quadrature rather than as a minimum for the same reason a minimum failed: it jumps to the next length whenever the field it is taken from flattens, whereas in quadrature a flat field contributes zero and L still cannot exceed the shortest individual scale. The velocity enters as $|dv/dr|/c_s$ and not $|(1/v)dv/dr|$ because the term the closure drops is the viscous stress, whose size relative to the pressure is $\sim \lambda |dv/dr|/\bar{v}$: a bulk-velocity change distorts the distribution in proportion to the thermal speed, not to the local bulk speed. That removes the Mach-number floor the old L_v needed. The floor at the local cell width says that a gradient claiming structure below one cell is measuring the mesh; it binds in three cells of one of the four runs, all inside $1.002 R_p$, and in no critical region. The *exobase* is the first crossing of $\text{Kn}_{\text{bulk}} = 1$ from below, never extrapolated beyond the domain. The *critical point* is the sonic point taken with the code’s own sound speed, $c_s = \sqrt{\gamma p / \rho}$ at $\gamma = 5/3$, and not a re-definition. The *critical region* runs from the peak of volumetric heating to the sonic point. “Collisional” means $\max \text{Kn} < 0.1$ across that region, over the bulk and over every species carrying at least 10^{-3} of the mass; where it fails the solution is reported **unvalidated**, not overestimated, because which way a collisionless critical region moves the answer is not something the continuum solution that assumed it away can decide.

12.2 The result

Table 16: Collisional validity of three representative LHS 1140 b solutions and of an HD 209458 b control, read from the advection-corrected profiles. Radii in R_p ; the LHS 1140 b domain ends at $29.05 R_p$, the HD 209458 b one at $4.15 R_p$. “Max Kn” is over the critical region, i.e. from peak heating to the sonic point, or to the outer boundary where there is no sonic point. Vintage of the four runs: heh0p55 and heh2p13_diff_kzz1e9 carry the He 2^3S Penning coefficient of García Muñoz (2025) (Update_EXHALE.md Sect. 87) but were post-processed before the advection fix of Sect. 88; flux_closure/hi/k06 and the HD 209458 b control were re-solved for neither and stand on the retired two-branch coefficient.

case	critical point	exobase	Kn = 0.1 at	max Kn	verdict
heh0p55 (no diffusion)	none, $\mathcal{M}_{\max} = 0.538$	23.57	6.69	2.36	unvalidated
heh2p13_diff_kzz1e9	none, $\mathcal{M}_{\max} = 0.554$	25.87	7.14	1.95	unvalidated
flux_closure/hi/k06	none, $\mathcal{M}_{\max} = 0.550$	18.84	5.80	3.14	unvalidated
HD 209458 b control	4.085	above 4.15	—	0.019	validated

LHS 1140 b’s wind never reaches its critical point inside the computational domain. All three solutions stay subsonic out to $29 R_p$, at $\mathcal{M}_{\max} = 0.54\text{--}0.55$. A subsonic solution is not fixed by a critical point: its mass flux is set by the outer boundary condition — and at that boundary Kn_{bulk} is already 1.35–2.38, with the exobase at 18.8–25.9 R_p , inside the domain. The continuum equations were therefore integrated through, and closed at, a region in which they do not hold. The escape rates quoted throughout this memo, $\log_{10} \dot{M} \simeq 7.5\text{--}7.8$, are **unvalidated in the collisional sense**.

Against the p-winds retrieval, whose isothermal Parker solution places its sonic point at 8–9.5 R_p : that radius lies *inside* the exobase measured here, but Kn_{bulk} there is 0.13–0.28, well past the threshold. So the finding is neither “the sonic point is above the exobase” nor “the critical region is collisional”: the critical region of LHS 1140 b is *transitional*. The scale separation the continuum equations need has been lost by the radius at which either code sets the mass flux, while the gas is not yet free molecular.

It is the neutral hydrogen that decides it, and the line does not care. In the critical region of the well-mixed solution the species split is H I 2.36, He I 1.68, H II 3.8×10^{-5} , He II 1.9×10^{-5} , He III 5.1×10^{-6} , e 7.0×10^{-4} (Fig. 20b). The neutrals carry the mass and the long mean free path; the charged species are Coulomb-held. That matters for reading the rest of this memo, because the He 10830 metastable population forms at 1.1–3 R_p , where Kn_{bulk} is $2.5\text{--}4.2 \times 10^{-4}$ at 1.1 R_p , 0.004–0.006 at 2 R_p and 0.012–0.019 at 3 R_p in every case here. **The line-forming layer is collisional by two orders of magnitude.** What is unvalidated is the wind solution’s mass flux, not the absorption Sects. 7.–8. measure, and not the 22 km s^{-1} of missing width, which is asked for at radii where the continuum description is secure.

The single-temperature energy equation is not what fails. Electron–ion energy coupling is fast wherever it matters: the largest $\tau_{ei}^E / \tau_{\text{heat}}$ over the critical region is 0.031, 0.021 and 0.018 in the three LHS 1140 b solutions and 8.3×10^{-5} in the control. Momentum coupling times are likewise short compared with the flow time — $\tau_{\text{HI}} = 924 \text{ s}$ at 8 R_p against a flow time of $1.9 \times 10^5 \text{ s}$ — which is not a contradiction but a different test: a strongly subsonic flow gives hundreds

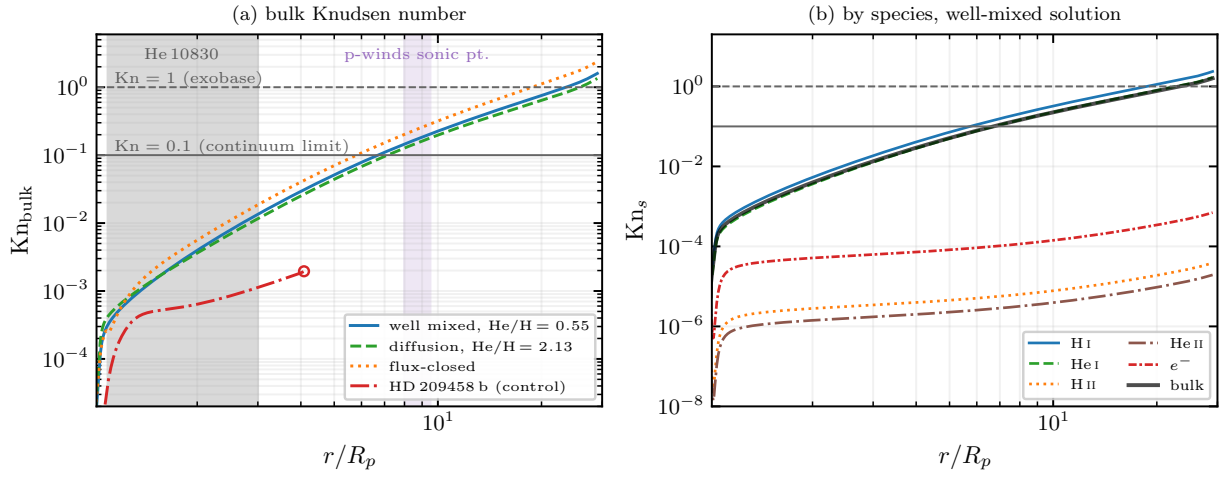


Figure 20: **Where the continuum description holds.** (a) The bulk Knudsen number of the three representative LHS 1140 b solutions and of the HD 209458 b control (open circle: its sonic point at $4.085 R_p$). Shaded: the He 10830 line-forming region, $1.1\text{--}3 R_p$, and the $8\text{--}9.5 R_p$ at which the p-winds retrieval places its sonic point. The LHS 1140 b solutions cross $Kn = 0.1$ at $5.8\text{--}7.1 R_p$ and $Kn = 1$ at $18.8\text{--}25.9 R_p$, both well inside the domain; the control stays below $Kn_{\text{bulk}} = 2.1 \times 10^{-3}$ everywhere and is collisional through its critical point, over which its largest Kn is 0.019 against 1.95–3.14 for the LHS 1140 b solutions — a factor of about 100. (b) The species split of the well-mixed solution: neutral hydrogen and neutral helium set the number, while the ions are held by Coulomb collisions 2.5–4.9 orders of magnitude more tightly and the electrons 1.2–3.4.

of collisions per flow time even at $\text{Kn} \sim 1$, and it is the Knudsen number that decides continuum validity, because the terms the closure drops (viscous stress, heat flux, the non-Maxwellian tail) scale with the gradient over a mean free path rather than with the collision count.

The kinetic scale, and why it is not a bound. Jeans escape through the measured exobase (Chamberlain & Hunten 1987, eq. 7.2.5) gives $1.0\text{--}2.0 \times 10^7 \text{ g s}^{-1}$, a factor about three below the continuum $\dot{M}$. That is a *scale*, not a bound. Jeans escape bounds a continuum rate from below only when the exobase is well bound, and here $\lambda_J(\text{H}) = 0.82\text{--}1.53$: the exobase is barely gravitationally bound, the Jeans integral is most of the Maxwellian rather than its escaping tail, and the atmosphere is in blow-off. What the number does say is that the continuum result sits within a factor of a few of what a collisionless outer atmosphere at the same density and temperature would lose, so the failure of the assumption is not hiding an order of magnitude — it is hiding a factor of a few, in an undetermined direction. On HD 209458 b, where $\lambda_J = 10.85$ and the solution is validated through its critical point, the hydrodynamic rate exceeds Jeans by 10^4 : the signature of a driven wind rather than an evaporating atmosphere.

Which profile pair is read moves the exobase, not the verdict. On the equilibrium profiles instead of the advection-corrected ones the LHS 1140 b solutions are still subsonic across the domain ($\mathcal{M}_{\text{max}} = 0.82$), still reach $\text{Kn} \sim 1$ where the outer boundary sets the flux, and the control is still collisional through its critical point. On that pair the well-mixed solution no longer crosses $\text{Kn}_{\text{bulk}} = 0.1$ at all inside the domain, reaching 0.097 at the top against 0.104, 0.191 and 5.2×10^{-4} for the other three, and the largest species Knudsen numbers are 0.99, 0.85, 1.25 and 0.013. The exobase, however, moves above $30R_p$, because the equilibrium far field is more ionized and Coulomb collisions there are far stronger than neutral ones. An exobase radius quoted from this diagnostic has to state which pair it came from; the numbers above are the advection-corrected pair throughout.

12.3 What this changes, and what it leaves open

- **The outer boundary condition closes this wind, not a critical point.** For a transonic solution the critical point fixes the mass flux and the outer boundary is passive. Here it is the other way round, and the boundary sits in gas that is already collisionless. Any $\dot{M}$ quoted for LHS 1140 b from these solutions — including the $\log_{10} \dot{M} = 7.500$ of the flux closure (Sect. 11.) and the comparison with the rate the retrieval imposes — has to carry that warning. The same applies to the p-winds number, whose sonic point sits at $\text{Kn} = 0.13\text{--}0.28$.
- **Two converged solutions are free to differ out there, and neither convergence test separates them.** The discarded 12.01 rung of Sect. 11.8 matches its control at the base, in matching-level composition to seven digits and in residual norm, and differs only outside $12R_p$ — beyond its own exobase, measured here at $11.8R_p$. Its equivalent width is 18 per cent higher for that reason alone. So this section also bounds what the flux closure and the JFNK residual can certify: neither reaches the region where the two solutions disagree, and the region they cannot reach is the one the continuum equations do not hold in.
- **The rate itself is not settled here, and cannot be.** What it would take is a kinetic or transitional-flow calculation — DSMC, or a 13-moment closure carried through the transitional region — and this repository contains neither.

It is recorded as open, not estimated.

- **The line analysis stands.** The He 10830 layer is collisional by two orders of magnitude, so the equivalent-width matching, the composition inference conditional on it, and the invariant 22 km s^{-1} width deficit are unaffected by this section. It constrains what the escape rate may be quoted as, not what the line says.

13. Status against the goal

The line is not reproduced. What stands:

- **Amount of absorption: matched, and the composition it takes is a pair set by the lower boundary** — He/H = 1.63 above a prescribed metal-free scalar base, 9.05 above a lower atmosphere solved as a photochemical column and handed over as a profile (Sect. 11.7). Both carry H/He element diffusion at the same adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$, so the base is what separates them, not the mixing. Inside the prescribed-base branch the answer is further conditional: 0.0479 well mixed on the GJ 699 proxy, 0.4245 well mixed on the GJ 1132 proxy, 3.81 wherever $K_{zz} \leq 10^4$ leaves the molecular coefficient alone. Every one of those is read inside a bracket of two solved arms, a chord and a quadratic through each ladder agree on it to 0.06 per cent or better, and each has since been solved at its own composition and returns the measured equivalent width to within 0.04σ (LHS1140b/exhale/confirm_gm25); the confirmation holds on the lineage each crossing was read on, and reseeded from the nearest neighbouring arm instead the same compositions move by about 1 per cent in He/H, the path spread quoted elsewhere here. The match is always available, so it constrains the composition only as far as the assumptions behind it are secure.
- **Those five numbers are all 0.74–0.81 of what the earlier draft carried, and the structure between them is unchanged.** The cause is one rate coefficient, replaced because it was discontinuous in temperature (Sect. 2.5). The ratios that the memo's argument is made of survive: the two lower boundaries still differ by 5.6, the well-mixed and $K_{zz} = 10^9$ crossings by 3.8, the two proxy spectra by 8.9, and the exponent $d\log(\text{He}/\text{H})/d\log K_{zz}$ decade by decade to within 0.01–0.02. *What a wrong rate coefficient moved was the level of the inference, by about a fifth; it did not move its shape.*
- **Line shape: not matched**, by a factor 3.3 in depth and width alike, and not fixable by composition. The missing ingredient is 22.22 km s^{-1} FWHM of velocity spread ($\sigma = 9.44 \text{ km s}^{-1}$), measured in Sect. 8.2 and Fig. 5; supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, but its physical origin is unidentified. The demand is unchanged under diffusion and the adopted K_{zz} : measured the same way on the EW-matched diffusive solution it is 22.23 km s^{-1} , 0.01 km s^{-1} away (Sect. 8.4); it is 21.95 km s^{-1} at the flux-closure crossing (Sect. 11.7); and it is 21.97 km s^{-1} on the $5.1\times$ brighter proxy spectrum (Sect. 10.) — 1.3 per cent of spread over four solutions that differ by a factor 189 in composition. The adiabatic metastable bump, which would have supplied it without a free parameter, is present and measured (Sect. 8.6) but sits too slow to contribute.

- **The width shortfall is not a measurement artifact, and it is not this code's alone.** Audited term by term, the instrument profile, the triplet splitting, stellar rotation, planetary rotation and orbital smearing account for at most 0.4 km s^{-1} of it; deconvolving both sides from their own instrument kernels puts the like-for-like requirement at 22.00 km s^{-1} FWHM (Sect. 8.3, docs/lhs1140b_width_measurement_audit.md). The paper's own p-winds fit, measured the same way, is itself 1.5 times narrower than the width printed in the same paper, so the shortfall appears to be a property of one-dimensional steady-state wind models rather than of this one.
- **Escape rate: an output, not a fit, and SED-set.** $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$ on the GJ 1132 proxy, a third of the value the retrieval imposes, and nearly independent of composition. On the GJ 699 proxy, $5.1 \times$ brighter in XUV, it is $3.8 \times 10^8 \text{ g s}^{-1}$ — a factor 7 higher and within 0.05 dex of the value the paper retrieves on that same proxy (Sect. 10.).
- **That escape rate is unvalidated in the collisional sense; the line is not.** All three representative solutions are subsonic to $29 R_p$ ($\mathcal{M}_{\text{max}} = 0.54\text{--}0.55$), so the mass flux is set at the outer boundary, where $\text{Kn}_{\text{bulk}} = 1.35\text{--}2.38$; the exobase is at $18.8\text{--}25.9 R_p$ and the p-winds sonic point at $8\text{--}9.5 R_p$ sits at $\text{Kn} = 0.13\text{--}0.28$, so the critical region of this planet is transitional for both codes. The He 10830 layer at $1.1\text{--}3 R_p$ is collisional by two orders of magnitude ($\text{Kn} \leq 0.019$), and an HD 209458 b control validates cleanly at a factor 100 lower Kn (Sect. 12.). What the true rate is needs a kinetic or transitional-flow calculation, which this repository does not contain.
- **Composition: base-limited and SED-limited, and weakly K_{zz} -limited.** In the well-mixed limit the equivalent-width crossing is $\text{He}/\text{H} = 0.4245$ on one proxy spectrum and 0.0479 on the other, helium-rich against helium-poor, with the measurement unable to separate them. With binary H/He diffusion active the crossing rises, and the whole K_{zz} axis has now been walked: $\text{He}/\text{H} = 3.81$ on the plateau at $K_{zz} \leq 10^4$, then 3.79, 3.62, 3.08, 2.28, 1.63, 1.14 and 0.85 at 10^5 through $10^{11} \text{ cm}^2 \text{ s}^{-1}$ (Table 2, Fig. 6) — a factor 2.0 across the range the literature leaves open, 4.5 across the entire axis, against the factor 8.9 the SED carries. Those are all conditions inside the prescribed-base branch. Replacing that base with the solved photochemical column moves the crossing to 9.05 (Sect. 11.7), so end to end the inference spans 0.0479 to 9.05, a factor 189. So any composition read off this line, the paper's included, is strongly conditional on the assumed XUV spectrum and on what is placed at the lower boundary, and weakly conditional on the eddy coefficient, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all.
- **Composition as an output, on one axis, and the lower boundary raised to an axis of its own in its place.** With the lower atmosphere solved as a photochemical column and the elemental fluxes iterated to continuity across the $1 \mu\text{bar}$ match, He/H at the match is a solution: 2.09235, single-valued to 5.5×10^{-6} from trial fluxes a factor of ten apart, at $\log_{10} \dot{M} = 7.500$ (Sect. 11.). At the adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ that output is the well-mixed value — only 2×10^{-4} of it comes from the escape flux — so the conditionality moves onto K_{zz} rather than away. Above the match the wind still fractionates strongly, $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir's 8.31. The handoff also brings the column's C, N and O, which take $\log_{10} \dot{M}$ from 7.80 to 7.48 and the red-pair depth down by a factor 2.6; **at that base the flux-closed solution at 2.09 no longer reproduces the equivalent width**, 0.522 against the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$ (the comparison is at fixed reservoir).
- **The composition the line needs above a photochemical base:** $\text{He}/\text{H} = 9.048$. A ladder of eleven flux-closed

solutions over reservoirs 8.45–11.0, all from one common parent and each arm’s wind re-solved to its own fixed point, has two rungs that straddle the measured equivalent width directly, 9.01 at $EW = 1.1069$ and 9.11 at 1.1092 against $1.108 \pm 0.030 \text{ } \%$ Å, and every rung from 8.51 to 11.01 is inside the measurement’s 1σ band. The crossing is 9.048, with 8.465–10.736 for that band, both edges of it bracketed by solved arms too (Sect. 11.7, Table 13). The crossing has since been solved directly at 9.048 and returns $EW = 1.10800 \text{ } \%$ Å, on the target to $1.3 \times 10^{-5} \text{ } \%$ Å; chord and quadratic readings of the bracket agree on it to 0.03 per cent (LHS1140b/exhale/confirm_gm25). *It is also the number that did not move when the chemistry was corrected:* solved on Photochem 0.8.4 and again on 0.9.0 with the local solver corrections, which take the deepest-level elemental residual from 2×10^{-5} to 10^{-14} , the crossing comes back 9.0484 and 9.0461 both times. Two things follow: the closure condition alone does not select a reservoir here — every rung satisfies it at 2.3–4.9% against its 5% tolerance, so the line selects, and what the closure adds is consistency with a photochemical lower atmosphere carrying its own C, N and O — and $\dot{M}$ is flat across the whole ladder, $\log_{10} \dot{M} = 7.39\text{--}7.40$, i.e. an eighth of the retrieval value at the crossing. Both numbers carry the width shortfall (FWHM 0.280 against 0.841 Å) and the collisional caveat above, and a seed systematic of order half a unit in reservoir that no test the driver applies can see. This is the answer on the solved-column branch; 1.63 is the answer to the same question above a prescribed metal-free base, and neither is an independent prediction of the model.

- **A wind inherits its outer region from its seed, and neither convergence test sees it.** Two solutions at the same 12.01 reservoir, one seeded across a $1.50\times$ jump and one stepped up by $1.08\times$, agree at the base, in matching-level composition to seven digits and in cold trap, both converge at $\text{info} = 0$, and differ by 18 per cent in equivalent width because one carries a hotter and more extended atmosphere outside $12R_p$. The first is discarded here on trend continuity, not on a physical criterion, because the discarded solution’s own exobase sits at $11.8R_p$, inside that range, and the continuum solution is not valid beyond it (Sects. 11.8, 12.). The same argument puts a systematic error on the absolute equivalent widths of the whole ladder that the flux-closure tolerance does not catch — small, because the outer region is about one per cent of the metastable column, but not zero. Practically: a ladder scan is climbed from one common parent, which is how the ladder of Table 13 was run.
- **Why the same helium absorbs differently over the two bases, measured.** Adding the column’s C, N and O at fixed reservoir takes radiative cooling from 6.6–9.6% of the heating to 71% at $1.2R_p$ (95% of it metal-line cooling, largest single contributor neutral C I) and T_{max} from 5498 to 3943 K; the metastable peak rises, the column inside $2R_p$ falls by 16% ($34.6 \rightarrow 29.1$), and the column outside $2R_p$ falls by 2.5, close to the 2.6 the equivalent width falls by. Raising the reservoir to 10.31 on that same base returns T_{max} to 5914 K — past the metal-free 5498 — with the mean mass per particle at $2R_p$ up by 1.68 and the heating rate divided by the particle density up by 2.01, and with the C/N/O density diluted along with the hydrogen it is referred to. In the metastable balance the peak ratio $57.7 \rightarrow 163.4$ splits as a source up by 1.77 against a lifetime up by 1.60, the latter because Penning ionization on H I — 88% of the sink at 2.09, 54% at 10.31 — slows as hydrogen is thinned (Sect. 11.6, Table 12, Fig. 17). All three solutions are re-solved on the current Penning coefficient. *Read as an interpretation*, the ladder is a thermal trade between the metal content of the base and the helium content above it, and one line cannot separate them.

- **The 2025 non-detection sits at 0.32 of the fiducial XUV in one model and 0.42 in the other.** Scaling the irradiation with the lower atmosphere held fixed, the red-pair depth falls below the 0.6 per cent limit of the 2025 epoch at $\simeq 0.32$ of fiducial in the scalar-base model and at $\simeq 0.42$ in the flux-closed one (Sect. 11.9, Table 15) — about $10\text{--}14 \text{ erg cm}^{-2} \text{ s}^{-1}$ or less. Both grids are now on the current Penning coefficient. The two models are a factor 1.31 apart there, having differed by 1.2 before the re-solve, so they no longer agree closely enough for the crossing to be quoted as a single number; and they do not agree at all on the slope with which they get there, which is a property of each model and not a result.
- **Not verified: the second-chemistry-code cross-check.** VULCAN did not converge on this cold helium-rich column — its convergence measure rose against the criterion with the timestep pinned, the limiting cell being H_2O at the cold trap — so no composition from a second code, and no cross-code spread, is quoted (Sect. 11.).
- **One solution is not converged to tolerance.** $\text{He}/\text{H} = 100$ stagnates at $\|R\| = 1.18 \times 10^{-3}$ from every seed tried; neighbours bracket it smoothly, so it is retained with the residual recorded.

Four axes have now been walked — the assumed XUV spectrum, the mixing assumption with its eddy coefficient, composition itself, and what is placed at the lower boundary — and each separates the two questions rather than answering them together. The last of them is the one that splits the answer in two rather than shifting it: 1.63 above a prescribed metal-free base, 9.05 above a solved photochemical column. Every one of them moves the composition the equivalent width implies, two of them by a large factor; none of them moves the width. What is left as a possible source of the missing 22 km s^{-1} is therefore what it was: velocity fields that are three-dimensional, or heating a 1D hydrodynamic energy equation does not carry. The adiabatic-cooling metastable bump, the one candidate that would have needed no free parameter, has been looked for on both spectra and does not supply it for this host (Sect. 8.6).

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